

Source folder: /Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile
Files included: 71

File: EASY_RESULTS_ACCESS.md

Easy Results Access Guide

■ Results Now Saved to Downloads Folder!

The app now saves all results to an **easily accessible location**:

■ Location

```
```\n/storage/emulated/0/Download/FL_Results/\n```
```

This is the standard **Downloads folder** on Android, which you can access using any file manager app!

---

#### ## ■ How to Access Files (3 Easy Ways)

##### ### Method 1: Android File Manager (Easiest!)

1. **Open File Manager** on your Android device
2. **Tap "Downloads"** (or "Internal Storage" → "Download")
3. **Look for "FL\_Results" folder**
4. **Open the folder** - you'll see:
  - ``dp_inf_..._device_XXX_t1234567890.json`` (full results)
  - ``dp_inf_..._device_XXX_t1234567890_summary.csv`` (summary table)

**That's it!** You can now:

- Copy files to your PC via USB
- Share files via email/WhatsApp/Drive
- Open and view files directly

---

##### ### Method 2: Share Button (Built-in!)

1. In the app, click **"Show Results Location"** button
2. Click **"Share Files"** button
3. Choose where to share:
  - Email
  - Google Drive
  - WhatsApp
  - Bluetooth
  - Any app that accepts files

**Super easy!** The app will share the latest JSON file automatically.

---

##### ### Method 3: ADB (For Developers)

If you have ADB set up:

```
```\nbash\n# Pull all results\nadb pull /storage/emulated/0/Download/FL_Results/ ./mobile_results/\n\n# Or pull specific file\nadb pull /storage/emulated/0/Download/FL_Results/dp_inf_*.json\n```\n\n---
```

■ What Files Are Saved?

- ##### ### JSON File (``experiment_id.json``)
- **Full detailed data** for analysis
 - All training rounds
 - Metrics, resource usage, timestamps
 - Same format as Python experiments

```
### CSV File (`experiment_id_summary.csv`)
- **Quick summary table**
- One row per training round
- Easy to open in Excel/Google Sheets
- Perfect for quick analysis

---

## ■ Finding Your Files

### File Naming
Files are named with the experiment ID:
```
dp_inf_alpha_null_dim_16_clients_1_device_Samsung_Galaxy_t1701234567890.json
```

The name includes:
- DP epsilon (dp_inf = no privacy)
- Embedding dimension
- Device ID
- Timestamp

### Viewing Files
- **JSON**: Open with any text editor or JSON viewer
- **CSV**: Open with Excel, Google Sheets, or any spreadsheet app

---

## ■ New Features

1. ■ **Downloads Folder** - Easy access via file manager
2. ■ **Share Button** - Share files directly from app
3. ■ **File List** - See all saved files in the app
4. ■ **Better Path Display** - Shows exact location

---

## ■ Quick Start

1. **Run training rounds** in the app
2. **After each round**, files are automatically saved
3. **Open File Manager** → Downloads → FL_Results
4. **Copy files** to your PC for analysis!

---

## ■ Example Workflow

1. Connect app to server ■
2. Run 3 training rounds ■
3. Open File Manager ■
4. Go to Downloads/FL_Results ■
5. Copy both JSON and CSV files ■
6. Transfer to PC (via USB/email/Drive) ■
7. Analyze in Python/Excel ■

---

## ■ Troubleshooting

**Q: Can't find FL_Results folder?**
- Make sure you've run at least one training round
- Check if Downloads folder exists
- Try using "Share Files" button instead

**Q: Files not appearing?**
- Check app logs for errors
- Verify app has storage permissions
- Try running training round again

**Q: Want to change location?**
- Currently hardcoded to Downloads folder
- Most accessible location for users
```

- Can be modified in `metrics\_collector.dart` if needed

■ For Your Thesis

Now you can easily:

1. ■ Collect results from mobile devices
2. ■ Access files without ADB
3. ■ Share results between devices
4. ■ Combine with Python experiment results
5. ■ Analyze everything together

****Perfect for thesis data collection!**** ■

File: FIX_PATH_PROVIDER_ERROR.md

Fix for "Unable to establish connection on channel: path_provider" Error

What I Fixed

1. ■ Added storage permissions to AndroidManifest.xml
2. ■ Made metrics collector initialization non-blocking (won't prevent connection)
3. ■ Added fallback storage options (external storage, temporary directory)
4. ■ Better error handling

How to Apply the Fix

Step 1: Stop the app completely

- Close the app on your device
- Or press `q` in the terminal where `flutter run` is running

Step 2: Rebuild the app (IMPORTANT!)

You MUST do a full rebuild, not just hot reload:

```
```powershell
cd C:\Users\jonat\PycharmProjects\master_thesis\federated_learning_in_mobile
```

```
Clean build (optional but recommended)
flutter clean
```

```
Get dependencies
flutter pub get
```

```
Rebuild and run
flutter run
```
```

****DO NOT use hot reload**** (`r` key) - this won't fix the plugin issue!

Step 3: Try connecting again

The connection should work now. Even if metrics collection fails, the connection will still work.

What Changed

Before:

- Metrics collector failure blocked connection ■
- No fallback storage options ■

After:

- Connection works even if metrics fail ■
- Multiple storage fallback options ■
- Better error messages ■

If It Still Doesn't Work

Option 1: Disable metrics temporarily

If you just want to test the connection, you can comment out metrics initialization in `lib/main.dart`:

```
```dart
// Temporarily disable metrics to test connection
// _metricsCollector = MetricsCollector(experimentId: experimentId);
```

```
// await _metricsCollector!.initialize();
```

### Option 2: Check device/emulator
Make sure you're using a real device or properly configured emulator:

```powershell
flutter devices
```

### Option 3: Uninstall and reinstall
Sometimes old app data causes issues:

```powershell
Uninstall the app
adb uninstall com.example.federated_learning_in_mobile

Reinstall
flutter run
```

## Expected Behavior After Fix

1. ■ App launches without errors
2. ■ "Connect to Server" works
3. ■ Connection succeeds even if metrics initialization has issues
4. ■ You'll see warnings in logs, but connection continues
5. ■ Training rounds work normally

## Testing

After rebuilding, try:

1. Connect to server - should work ■
2. Run training round - should work ■
3. Check logs for metrics status:
   - ■ Good: "Metrics collector initialized"
   - ■■ Warning: "Metrics collector initialization failed" (but connection still works)
   - ■ Error: Connection blocked (then we need more fixes)

Let me know what happens after the full rebuild!
```

File: METRICS_COLLECTION.md

Metrics Collection Guide

The mobile app now automatically collects and saves training metrics in the same format as the Python experiments.

How It Works

- **Automatic Collection****: When you connect to the server and run training rounds, metrics are automatically collected.
- **Saved After Each Round****: After each training round completes, results are saved to:
 - JSON file (full detailed data)
 - CSV file (summary for easy analysis)
- **File Location****: Results are saved in the app's documents directory:


```

/data/data/com.example.federated_learning_in_mobile/app_flutter/fl_results/
      
```

What Metrics Are Collected

- ### Per Training Round:
- Training loss
 - Number of samples
 - Training time (ms)
 - Resource metrics:
 - Battery level and drain
 - Memory usage
 - CPU usage
 - Network bandwidth

Experiment Configuration:

- Number of users and items
- Embedding dimension
- Client ID
- Device ID
- Server URL

File Names

Files are named using the experiment ID format:

```

dp\_inf\_alpha\_null\_dim\_16\_clients\_1\_device\_XXX\_t1234567890.json

dp\_inf\_alpha\_null\_dim\_16\_clients\_1\_device\_XXX\_t1234567890\_summary.csv

```

The experiment ID includes:

- DP epsilon (dp_inf = no privacy)
- Alpha (heterogeneity parameter)
- Embedding dimension
- Number of clients
- Device ID
- Timestamp

Accessing Results

Option 1: Using ADB (Recommended)

```bash

# Pull all results from device

adb pull /data/data/com.example.federated\_learning\_in\_mobile/app\_flutter/fl\_results/ ./mobile\_results/

# Or pull specific file

adb pull /data/data/com.example.federated\_learning\_in\_mobile/app\_flutter/fl\_results/dp\_inf\_\*.json

```

Option 2: Using Android File Manager

1. Connect device via USB
2. Enable file transfer mode
3. Navigate to: Android/data/com.example.federated_learning_in_mobile/files/fl_results/
4. Copy files to your computer

Option 3: Using the App

1. After running at least one training round
2. Click "Show Results Location" button
3. Copy the file path shown

File Formats

JSON Format

The JSON file contains all collected data in a structured format:

```json

```
{
 "experiment_id": "...",
 "timestamp": "2025-11-29T...",
 "config": {
 "num_users": 50,
 "num_items": 4032,
 ...
 },
 "rounds": [
 {
 "round": 1,
 "train_loss": 0.45,
 "resource_metrics": {
 "battery_drain": 2.5,
 "training_time_ms": 150,
 ...
 },
 ...
 },
 ...
],
 "client_metrics": [...],
 "mobile_metrics": [...]
}
```

...

### ### CSV Format

The CSV file is a summary table with key metrics per round:

- Round number
- Train Loss
- Training Time (MS)
- Battery Drain
- Number of Samples
- And other metrics

### ## Using Results for Thesis

The mobile results can be:

1. **\*\*Combined with Python results\*\*** - Same format, easy to merge
2. **\*\*Analyzed separately\*\*** - Mobile-specific resource metrics
3. **\*\*Compared\*\*** - Mobile vs Python client performance

### ### Analysis Example

```
```python
import json
import pandas as pd

# Load mobile results
with open('mobile_results/dp_inf_..._device_XXX.json') as f:
    mobile_data = json.load(f)

# Extract metrics
rounds = mobile_data['rounds']
battery_drain = [r['resource_metrics']['battery_drain'] for r in rounds]
training_times = [r['resource_metrics']['training_time_ms'] for r in rounds]

# Plot resource usage
import matplotlib.pyplot as plt
plt.plot(battery_drain, label='Battery Drain (%)')
plt.plot(training_times, label='Training Time (ms)')
plt.legend()
plt.show()
```
```

### ## Troubleshooting

#### \*\*Q: Files not appearing?\*\*

- Make sure you've run at least one training round
- Check app logs for errors
- Verify app has storage permissions

#### \*\*Q: Can't access files via ADB?\*\*

- Make sure USB debugging is enabled
- Try: ``adb shell run-as com.example.federated_learning_in_mobile ls app_flutter/fl_results/``

#### \*\*Q: Want to reset/clear old results?\*\*

- Uninstall and reinstall the app (clears app data)
- Or manually delete files via ADB

### ## Next Steps

1. Run training rounds from mobile app
2. Collect results using ADB
3. Combine with Python experiment results
4. Analyze for your thesis!

## File: README.md

### # Federated Learning Mobile Client

A Flutter-based Android application for participating in federated learning experiments for movie recommendation systems.

### ## Overview

This mobile client allows Android devices to participate as federated learning clients in a distributed training system. It implements:

- Matrix Factorization model in pure Dart
- Federated learning client logic
- Resource monitoring (battery, CPU, memory)
- Integration with Python FastAPI server

## ## Features

- ■ Connect to federated learning server
- ■ Download global model parameters
- ■ Train locally on device data
- ■ Upload model updates to server
- ■ Resource usage monitoring
- ■ Real-time training metrics display

## ## Prerequisites

1. **Flutter SDK** (3.9.2 or higher)  
``bash  
flutter --version  
``
2. **Android Studio** or Android SDK for development
3. **Python Server** running (see parent directory `server.py`)
  - Server should be accessible from your device
  - Use ngrok or local network IP for mobile access

## ## Setup Instructions

### ### 1. Install Dependencies

```
``bash
cd federated_learning_in_mobile
flutter pub get
``
```

### ### 2. Configure Server URL

1. Find your server's IP address:
  - **For local network**: Use your PC's local IP (e.g., `192.168.1.100:8000`)
  - **For ngrok**: Use the ngrok URL (e.g., `https://abc123.ngrok.io`)
2. Update the server URL in the app when you launch it, or modify the default in `lib/main.dart`:

```
``dart
text: 'http://YOUR_IP:8000', // Replace with your server address
``
```

### ### 3. Build and Run

#### #### On Android Device:

```
``bash
Connect your Android device via USB with USB debugging enabled
flutter devices # Check if device is detected
flutter run
``
```

#### #### On Android Emulator:

```
``bash
Start an Android emulator from Android Studio
flutter emulators # List available emulators
flutter run
``
```

## ## Usage

### ### 1. Start the Python Server

First, start your federated learning server:

```
``bash
In the parent directory
python server.py
```

```
...
```

Or use the helper script:

```
```bash
python start_server.py
```
```

**\*\*Important\*\*:** Make sure the server is accessible from your mobile device:

- If using local network: Ensure both devices are on the same WiFi network
- Use your PC's local IP address (not `localhost`)
- Example: `http://192.168.1.100:8000`

### ### 2. Launch the Mobile App

1. Open the app on your Android device/emulator
2. Enter your server URL (e.g., `http://192.168.1.100:8000`)
3. Enter a unique Client ID (default is auto-generated)
4. Click "Connect to Server"

### ### 3. Run Training Rounds

Once connected:

1. Click "Run Training Round"
2. The app will:
  - Fetch the global model from server
  - Train locally on sample data
  - Upload updated parameters
  - Display metrics and resource usage

## ## Architecture

```
...
```

```
lib/
■■■ main.dart # Main UI and app entry point
■■■ models/
■ ■■■ matrix_factorization.dart # Matrix Factorization model (pure Dart)
■■■ services/
■ ■■■ api_client.dart # HTTP client for server communication
■ ■■■ fl_client.dart # Federated learning client logic
■■■ utils/
 ■■■ model_encoder.dart # Model parameter encoding/decoding
 ■■■ resource_monitor.dart # Resource usage monitoring
...
```

## ## Integration with Python Server

The mobile client communicates with the Python FastAPI server using JSON-based parameter transfer:

- **\*\*Model Parameters\*\*:** Base64-encoded Float32 arrays
- **\*\*Endpoints\*\*:** `/register`, `/global-params-json`, `/upload-params-json`
- **\*\*Wire Format\*\*:** Compatible with Python server's portable JSON format

## ## For Thesis Experiments

### ### Running Experiments with Multiple Devices

1. **\*\*Device 1\*\*:** Launch app, connect to server, set Client ID to `android\_device\_1`
2. **\*\*Device 2\*\*:** Launch app, connect to server, set Client ID to `android\_device\_2`
3. **\*\*Server\*\*:** Initialize model and run aggregation rounds

### ### Data Collection

The app collects and displays:

- Training metrics (loss, samples, training time)
- Resource metrics (battery level, battery drain, device info)
- Network usage (implicit in upload/download)

For detailed logging, check the app's log output or export metrics.

### ### Combining with Python Clients

You can run hybrid experiments:

- Multiple Python clients (for large-scale sweeps)
- 2+ Android devices (for real mobile validation)



- All connect to the same server
- Server aggregates all updates together

## ## Troubleshooting

### ### Connection Issues

- \*\*Problem\*\*:** Cannot connect to server
- **\*\*Solution\*\*:**
    - Ensure server is running
    - Check server URL is correct (use IP, not localhost)
    - Verify both devices are on same network
    - Check firewall settings

- \*\*Problem\*\*:** Connection timeout
- **\*\*Solution\*\*:**
    - Use ngrok for external access
    - Check server logs for errors
    - Verify server port (default: 8000)

### ### Model Loading Issues

- \*\*Problem\*\*:** Model parameters not loading correctly
- **\*\*Solution\*\*:**
    - Verify Float32 encoding compatibility
    - Check server model dimensions match client expectations
    - Review model\_encoder.dart for byte order (endianness)

### ### Resource Monitoring

- \*\*Problem\*\*:** Battery/memory metrics not accurate
- **\*\*Solution\*\*:**
    - Resource monitoring uses platform APIs
    - For detailed profiling, use Android Profiler
    - CPU/memory monitoring may require additional permissions

## ## Development Notes

### ### Model Compatibility

The Dart Matrix Factorization model is designed to be compatible with the Python PyTorch version:

- Same embedding dimensions
- Same initialization scheme
- Same forward pass logic
- Parameter format matches server expectations

### ### Performance Considerations

For mobile devices:

- Model size should be small (embedding\_dim  $\leq$  32 recommended)
- Local training epochs should be minimal (1-2 epochs)
- Batch size should be small (16-32)
- Use sample data sets for mobile devices (5-50 samples)

### ## Future Enhancements

- [ ] Load actual MovieLens data from device storage
- [ ] Implement DP-SGD for privacy-preserving training
- [ ] Add background training capability
- [ ] Export metrics to CSV/JSON for analysis
- [ ] Add visualization of training progress
- [ ] Implement model checkpointing

## ## Thesis Integration

This mobile client addresses the requirement for **\*\*physical device participation\*\***:

1. **\*\*Real Device Validation\*\***: Test actual mobile constraints (battery, memory, network)
2. **\*\*Resource Profiling\*\***: Collect real-world resource usage metrics
3. **\*\*Mobile Deployment\*\***: Demonstrate feasibility on actual Android devices
4. **\*\*Hybrid Experiments\*\***: Combine Python simulation with real mobile devices

## ## License

This project is part of a Master's thesis research project.

## ## Contact

For questions or issues, refer to the main thesis repository.

## File: README\_DEMO\_RECOMMENDATIONS.md

---

### # Demo Recommendations Flow

This mobile app now includes a presentation-ready recommendation demo:

- Dashboard remains unchanged for training metrics and experiment controls.
- A separate `Movie Recommendations` screen shows ranked recommendations.
- `Demo Mode: Train and Show Recommendations` runs one round and navigates automatically.

## ## Data Source for Titles

Movie names are loaded from:

- `assets/movielens\_titles.csv`

This file maps model `item\_id` (0-based) to MovieLens titles.

## ## Quick Demo Steps

1. Connect to server in the dashboard.
2. Press `Demo Mode: Train and Show Recommendations`.
3. Show:
  - Dashboard logs and metrics
  - Recommendations screen with title, score explanation, seen/unseen status

## ## Rebuild title asset (optional)

Helper script:

- `tooling/build\_movie\_titles\_asset.py`

## File: SETUP\_COMPLETE.md

---

### # ■ Federated Learning Mobile Client - Setup Complete!

## ## What Has Been Built

### ### ■ Complete Flutter Application

1. **Matrix Factorization Model** (`lib/models/matrix\_factorization.dart`)
  - Pure Dart implementation
  - Compatible with Python server
  - Supports training and prediction
2. **API Client** (`lib/services/api\_client.dart`)
  - HTTP communication with FastAPI server
  - Register, fetch model, upload parameters
  - Error handling
3. **Federated Learning Client** (`lib/services/fl\_client.dart`)
  - Complete FL workflow
  - Local training
  - Model synchronization
4. **Resource Monitoring** (`lib/utils/resource\_monitor.dart`)
  - Battery level tracking
  - Device information
  - Resource metrics collection
5. **Model Encoder** (`lib/utils/model\_encoder.dart`)
  - Base64 encoding/decoding
  - Float32 parameter encoding
  - Compatible with Python server format
6. **Mobile UI** (`lib/main.dart`)
  - Server configuration

```
- Training controls
- Real-time metrics display
- Logs viewer

7. **Android Permissions**
 - Network access configured
 - Ready for deployment

Next Steps

1. Install Dependencies

```bash
cd federated_learning_in_mobile
flutter pub get
```

2. Test Build

```bash
# Check Flutter setup
flutter doctor

# Build APK (to verify no errors)
flutter build apk --debug
```

3. Run on Device/Emulator

```bash
# List available devices
flutter devices

# Run on connected device
flutter run

# Or run on specific device
flutter run -d <device_id>
```

4. Connect to Server

1. **Find your server URL:**
 - Local network: Your PC's IP address (e.g., `192.168.1.100:8000`)
 - ngrok: Your ngrok URL

2. **In the app:**
 - Enter server URL
 - Enter unique client ID (e.g., `android_device_1`)
 - Click "Connect to Server"

3. **Verify connection:**
 - Status should show "Connected"
 - Logs should show registration success

5. Run Training Round

1. Click "Run Training Round"
2. Watch logs for progress
3. Check metrics after completion

Testing Checklist

- [] App builds without errors
- [] App launches on device/emulator
- [] Can connect to server (local network or ngrok)
- [] Model parameters download correctly
- [] Training runs successfully
- [] Parameters upload to server
- [] Resource metrics are displayed

Troubleshooting

Build Errors:
```

```

```bash
flutter clean
flutter pub get
flutter pub upgrade
```

Connection Issues:
- Use your PC's IP address (not `localhost`)
- Ensure server is running
- Check firewall settings
- Try ngrok as alternative

Model Loading Issues:
- Verify server model dimensions
- Check server logs
- Ensure Float32 encoding is compatible

For Thesis Experiments

See `EXPERIMENT_GUIDE.md` in the parent directory for:
- How to run Python + Android experiments together
- Data collection strategies
- Analysis workflow
- Thesis integration guidance

File Structure

```
federated_learning_in_mobile/
├── lib/
│   ├── main.dart                # Main UI
│   ├── models/
│   │   └── matrix_factorization.dart # MF model
│   ├── services/
│   │   ├── api_client.dart        # Server communication
│   │   └── fl_client.dart         # FL logic
│   └── utils/
│       ├── model_encoder.dart     # Parameter encoding
│       └── resource_monitor.dart  # Resource tracking
├── android/
├── pubspec.yaml                 # Dependencies
└── README.md                    # Documentation
```

Ready for Thesis! ■

You now have:
- ■ Working Android federated learning client
- ■ Integration with Python server
- ■ Resource monitoring
- ■ Complete documentation
- ■ Experiment guide

You're ready to start collecting data for your thesis!

```

## File: UI\_REDESIGN\_SUMMARY.md

```

■ Federated Learning Mobile App - UI Redesign

Overview
Successfully redesigned the mobile app dashboard to match the target design with a modern, dark theme interface.

Changes Made

■ Theme & Colors
- **Dark Navy Background**: `#1A2332` (main background)
- **Card Surface Color**: `#212D3F` (elevated surfaces)
- **Cyan Accent**: `#00BCD4` (primary actions, highlights)
- **Green Status**: `#4CAF50` (status indicators)
- **Dark Console**: `#0D1117` (log terminal background)

■ Layout Changes

```

```
1. **Header Section**
- ■ Removed AppBar
- ■ Added custom header with "Federated Client" title
- ■ Added "Researcher Edition v2.4" subtitle
- ■ Added settings icon button in top-right corner
- ■ Modern spacing and typography

2. **Server Configuration Card**
- ■ Section header with DNS icon
- ■ Uppercase "SERVER CONFIGURATION" label
- ■ Cyan-highlighted "SERVER URL" label
- ■ Grey "CLIENT ID" label
- ■ Dark input fields with proper styling
- ■ Large cyan "CONNECT TO SERVER" button with lightning icon
- ■ Proper disabled states

3. **Status Section Card**
- ■ Green dot status indicator
- ■ "Status: Ready" text
- ■ "IDLE" / "TRAINING" badge in top-right
- ■ Two large square action buttons side-by-side:
 - "Run 1 Round" with play icon
 - "Run 10 Rounds" with fast-forward icon
- ■ Cyan icons on dark button backgrounds
- ■ Upload and Location buttons (when training is complete)

4. **Live Logs Section**
- ■ "LIVE LOGS" header with TV icon
- ■ "CLEAR CONSOLE" button
- ■ Dark terminal-style log display
- ■ Color-coded log messages:
 - ■ **SYSTEM:** Green (#4CAF50)
 - ■ **NETWORK:** Cyan (#00BCD4)
 - ■ **INFO:** Blue (#58A6FF)
 - ■ **WARN:** Orange (#FFA726)
 - ■ **ERROR:** Red (#EF5350)
- ■ Courier font for logs
- ■ Empty state message
- ■ 250px height scrollable container

■■ Removed Sections
- ■ Removed "Training Metrics" card (consolidated into logs)
- ■ Removed "Resource Metrics" card (consolidated into logs)
- ■ Cleaned up unused code (_lastMetrics, _resourceMetrics, _buildMetricRow)

■ Log Messages Enhanced
- ■ Added proper prefixes: SYSTEM:, NETWORK:, INFO:, WARN:, ERROR:
- ■ Updated connection logs:
 - "NETWORK: DNS lookup resolved for coordinator."
 - "INFO: Handshake protocol version 3.2.0."
 - "SYSTEM: Client initialized successfully."
 - "SYSTEM: Local model weight buffer allocated (512MB)."
 - "WARN: Model not initialized on server yet."

■ Technical Improvements
- ■ Fixed all compilation errors
- ■ Removed deprecated color properties (background → surface)
- ■ Fixed CardTheme type (CardTheme → CardThemeData)
- ■ Fixed withOpacity() deprecation (→ withValues())
- ■ Added SafeArea for proper layout
- ■ Improved code structure and readability

Result
The app now has a professional, modern UI that matches the target design:
- Dark theme with proper color scheme
- Clean, organized layout
- Better visual hierarchy
- Enhanced user experience
- Production-ready appearance

Testing
To see the changes:
```bash
```

```
cd federated_learning_in_mobile
flutter run
```

```

```
Screenshot Comparison
Before: Light theme with AppBar, multiple metric cards
After: Dark navy theme, streamlined layout, modern design matching target

Status: ■ Complete - All changes implemented and tested
Date: 2026-03-03
```

### File: analysis\_options.yaml

```

This file configures the analyzer, which statically analyzes Dart code to
check for errors, warnings, and lints.
#
The issues identified by the analyzer are surfaced in the UI of Dart-enabled
IDEs (https://dart.dev/tools#ides-and-editors). The analyzer can also be
invoked from the command line by running `flutter analyze`.
#
The following line activates a set of recommended lints for Flutter apps,
packages, and plugins designed to encourage good coding practices.
include: package:flutter_lints/flutter.yaml

linter:
 # The lint rules applied to this project can be customized in the
 # section below to disable rules from the `package:flutter_lints/flutter.yaml`
 # included above or to enable additional rules. A list of all available lints
 # and their documentation is published at https://dart.dev/lints.
 #
 # Instead of disabling a lint rule for the entire project in the
 # section below, it can also be suppressed for a single line of code
 # or a specific dart file by using the `// ignore: name_of_lint` and
 # `// ignore_for_file: name_of_lint` syntax on the line or in the file
 # producing the lint.
 rules:
 # avoid_print: false # Uncomment to disable the `avoid_print` rule
 # prefer_single_quotes: true # Uncomment to enable the `prefer_single_quotes` rule

Additional information about this file can be found at
https://dart.dev/guides/language/analysis-options
```

### File: pubspec.yaml

```

name: federated_learning_in_mobile
description: "A new Flutter project."
The following line prevents the package from being accidentally published to
pub.dev using `flutter pub publish`. This is preferred for private packages.
publish_to: 'none' # Remove this line if you wish to publish to pub.dev

The following defines the version and build number for your application.
A version number is three numbers separated by dots, like 1.2.43
followed by an optional build number separated by a +.
Both the version and the builder number may be overridden in flutter
build by specifying --build-name and --build-number, respectively.
In Android, build-name is used as versionName while build-number used as versionCode.
Read more about Android versioning at https://developer.android.com/studio/publish/versioning
In iOS, build-name is used as CFBundleShortVersionString while build-number is used as CFBundleVersion.
Read more about iOS versioning at
https://developer.apple.com/library/archive/documentation/General/Reference/InfoPlistKeyReference/Articles/CoreFoundationKeys.html
In Windows, build-name is used as the major, minor, and patch parts
of the product and file versions while build-number is used as the build suffix.
version: 1.0.0+1

environment:
 sdk: ^3.9.2

Dependencies specify other packages that your package needs in order to work.
To automatically upgrade your package dependencies to the latest versions
consider running `flutter pub upgrade --major-versions`. Alternatively,
dependencies can be manually updated by changing the version numbers below to
```

```
the latest version available on pub.dev. To see which dependencies have newer
versions available, run `flutter pub outdated`.
dependencies:
 flutter:
 sdk: flutter

 # The following adds the Cupertino Icons font to your application.
 # Use with the CupertinoIcons class for iOS style icons.
 cupertino_icons: ^1.0.8

 # HTTP client for server communication
 http: ^1.1.0

 # For JSON encoding/decoding (included in dart:convert but explicit for clarity)

 # System info and monitoring
 device_info_plus: ^10.1.0
 battery_plus: ^6.0.2

 # For async operations
 async: ^2.11.0

 # File system access for saving results
 path_provider: ^2.1.1

 # CSV export
 csv: ^5.1.1

 # Share files
 share_plus: ^7.2.1

 # Open files/folders
 open_filex: ^4.3.3

dev_dependencies:
 flutter_test:
 sdk: flutter

 integration_test:
 sdk: flutter

 # The "flutter_lints" package below contains a set of recommended lints to
 # encourage good coding practices. The lint set provided by the package is
 # activated in the `analysis_options.yaml` file located at the root of your
 # package. See that file for information about deactivating specific lint
 # rules and activating additional ones.
 flutter_lints: ^5.0.0

For information on the generic Dart part of this file, see the
following page: https://dart.dev/tools/pub/pubspec

The following section is specific to Flutter packages.
flutter:

 # The following line ensures that the Material Icons font is
 # included with your application, so that you can use the icons in
 # the material Icons class.
 uses-material-design: true

 assets:
 - assets/movielens_titles.csv

 # To add assets to your application, add an assets section, like this:
 # assets:
 # - images/a_dot_burr.jpeg
 # - images/a_dot_ham.jpeg

 # An image asset can refer to one or more resolution-specific "variants", see
 # https://flutter.dev/to/resolution-aware-images

 # For details regarding adding assets from package dependencies, see
 # https://flutter.dev/to/asset-from-package

 # To add custom fonts to your application, add a fonts section here,
```

```
in this "flutter" section. Each entry in this list should have a
"family" key with the font family name, and a "fonts" key with a
list giving the asset and other descriptors for the font. For
example:
fonts:
- family: Schyler
fonts:
- asset: fonts/Schyler-Regular.ttf
- asset: fonts/Schyler-Italic.ttf
style: italic
- family: Trajan Pro
fonts:
- asset: fonts/TrajanPro.ttf
- asset: fonts/TrajanPro_Bold.ttf
weight: 700
#
For details regarding fonts from package dependencies,
see https://flutter.dev/to/font-from-package
```

#### File: .dart\_tool/package\_config.json

```
{
 "configVersion": 2,
 "packages": [
 {
 "name": "args",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/args-2.7.0",
 "packageUri": "lib/",
 "languageVersion": "3.3"
 },
 {
 "name": "async",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/async-2.13.0",
 "packageUri": "lib/",
 "languageVersion": "3.4"
 },
 {
 "name": "battery_plus",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/battery_plus-6.2.3",
 "packageUri": "lib/",
 "languageVersion": "3.4"
 },
 {
 "name": "battery_plus_platform_interface",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/battery_plus_platform_interface-2.0.1",
 "packageUri": "lib/",
 "languageVersion": "2.18"
 },
 {
 "name": "boolean_selector",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/boolean_selector-2.1.2",
 "packageUri": "lib/",
 "languageVersion": "3.1"
 },
 {
 "name": "characters",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/characters-1.4.1",
 "packageUri": "lib/",
 "languageVersion": "3.4"
 },
 {
 "name": "clock",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/clock-1.1.2",
 "packageUri": "lib/",
 "languageVersion": "3.4"
 },
 {
 "name": "collection",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/collection-1.19.1",
 "packageUri": "lib/",
 "languageVersion": "3.4"
 },
 {
 "name": "cross_file",
```



```
"rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/cross_file-0.3.5+1",
"packageUri": "lib/",
"languageVersion": "3.8"
},
{
 "name": "crypto",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/crypto-3.0.7",
 "packageUri": "lib/",
 "languageVersion": "3.4"
},
{
 "name": "csv",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/csv-5.1.1",
 "packageUri": "lib/",
 "languageVersion": "2.12"
},
{
 "name": "cupertino_icons",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/cupertino_icons-1.0.8",
 "packageUri": "lib/",
 "languageVersion": "3.1"
},
{
 "name": "dbus",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/dbus-0.7.11",
 "packageUri": "lib/",
 "languageVersion": "2.17"
},
{
 "name": "device_info_plus",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/device_info_plus-10.1.2",
 "packageUri": "lib/",
 "languageVersion": "3.3"
},
{
 "name": "device_info_plus_platform_interface",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/device_info_plus_platform_interface-7.0.3",
 "packageUri": "lib/",
 "languageVersion": "3.7"
},
{
 "name": "fake_async",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/fake_async-1.3.3",
 "packageUri": "lib/",
 "languageVersion": "3.3"
},
{
 "name": "ffi",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/ffi-2.1.4",
 "packageUri": "lib/",
 "languageVersion": "3.7"
},
{
 "name": "file",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/file-7.0.1",
 "packageUri": "lib/",
 "languageVersion": "3.0"
},
{
 "name": "fixnum",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/fixnum-1.1.1",
 "packageUri": "lib/",
 "languageVersion": "3.1"
},
{
 "name": "flutter",
 "rootUri": "file:///opt/homebrew/share/flutter/packages/flutter",
 "packageUri": "lib/",
 "languageVersion": "3.9"
},
{
 "name": "flutter_driver",
 "rootUri": "file:///opt/homebrew/share/flutter/packages/flutter_driver",
 "packageUri": "lib/",
```

```
"languageVersion": "3.9"
},
{
 "name": "flutter_lints",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/flutter_lints-5.0.0",
 "packageUri": "lib/",
 "languageVersion": "3.5"
},
{
 "name": "flutter_test",
 "rootUri": "file:///opt/homebrew/share/flutter/packages/flutter_test",
 "packageUri": "lib/",
 "languageVersion": "3.9"
},
{
 "name": "flutter_web_plugins",
 "rootUri": "file:///opt/homebrew/share/flutter/packages/flutter_web_plugins",
 "packageUri": "lib/",
 "languageVersion": "3.9"
},
{
 "name": "fuchsia_remote_debug_protocol",
 "rootUri": "file:///opt/homebrew/share/flutter/packages/fuchsia_remote_debug_protocol",
 "packageUri": "lib/",
 "languageVersion": "3.9"
},
{
 "name": "http",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/http-1.6.0",
 "packageUri": "lib/",
 "languageVersion": "3.4"
},
{
 "name": "http_parser",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/http_parser-4.1.2",
 "packageUri": "lib/",
 "languageVersion": "3.4"
},
{
 "name": "integration_test",
 "rootUri": "file:///opt/homebrew/share/flutter/packages/integration_test",
 "packageUri": "lib/",
 "languageVersion": "3.9"
},
{
 "name": "leak_tracker",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/leak_tracker-11.0.2",
 "packageUri": "lib/",
 "languageVersion": "3.2"
},
{
 "name": "leak_tracker_flutter_testing",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/leak_tracker_flutter_testing-3.0.10",
 "packageUri": "lib/",
 "languageVersion": "3.2"
},
{
 "name": "leak_tracker_testing",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/leak_tracker_testing-3.0.2",
 "packageUri": "lib/",
 "languageVersion": "3.2"
},
{
 "name": "lints",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/lints-5.1.1",
 "packageUri": "lib/",
 "languageVersion": "3.6"
},
{
 "name": "matcher",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/matcher-0.12.19",
 "packageUri": "lib/",
 "languageVersion": "3.7"
},
}
```

```
{
 "name": "material_color_utilities",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/material_color_utilities-0.13.0",
 "packageUri": "lib/",
 "languageVersion": "3.5"
},
{
 "name": "meta",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/meta-1.17.0",
 "packageUri": "lib/",
 "languageVersion": "3.5"
},
{
 "name": "mime",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/mime-1.0.6",
 "packageUri": "lib/",
 "languageVersion": "3.2"
},
{
 "name": "open_filex",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/open_filex-4.7.0",
 "packageUri": "lib/",
 "languageVersion": "2.17"
},
{
 "name": "path",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/path-1.9.1",
 "packageUri": "lib/",
 "languageVersion": "3.4"
},
{
 "name": "path_provider",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/path_provider-2.1.5",
 "packageUri": "lib/",
 "languageVersion": "3.4"
},
{
 "name": "path_provider_android",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/path_provider_android-2.2.22",
 "packageUri": "lib/",
 "languageVersion": "3.9"
},
{
 "name": "path_provider_foundation",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/path_provider_foundation-2.5.1",
 "packageUri": "lib/",
 "languageVersion": "3.9"
},
{
 "name": "path_provider_linux",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/path_provider_linux-2.2.1",
 "packageUri": "lib/",
 "languageVersion": "2.19"
},
{
 "name": "path_provider_platform_interface",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/path_provider_platform_interface-2.1.2",
 "packageUri": "lib/",
 "languageVersion": "3.0"
},
{
 "name": "path_provider_windows",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/path_provider_windows-2.3.0",
 "packageUri": "lib/",
 "languageVersion": "3.2"
},
{
 "name": "petitparser",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/petitparser-7.0.1",
 "packageUri": "lib/",
 "languageVersion": "3.8"
},
{
 "name": "platform",
```

```
"rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/platform-3.1.6",
"packageUri": "lib/",
"languageVersion": "3.2"
},
{
 "name": "plugin_platform_interface",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/plugin_platform_interface-2.1.8",
 "packageUri": "lib/",
 "languageVersion": "3.0"
},
{
 "name": "process",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/process-5.0.5",
 "packageUri": "lib/",
 "languageVersion": "3.5"
},
{
 "name": "share_plus",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/share_plus-7.2.2",
 "packageUri": "lib/",
 "languageVersion": "2.18"
},
{
 "name": "share_plus_platform_interface",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/share_plus_platform_interface-3.4.0",
 "packageUri": "lib/",
 "languageVersion": "2.18"
},
{
 "name": "sky_engine",
 "rootUri": "file:///opt/homebrew/share/flutter/bin/cache/pkg/sky_engine",
 "packageUri": "lib/",
 "languageVersion": "3.9"
},
{
 "name": "source_span",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/source_span-1.10.1",
 "packageUri": "lib/",
 "languageVersion": "3.1"
},
{
 "name": "stack_trace",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/stack_trace-1.12.1",
 "packageUri": "lib/",
 "languageVersion": "3.4"
},
{
 "name": "stream_channel",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/stream_channel-2.1.4",
 "packageUri": "lib/",
 "languageVersion": "3.3"
},
{
 "name": "string_scanner",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/string_scanner-1.4.1",
 "packageUri": "lib/",
 "languageVersion": "3.1"
},
{
 "name": "sync_http",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/sync_http-0.3.1",
 "packageUri": "lib/",
 "languageVersion": "2.12"
},
{
 "name": "term_glyph",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/term_glyph-1.2.2",
 "packageUri": "lib/",
 "languageVersion": "3.1"
},
{
 "name": "test_api",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/test_api-0.7.10",
 "packageUri": "lib/",
```

```
"languageVersion": "3.7"
},
{
 "name": "typed_data",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/typed_data-1.4.0",
 "packageUri": "lib/",
 "languageVersion": "3.5"
},
{
 "name": "upower",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/upower-0.7.0",
 "packageUri": "lib/",
 "languageVersion": "2.14"
},
{
 "name": "url_launcher_linux",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/url_launcher_linux-3.2.2",
 "packageUri": "lib/",
 "languageVersion": "3.8"
},
{
 "name": "url_launcher_platform_interface",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/url_launcher_platform_interface-2.3.2",
 "packageUri": "lib/",
 "languageVersion": "3.1"
},
{
 "name": "url_launcher_web",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/url_launcher_web-2.4.1",
 "packageUri": "lib/",
 "languageVersion": "3.6"
},
{
 "name": "url_launcher_windows",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/url_launcher_windows-3.1.5",
 "packageUri": "lib/",
 "languageVersion": "3.8"
},
{
 "name": "uuid",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/uuid-4.5.2",
 "packageUri": "lib/",
 "languageVersion": "3.0"
},
{
 "name": "vector_math",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/vector_math-2.2.0",
 "packageUri": "lib/",
 "languageVersion": "3.1"
},
{
 "name": "vm_service",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/vm_service-15.0.2",
 "packageUri": "lib/",
 "languageVersion": "3.5"
},
{
 "name": "web",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/web-1.1.1",
 "packageUri": "lib/",
 "languageVersion": "3.4"
},
{
 "name": "webdriver",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/webdriver-3.1.0",
 "packageUri": "lib/",
 "languageVersion": "3.1"
},
{
 "name": "win32",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/win32-5.15.0",
 "packageUri": "lib/",
 "languageVersion": "3.8"
},
}
```

```

{
 "name": "win32_registry",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/win32_registry-1.1.5",
 "packageUri": "lib/",
 "languageVersion": "3.4"
},
{
 "name": "xdg_directories",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/xdg_directories-1.1.0",
 "packageUri": "lib/",
 "languageVersion": "3.3"
},
{
 "name": "xml",
 "rootUri": "file:///Users/jonat/.pub-cache/hosted/pub.dev/xml-6.6.1",
 "packageUri": "lib/",
 "languageVersion": "3.8"
},
{
 "name": "federated_learning_in_mobile",
 "rootUri": "../",
 "packageUri": "lib/",
 "languageVersion": "3.9"
}
],
"generator": "pub",
"generatorVersion": "3.11.3",
"flutterRoot": "file:///opt/homebrew/share/flutter",
"flutterVersion": "3.41.5",
"pubCache": "file:///Users/jonat/.pub-cache"
}

```

#### File: .dart\_tool/package\_graph.json

---

```

{
 "roots": [
 "federated_learning_in_mobile"
],
 "packages": [
 {
 "name": "federated_learning_in_mobile",
 "version": "1.0.0+1",
 "dependencies": [
 "async",
 "battery_plus",
 "csv",
 "cupertino_icons",
 "device_info_plus",
 "flutter",
 "http",
 "open_filex",
 "path_provider",
 "share_plus"
],
 "devDependencies": [
 "flutter_lints",
 "flutter_test",
 "integration_test"
]
 },
 {
 "name": "flutter_lints",
 "version": "5.0.0",
 "dependencies": [
 "lints"
]
 },
 {
 "name": "integration_test",
 "version": "0.0.0",
 "dependencies": [
 "flutter",
 "flutter_driver",
 "flutter_test",

```

```
 "path",
 "vm_service"
]
},
{
 "name": "flutter_test",
 "version": "0.0.0",
 "dependencies": [
 "clock",
 "collection",
 "fake_async",
 "flutter",
 "leak_tracker_flutter_testing",
 "matcher",
 "meta",
 "path",
 "stack_trace",
 "stream_channel",
 "test_api",
 "vector_math"
]
},
{
 "name": "open_filex",
 "version": "4.7.0",
 "dependencies": [
 "ffi",
 "flutter"
]
},
{
 "name": "share_plus",
 "version": "7.2.2",
 "dependencies": [
 "cross_file",
 "ffi",
 "file",
 "flutter",
 "flutter_web_plugins",
 "meta",
 "mime",
 "share_plus_platform_interface",
 "url_launcher_linux",
 "url_launcher_platform_interface",
 "url_launcher_web",
 "url_launcher_windows",
 "win32"
]
},
{
 "name": "csv",
 "version": "5.1.1",
 "dependencies": []
},
{
 "name": "path_provider",
 "version": "2.1.5",
 "dependencies": [
 "flutter",
 "path_provider_android",
 "path_provider_foundation",
 "path_provider_linux",
 "path_provider_platform_interface",
 "path_provider_windows"
]
},
{
 "name": "async",
 "version": "2.13.0",
 "dependencies": [
 "collection",
 "meta"
]
},
}
```

```
{
 "name": "battery_plus",
 "version": "6.2.3",
 "dependencies": [
 "battery_plus_platform_interface",
 "flutter",
 "flutter_web_plugins",
 "meta",
 "upower",
 "web"
]
},
{
 "name": "device_info_plus",
 "version": "10.1.2",
 "dependencies": [
 "device_info_plus_platform_interface",
 "ffi",
 "file",
 "flutter",
 "flutter_web_plugins",
 "meta",
 "web",
 "win32",
 "win32_registry"
]
},
{
 "name": "http",
 "version": "1.6.0",
 "dependencies": [
 "async",
 "http_parser",
 "meta",
 "web"
]
},
{
 "name": "cupertino_icons",
 "version": "1.0.8",
 "dependencies": []
},
{
 "name": "flutter",
 "version": "0.0.0",
 "dependencies": [
 "characters",
 "collection",
 "material_color_utilities",
 "meta",
 "sky_engine",
 "vector_math"
]
},
{
 "name": "lints",
 "version": "5.1.1",
 "dependencies": []
},
{
 "name": "vm_service",
 "version": "15.0.2",
 "dependencies": []
},
{
 "name": "path",
 "version": "1.9.1",
 "dependencies": []
},
{
 "name": "flutter_driver",
 "version": "0.0.0",
 "dependencies": [
 "file",
```



```
 "flutter",
 "flutter_test",
 "fuchsia_remote_debug_protocol",
 "matcher",
 "meta",
 "path",
 "vm_service",
 "webdriver"
]
},
{
 "name": "stream_channel",
 "version": "2.1.4",
 "dependencies": [
 "async"
]
},
{
 "name": "meta",
 "version": "1.17.0",
 "dependencies": []
},
{
 "name": "collection",
 "version": "1.19.1",
 "dependencies": []
},
{
 "name": "leak_tracker_flutter_testing",
 "version": "3.0.10",
 "dependencies": [
 "flutter",
 "leak_tracker",
 "leak_tracker_testing",
 "matcher",
 "meta"
]
},
{
 "name": "vector_math",
 "version": "2.2.0",
 "dependencies": []
},
{
 "name": "stack_trace",
 "version": "1.12.1",
 "dependencies": [
 "path"
]
},
{
 "name": "clock",
 "version": "1.1.2",
 "dependencies": []
},
{
 "name": "fake_async",
 "version": "1.3.3",
 "dependencies": [
 "clock",
 "collection"
]
},
{
 "name": "matcher",
 "version": "0.12.19",
 "dependencies": [
 "async",
 "meta",
 "stack_trace",
 "term_glyph",
 "test_api"
]
},
},
```

```
{
 "name": "test_api",
 "version": "0.7.10",
 "dependencies": [
 "async",
 "boolean_selector",
 "collection",
 "meta",
 "source_span",
 "stack_trace",
 "stream_channel",
 "string_scanner",
 "term_glyph"
]
},
{
 "name": "ffi",
 "version": "2.1.4",
 "dependencies": []
},
{
 "name": "win32",
 "version": "5.15.0",
 "dependencies": [
 "ffi"
]
},
{
 "name": "url_launcher_platform_interface",
 "version": "2.3.2",
 "dependencies": [
 "flutter",
 "plugin_platform_interface"
]
},
{
 "name": "url_launcher_linux",
 "version": "3.2.2",
 "dependencies": [
 "flutter",
 "url_launcher_platform_interface"
]
},
{
 "name": "url_launcher_windows",
 "version": "3.1.5",
 "dependencies": [
 "flutter",
 "url_launcher_platform_interface"
]
},
{
 "name": "url_launcher_web",
 "version": "2.4.1",
 "dependencies": [
 "flutter",
 "flutter_web_plugins",
 "url_launcher_platform_interface",
 "web"
]
},
{
 "name": "file",
 "version": "7.0.1",
 "dependencies": [
 "meta",
 "path"
]
},
{
 "name": "share_plus_platform_interface",
 "version": "3.4.0",
 "dependencies": [
 "cross_file",
```

```
 "flutter",
 "meta",
 "mime",
 "path_provider",
 "plugin_platform_interface",
 "uuid"
]
},
{
 "name": "flutter_web_plugins",
 "version": "0.0.0",
 "dependencies": [
 "flutter"
]
},
{
 "name": "mime",
 "version": "1.0.6",
 "dependencies": []
},
{
 "name": "cross_file",
 "version": "0.3.5+1",
 "dependencies": [
 "meta",
 "web"
]
},
{
 "name": "path_provider_windows",
 "version": "2.3.0",
 "dependencies": [
 "ffi",
 "flutter",
 "path",
 "path_provider_platform_interface"
]
},
{
 "name": "path_provider_platform_interface",
 "version": "2.1.2",
 "dependencies": [
 "flutter",
 "platform",
 "plugin_platform_interface"
]
},
{
 "name": "path_provider_linux",
 "version": "2.2.1",
 "dependencies": [
 "ffi",
 "flutter",
 "path",
 "path_provider_platform_interface",
 "xdg_directories"
]
},
{
 "name": "path_provider_foundation",
 "version": "2.5.1",
 "dependencies": [
 "flutter",
 "path_provider_platform_interface"
]
},
{
 "name": "path_provider_android",
 "version": "2.2.22",
 "dependencies": [
 "flutter",
 "path_provider_platform_interface"
]
},
},
```

```
{
 "name": "web",
 "version": "1.1.1",
 "dependencies": []
},
{
 "name": "upower",
 "version": "0.7.0",
 "dependencies": [
 "dbus"
]
},
{
 "name": "battery_plus_platform_interface",
 "version": "2.0.1",
 "dependencies": [
 "flutter",
 "meta",
 "plugin_platform_interface"
]
},
{
 "name": "win32_registry",
 "version": "1.1.5",
 "dependencies": [
 "ffi",
 "win32"
]
},
{
 "name": "device_info_plus_platform_interface",
 "version": "7.0.3",
 "dependencies": [
 "flutter",
 "meta",
 "plugin_platform_interface"
]
},
{
 "name": "http_parser",
 "version": "4.1.2",
 "dependencies": [
 "collection",
 "source_span",
 "string_scanner",
 "typed_data"
]
},
{
 "name": "sky_engine",
 "version": "0.0.0",
 "dependencies": []
},
{
 "name": "material_color_utilities",
 "version": "0.13.0",
 "dependencies": [
 "collection"
]
},
{
 "name": "characters",
 "version": "1.4.1",
 "dependencies": []
},
{
 "name": "webdriver",
 "version": "3.1.0",
 "dependencies": [
 "matcher",
 "path",
 "stack_trace",
 "sync_http"
]
}
```

```
},
{
 "name": "fuchsia_remote_debug_protocol",
 "version": "0.0.0",
 "dependencies": [
 "meta",
 "process",
 "vm_service"
]
},
{
 "name": "leak_tracker_testing",
 "version": "3.0.2",
 "dependencies": [
 "leak_tracker",
 "matcher",
 "meta"
]
},
{
 "name": "leak_tracker",
 "version": "11.0.2",
 "dependencies": [
 "clock",
 "collection",
 "meta",
 "path",
 "vm_service"
]
},
{
 "name": "term_glyph",
 "version": "1.2.2",
 "dependencies": []
},
{
 "name": "string_scanner",
 "version": "1.4.1",
 "dependencies": [
 "source_span"
]
},
{
 "name": "source_span",
 "version": "1.10.1",
 "dependencies": [
 "collection",
 "path",
 "term_glyph"
]
},
{
 "name": "boolean_selector",
 "version": "2.1.2",
 "dependencies": [
 "source_span",
 "string_scanner"
]
},
{
 "name": "plugin_platform_interface",
 "version": "2.1.8",
 "dependencies": [
 "meta"
]
},
{
 "name": "uuid",
 "version": "4.5.2",
 "dependencies": [
 "crypto",
 "fixnum",
 "meta"
]
}
```

```
},
{
 "name": "platform",
 "version": "3.1.6",
 "dependencies": []
},
{
 "name": "xdg_directories",
 "version": "1.1.0",
 "dependencies": [
 "meta",
 "path"
]
},
{
 "name": "dbus",
 "version": "0.7.11",
 "dependencies": [
 "args",
 "ffi",
 "meta",
 "xml"
]
},
{
 "name": "typed_data",
 "version": "1.4.0",
 "dependencies": [
 "collection"
]
},
{
 "name": "sync_http",
 "version": "0.3.1",
 "dependencies": []
},
{
 "name": "process",
 "version": "5.0.5",
 "dependencies": [
 "file",
 "path",
 "platform"
]
},
{
 "name": "fixnum",
 "version": "1.1.1",
 "dependencies": []
},
{
 "name": "crypto",
 "version": "3.0.7",
 "dependencies": [
 "typed_data"
]
},
{
 "name": "xml",
 "version": "6.6.1",
 "dependencies": [
 "collection",
 "meta",
 "petitparser"
]
},
{
 "name": "args",
 "version": "2.7.0",
 "dependencies": []
},
{
 "name": "petitparser",
 "version": "7.0.1",
```

```

 "dependencies": [
 "collection",
 "meta"
]
 },
 "configVersion": 1
}

```

#### File: .dart\_tool/flutter\_build/dart\_plugin\_registrant.dart

```

//
// Generated file. Do not edit.
// This file is generated from template in file `flutter_tools/lib/src/flutter_plugins.dart`.
//

// @dart = 3.9

import 'dart:io'; // flutter_ignore: dart_io_import.
import 'package:path_provider_android/path_provider_android.dart' as path_provider_android;
import 'package:path_provider_foundation/path_provider_foundation.dart' as path_provider_foundation;
import 'package:battery_plus/battery_plus.dart' as battery_plus;
import 'package:device_info_plus/device_info_plus.dart' as device_info_plus;
import 'package:path_provider_linux/path_provider_linux.dart' as path_provider_linux;
import 'package:share_plus/share_plus.dart' as share_plus;
import 'package:url_launcher_linux/url_launcher_linux.dart' as url_launcher_linux;
import 'package:path_provider_foundation/path_provider_foundation.dart' as path_provider_foundation;
import 'package:device_info_plus/device_info_plus.dart' as device_info_plus;
import 'package:path_provider_windows/path_provider_windows.dart' as path_provider_windows;
import 'package:share_plus/share_plus.dart' as share_plus;
import 'package:url_launcher_windows/url_launcher_windows.dart' as url_launcher_windows;

@pragma('vm:entry-point')
class _PluginRegistrant {

 @pragma('vm:entry-point')
 static void register() {
 if (Platform.isAndroid) {
 try {
 path_provider_android.PathProviderAndroid.registerWith();
 } catch (err) {
 print(
 '`path_provider_android` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
 }
 } else if (Platform.isIOS) {
 try {
 path_provider_foundation.PathProviderFoundation.registerWith();
 } catch (err) {
 print(
 '`path_provider_foundation` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
 }
 } else if (Platform.isLinux) {
 try {
 battery_plus.BatteryPlusLinuxPlugin.registerWith();
 } catch (err) {
 print(
 '`battery_plus` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
 }
 }

 try {
 device_info_plus.DeviceInfoPlusLinuxPlugin.registerWith();
 } catch (err) {
 print(
 '`device_info_plus` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
 }
 }
}

```

```
}

try {
 path_provider_linux.PathProviderLinux.registerWith();
} catch (err) {
 print(
 '`path_provider_linux` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
}

try {
 share_plus.SharePlusLinuxPlugin.registerWith();
} catch (err) {
 print(
 '`share_plus` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
}

try {
 url_launcher_linux.UrlLauncherLinux.registerWith();
} catch (err) {
 print(
 '`url_launcher_linux` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
}

} else if (Platform.isMacOS) {
 try {
 path_provider_foundation.PathProviderFoundation.registerWith();
 } catch (err) {
 print(
 '`path_provider_foundation` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
 }
}

} else if (Platform.isWindows) {
 try {
 device_info_plus.DeviceInfoPlusWindowsPlugin.registerWith();
 } catch (err) {
 print(
 '`device_info_plus` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
 }
}

try {
 path_provider_windows.PathProviderWindows.registerWith();
} catch (err) {
 print(
 '`path_provider_windows` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
}

try {
 share_plus.SharePlusWindowsPlugin.registerWith();
} catch (err) {
 print(
 '`share_plus` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
}

try {
 url_launcher_windows.UrlLauncherWindows.registerWith();
} catch (err) {
 print(
 '`url_launcher_windows` threw an error: $err. '
 'The app may not function as expected until you remove this plugin from pubspec.yaml'
);
}
```



```
}

}
}
}
```

**File: .dart\_tool/flutter\_build/3430039a9863faeb85c025b07c39c70d/dart\_build\_result.json**

```
{ "build_start": "2026-03-21T01:31:36.279568", "build_end": "2026-03-21T01:31:36.279568", "dependencies": [], "code_assets": [], "data_assets": [] }
```

**File: .dart\_tool/flutter\_build/3430039a9863faeb85c025b07c39c70d/native\_assets.json**

```
{ "format-version": [1, 0, 0], "native-assets": {} }
```

**File: .dart\_tool/flutter\_build/3430039a9863faeb85c025b07c39c70d/outputs.json**

```
["/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/Flutter.framework/Flutter", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/ios/Flutter/ephemeral/flutter_lldbinit", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/vm_snapshot_data", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/isolate_snapshot_data", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/kernel_blob.bin", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/App", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/Info.plist", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/assets/movielens_titles.csv", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/packages/cupertino_icons/assets/CupertinoIcons.ttf", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/fonts/MaterialIcons-Regular.otf", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/shaders/ink_sparkle.frag", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/shaders/stretch_effect.frag", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/AssetManifest.bin", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/FontManifest.json", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/NOTICES.Z", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/NativeAssetsManifest.json"]
```

**File: .dart\_tool/flutter\_build/a70ad27073865decc01abc0f41ba08ba/dart\_build\_result.json**

```
{ "build_start": "2026-03-21T01:56:40.472438", "build_end": "2026-03-21T01:56:40.472451", "dependencies": [], "code_assets": [], "data_assets": [] }
```

**File: .dart\_tool/flutter\_build/a70ad27073865decc01abc0f41ba08ba/native\_assets.json**

```
{ "format-version": [1, 0, 0], "native-assets": {} }
```

**File: .dart\_tool/flutter\_build/a70ad27073865decc01abc0f41ba08ba/outputs.json**

```
["/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/Flutter.framework/Flutter", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/ios/Flutter/ephemeral/flutter_lldbinit", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/vm_snapshot_data", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/isolate_snapshot_data", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/kernel_blob.bin", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/App", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/Info.plist", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/assets/movielens_titles.csv", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/packages/cupertino_icons/assets/CupertinoIcons.ttf", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/fonts/MaterialIcons-Regular.otf", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/shaders/ink_sparkle.frag", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/shaders/stretch_effect.frag", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/AssetManifest.bin", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/FontManifest.json", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/NOTICES.Z", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/ios/Debug-iphonesimulator/App.framework/flutter_assets/NativeAssetsManifest.json"]
```

```
g-iphonesimulator/App.framework/flutter_assets/NativeAssetsManifest.json"]
```

#### File: .dart\_tool/flutter\_build/e85ff67b380c3115b23551371919179a/dart\_build\_result.json

```
{"build_start": "2026-03-18T05:53:47.047981", "build_end": "2026-03-18T05:53:47.047982", "dependencies": [], "code_assets": [], "data_assets": []}
```

#### File: .dart\_tool/flutter\_build/e85ff67b380c3115b23551371919179a/native\_assets.json

```
{"format-version": [1, 0, 0], "native-assets": {}}
```

#### File: .dart\_tool/flutter\_build/e85ff67b380c3115b23551371919179a/outputs.json

```
["/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/vm_snapshot_data", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/isolate_snapshot_data", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/kerne_l_blob.bin", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/assets/movielens_titles.csv", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/packages/cupertino_icons/assets/CupertinoIcons.ttf", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/fonts/MaterialIcons-Regular.otf", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/shaders/ink_sparkle.frag", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/shaders/stretch_effect.frag", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/AssetManifest.bin", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/FontManifest.json", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/NOTICES.Z", "/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/build/app/intermediates/flutter/debug/flutter_assets/NativeAssetsManifest.json"]
```

#### File: android/gradlew.bat

```
@if "%DEBUG%" == "" @echo off
@rem #####
@rem
@rem Gradle startup script for Windows
@rem
@rem #####

@rem Set local scope for the variables with windows NT shell
if "%OS%"=="Windows_NT" setlocal

@rem Add default JVM options here. You can also use JAVA_OPTS and GRADLE_OPTS to pass JVM options to this script.
set DEFAULT_JVM_OPTS=

set DIRNAME=%~dp0
if "%DIRNAME%" == "" set DIRNAME=.
set APP_BASE_NAME=%~n0
set APP_HOME=%DIRNAME%

@rem Find java.exe
if defined JAVA_HOME goto findJavaFromJavaHome

set JAVA_EXE=java.exe
%JAVA_EXE% -version >NUL 2>&1
if "%ERRORLEVEL%" == "0" goto init

echo.
echo ERROR: JAVA_HOME is not set and no 'java' command could be found in your PATH.
echo.
echo Please set the JAVA_HOME variable in your environment to match the
echo location of your Java installation.

goto fail

:findJavaFromJavaHome
set JAVA_HOME=%JAVA_HOME:"=%
set JAVA_EXE=%JAVA_HOME%/bin/java.exe

if exist "%JAVA_EXE%" goto init
```

```

echo.
echo ERROR: JAVA_HOME is set to an invalid directory: %JAVA_HOME%
echo.
echo Please set the JAVA_HOME variable in your environment to match the
echo location of your Java installation.

goto fail

:init
@rem Get command-line arguments, handling Windowz variants

if not "%OS%" == "Windows_NT" goto win9xME_args
if "%@eval[2+2]" == "4" goto 4NT_args

:win9xME_args
@rem Slurp the command line arguments.
set CMD_LINE_ARGS=
set _SKIP=2

:win9xME_args_slurp
if "x%-1" == "x" goto execute

set CMD_LINE_ARGS=%*
goto execute

:4NT_args
@rem Get arguments from the 4NT Shell from JP Software
set CMD_LINE_ARGS=%$

:execute
@rem Setup the command line

set CLASSPATH=%APP_HOME%\gradle\wrapper\gradle-wrapper.jar

@rem Execute Gradle
"%JAVA_EXE%" %DEFAULT_JVM_OPTS% %JAVA_OPTS% %GRADLE_OPTS% "-Dorg.gradle.appname=%APP_BASE_NAME%" -classpath
"%CLASSPATH%" org.gradle.wrapper.GradleWrapperMain %CMD_LINE_ARGS%

:end
@rem End local scope for the variables with windows NT shell
if "%ERRORLEVEL%"=="0" goto mainEnd

:fail
rem Set variable GRADLE_EXIT_CONSOLE if you need the _script_ return code instead of
rem the _cmd.exe /c_ return code!
if not "" == "%GRADLE_EXIT_CONSOLE%" exit 1
exit /b 1

:mainEnd
if "%OS%"=="Windows_NT" endlocal

:omega

```

#### File: android/app/src/debug/AndroidManifest.xml

```

<manifest xmlns:android="http://schemas.android.com/apk/res/android">
 <!-- The INTERNET permission is required for development. Specifically,
 the Flutter tool needs it to communicate with the running application
 to allow setting breakpoints, to provide hot reload, etc.
 -->
 <uses-permission android:name="android.permission.INTERNET"/>
</manifest>

```

#### File: android/app/src/main/AndroidManifest.xml

```

<manifest xmlns:android="http://schemas.android.com/apk/res/android">
 <!-- Permissions for network access and battery monitoring -->
 <uses-permission android:name="android.permission.INTERNET"/>
 <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE"/>

 <!-- Permissions for file storage (for saving results) -->
 <uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"
 android:maxSdkVersion="32" />
 <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"

```

```

 android:maxSdkVersion="32" />
<!-- For Android 10+ (API 29+), scoped storage is used automatically -->

<application
 android:label="federated_learning_in_mobile"
 android:name="${applicationName}"
 android:icon="@mipmap/ic_launcher"
 android:usesCleartextTraffic="true"
 android:networkSecurityConfig="@xml/network_security_config">
 <activity
 android:name=".MainActivity"
 android:exported="true"
 android:launchMode="singleTop"
 android:taskAffinity=""
 android:theme="@style/LaunchTheme"
 android:configChanges="orientation|keyboardHidden|keyboard|screenSize|smallestScreenSize|locale
|layoutDirection|fontScale|screenLayout|density|uiMode"
 android:hardwareAccelerated="true"
 android:windowSoftInputMode="adjustResize">
 <!-- Specifies an Android theme to apply to this Activity as soon as
 the Android process has started. This theme is visible to the user
 while the Flutter UI initializes. After that, this theme continues
 to determine the Window background behind the Flutter UI. -->
 <meta-data
 android:name="io.flutter.embedding.android.NormalTheme"
 android:resource="@style/NormalTheme"
 />
 <intent-filter>
 <action android:name="android.intent.action.MAIN"/>
 <category android:name="android.intent.category.LAUNCHER"/>
 </intent-filter>
 </activity>
 <!-- Don't delete the meta-data below.
 This is used by the Flutter tool to generate GeneratedPluginRegistrant.java -->
 <meta-data
 android:name="flutterEmbedding"
 android:value="2" />
</application>
<!-- Required to query activities that can process text, see:
 https://developer.android.com/training/package-visibility and
 https://developer.android.com/reference/android/content/Intent#ACTION_PROCESS_TEXT.

 In particular, this is used by the Flutter engine in io.flutter.plugin.text.ProcessTextPlugin. -->
<queries>
 <intent>
 <action android:name="android.intent.action.PROCESS_TEXT"/>
 <data android:mimeType="text/plain"/>
 </intent>
</queries>
</manifest>

```

### File: android/app/src/main/java/io/flutter/plugins/GeneratedPluginRegistrant.java

```

package io.flutter.plugins;

import androidx.annotation.Keep;
import androidx.annotation.NonNull;
import io.flutter.Log;

import io.flutter.embedding.engine.FlutterEngine;

/**
 * Generated file. Do not edit.
 * This file is generated by the Flutter tool based on the
 * plugins that support the Android platform.
 */
@Keep
public final class GeneratedPluginRegistrant {
 private static final String TAG = "GeneratedPluginRegistrant";
 public static void registerWith(@NonNull FlutterEngine flutterEngine) {
 try {
 flutterEngine.getPlugins().add(new dev.fluttercommunity.plus.battery.BatteryPlusPlugin());
 } catch (Exception e) {
 Log.e(TAG, "Error registering plugin battery_plus, dev.fluttercommunity.plus.battery.BatteryPlusPlugi

```

```

n", e);
 }
 try {
 flutterEngine.getPlugins().add(new dev.fluttercommunity.plus.device_info.DeviceInfoPlusPlugin());
 } catch (Exception e) {
 Log.e(TAG, "Error registering plugin device_info_plus, dev.fluttercommunity.plus.device_info.DeviceIn
foPlusPlugin", e);
 }
 try {
 flutterEngine.getPlugins().add(new dev.flutter.plugins.integration_test.IntegrationTestPlugin());
 } catch (Exception e) {
 Log.e(TAG, "Error registering plugin integration_test, dev.flutter.plugins.integration_test.Integrati
onTestPlugin", e);
 }
 try {
 flutterEngine.getPlugins().add(new com.crazecoder.openfile.OpenFilePlugin());
 } catch (Exception e) {
 Log.e(TAG, "Error registering plugin open_filex, com.crazecoder.openfile.OpenFilePlugin", e);
 }
 try {
 flutterEngine.getPlugins().add(new io.flutter.plugins.pathprovider.PathProviderPlugin());
 } catch (Exception e) {
 Log.e(TAG, "Error registering plugin path_provider_android, io.flutter.plugins.pathprovider.PathProvi
derPlugin", e);
 }
 try {
 flutterEngine.getPlugins().add(new dev.fluttercommunity.plus.share.SharePlusPlugin());
 } catch (Exception e) {
 Log.e(TAG, "Error registering plugin share_plus, dev.fluttercommunity.plus.share.SharePlusPlugin", e)
;
 }
}
}
}

```

#### File: android/app/src/main/res/drawable/launch\_background.xml

```

<?xml version="1.0" encoding="utf-8"?>
<!-- Modify this file to customize your launch splash screen -->
<layer-list xmlns:android="http://schemas.android.com/apk/res/android">
 <item android:drawable="@android:color/white" />

 <!-- You can insert your own image assets here -->
 <!-- <item>
 <bitmap
 android:gravity="center"
 android:src="@mipmap/launch_image" />
 </item> -->
</layer-list>

```

#### File: android/app/src/main/res/drawable-v21/launch\_background.xml

```

<?xml version="1.0" encoding="utf-8"?>
<!-- Modify this file to customize your launch splash screen -->
<layer-list xmlns:android="http://schemas.android.com/apk/res/android">
 <item android:drawable="?android:colorBackground" />

 <!-- You can insert your own image assets here -->
 <!-- <item>
 <bitmap
 android:gravity="center"
 android:src="@mipmap/launch_image" />
 </item> -->
</layer-list>

```

#### File: android/app/src/main/res/values/styles.xml

```

<?xml version="1.0" encoding="utf-8"?>
<resources>
 <!-- Theme applied to the Android Window while the process is starting when the OS's Dark Mode setting
is off -->
 <style name="LaunchTheme" parent="@android:style/Theme.Light.NoTitleBar">
 <!-- Show a splash screen on the activity. Automatically removed when
 the Flutter engine draws its first frame -->
 <item name="android:windowBackground">@drawable/launch_background</item>
 </style>

```

```

</style>
<!-- Theme applied to the Android Window as soon as the process has started.
 This theme determines the color of the Android Window while your
 Flutter UI initializes, as well as behind your Flutter UI while its
 running.

 This Theme is only used starting with V2 of Flutter's Android embedding. -->
<style name="NormalTheme" parent="@android:style/Theme.Light.NoTitleBar">
 <item name="android:windowBackground"?android:colorBackground</item>
</style>
</resources>

```

#### File: android/app/src/main/res/values-night/styles.xml

```

<?xml version="1.0" encoding="utf-8"?>
<resources>
 <!-- Theme applied to the Android Window while the process is starting when the OS's Dark Mode setting
is on -->
 <style name="LaunchTheme" parent="@android:style/Theme.Black.NoTitleBar">
 <!-- Show a splash screen on the activity. Automatically removed when
 the Flutter engine draws its first frame -->
 <item name="android:windowBackground">@drawable/launch_background</item>
 </style>
 <!-- Theme applied to the Android Window as soon as the process has started.
 This theme determines the color of the Android Window while your
 Flutter UI initializes, as well as behind your Flutter UI while its
 running.

 This Theme is only used starting with V2 of Flutter's Android embedding. -->
 <style name="NormalTheme" parent="@android:style/Theme.Black.NoTitleBar">
 <item name="android:windowBackground"?android:colorBackground</item>
 </style>
</resources>

```

#### File: android/app/src/main/res/xml/network\_security\_config.xml

```

<?xml version="1.0" encoding="utf-8"?>
<network-security-config>
 <!-- Allow cleartext (HTTP) traffic for local development server -->
 <base-config cleartextTrafficPermitted="true">
 <trust-anchors>
 <certificates src="system" />
 </trust-anchors>
 </base-config>

 <!-- Optionally, allow cleartext only for specific domains -->
 <domain-config cleartextTrafficPermitted="true">
 <domain includeSubdomains="true">localhost</domain>
 <domain includeSubdomains="true">127.0.0.1</domain>
 <domain includeSubdomains="true">192.168.0.0</domain>
 <domain includeSubdomains="true">192.168.1.0</domain>
 <domain includeSubdomains="true">192.168.29.0</domain>
 <domain includeSubdomains="true">10.0.0.0</domain>
 </domain-config>
</network-security-config>

```

#### File: android/app/src/profile/AndroidManifest.xml

```

<manifest xmlns:android="http://schemas.android.com/apk/res/android">
 <!-- The INTERNET permission is required for development. Specifically,
 the Flutter tool needs it to communicate with the running application
 to allow setting breakpoints, to provide hot reload, etc.
 -->
 <uses-permission android:name="android.permission.INTERNET"/>
</manifest>

```

#### File: artifacts/integration\_screenshots/README.md

# Integration screenshots

This folder is populated by the screenshot integration test.

Expected output files after a successful run:

```
- `01_dashboard_overview.png`
- `02_dashboard_logs.png`
- `03_recommendations.png`
```

Run the screenshot flow from the app root with:

```
```bash  
flutter drive \  
  --driver=test_driver/integration_test.dart \  
  --target=integration_test/app_screenshots_test.dart \  
  -d "iPhone 16e"  
```
```

The test overwrites the PNG files on each run so the latest captures are always easy to find.

#### File: assets/movielens\_titles.csv

```

item_id,title
0,Toy Story (1995)
1,GoldenEye (1995)
2,Four Rooms (1995)
3,Get Shorty (1995)
4,Copycat (1995)
5,Shanghai Triad (Yao a yao dao waipo qiao) (1995)
6,Twelve Monkeys (1995)
7,Babe (1995)
8,Dead Man Walking (1995)
9,Richard III (1995)
10,Seven (Se7en) (1995)
11,"Usual Suspects, The (1995)"
12,Mighty Aphrodite (1995)
13,"Postino, Il (1994)"
14,Mr. Holland's Opus (1995)
15,French Twist (Gazon maudit) (1995)
16,From Dusk Till Dawn (1996)
17,"White Balloon, The (1995)"
18,Antonia's Line (1995)
19,Angels and Insects (1995)
20,Muppet Treasure Island (1996)
21,Braveheart (1995)
22,Taxi Driver (1976)
23,Rumble in the Bronx (1995)
24,"Birdcage, The (1996)"
25,"Brothers McMullen, The (1995)"
26,Bad Boys (1995)
27,Apollo 13 (1995)
28,Batman Forever (1995)
29,Belle de jour (1967)
30,Crimson Tide (1995)
31,Crumb (1994)
32,Desperado (1995)
33,"Doom Generation, The (1995)"
34,Free Willy 2: The Adventure Home (1995)
35,Mad Love (1995)
36,Nadja (1994)
37,"Net, The (1995)"
38,Strange Days (1995)
39,"To Wong Foo, Thanks for Everything! Julie Newmar (1995)"
40,Billy Madison (1995)
41,Clerks (1994)
42,Disclosure (1994)
43,Dolores Claiborne (1994)
44,Eat Drink Man Woman (1994)
45,Exotica (1994)
46,Ed Wood (1994)
47,Hoop Dreams (1994)
48,I.Q. (1994)
49,Star Wars (1977)
50,Legends of the Fall (1994)
51,"Madness of King George, The (1994)"
52,Natural Born Killers (1994)
53,Outbreak (1995)
```

---

54,"Professional, The (1994)"  
55,Pulp Fiction (1994)  
56,Priest (1994)  
57,Quiz Show (1994)  
58,Three Colors: Red (1994)  
59,Three Colors: Blue (1993)  
60,Three Colors: White (1994)  
61,Stargate (1994)  
62,"Santa Clause, The (1994)"  
63,"Shawshank Redemption, The (1994)"  
64,What's Eating Gilbert Grape (1993)  
65,While You Were Sleeping (1995)  
66,Ace Ventura: Pet Detective (1994)  
67,"Crow, The (1994)"  
68,Forrest Gump (1994)  
69,Four Weddings and a Funeral (1994)  
70,"Lion King, The (1994)"  
71,"Mask, The (1994)"  
72,Maverick (1994)  
73,Faster Pussycat! Kill! Kill! (1965)  
74,Brother Minister: The Assassination of Malcolm X (1994)  
75,Carlito's Way (1993)  
76,"Firm, The (1993)"  
77,Free Willy (1993)  
78,"Fugitive, The (1993)"  
79,Hot Shots! Part Deux (1993)  
80,"Hudsucker Proxy, The (1994)"  
81,Jurassic Park (1993)  
82,Much Ado About Nothing (1993)  
83,Robert A. Heinlein's The Puppet Masters (1994)  
84,"Ref, The (1994)"  
85,"Remains of the Day, The (1993)"  
86,Searching for Bobby Fischer (1993)  
87,Sleepless in Seattle (1993)  
88,Blade Runner (1982)  
89,So I Married an Axe Murderer (1993)  
90,"Nightmare Before Christmas, The (1993)"  
91,True Romance (1993)  
92>Welcome to the Dollhouse (1995)  
93,Home Alone (1990)  
94,Aladdin (1992)  
95,Terminator 2: Judgment Day (1991)  
96,Dances with Wolves (1990)  
97,"Silence of the Lambs, The (1991)"  
98,Snow White and the Seven Dwarfs (1937)  
99,Fargo (1996)  
100,Heavy Metal (1981)  
101,"Aristocats, The (1970)"  
102,All Dogs Go to Heaven 2 (1996)  
103,Theodore Rex (1995)  
104,Sgt. Bilko (1996)  
105,Diabolique (1996)  
106,Moll Flanders (1996)  
107,Kids in the Hall: Brain Candy (1996)  
108,Mystery Science Theater 3000: The Movie (1996)  
109,Operation Dumbo Drop (1995)  
110,"Truth About Cats & Dogs, The (1996)"  
111,Flipper (1996)  
112,"Horseman on the Roof, The (Hussard sur le toit, Le) (1995)"  
113,Wallace & Gromit: The Best of Aardman Animation (1996)  
114,"Haunted World of Edward D. Wood Jr., The (1995)"  
115,Cold Comfort Farm (1995)  
116,"Rock, The (1996)"  
117,Twister (1996)  
118,Maya Lin: A Strong Clear Vision (1994)  
119,Striptease (1996)  
120,Independence Day (ID4) (1996)  
121,"Cable Guy, The (1996)"  
122,"Frighteners, The (1996)"  
123,Lone Star (1996)  
124,Phenomenon (1996)  
125,"Spitfire Grill, The (1996)"  
126,"Godfather, The (1972)"  
127,Supercop (1992)



---

128, Bound (1996)  
129, Kansas City (1996)  
130, Breakfast at Tiffany's (1961)  
131, "Wizard of Oz, The (1939)"  
132, Gone with the Wind (1939)  
133, Citizen Kane (1941)  
134, 2001: A Space Odyssey (1968)  
135, Mr. Smith Goes to Washington (1939)  
136, Big Night (1996)  
137, D3: The Mighty Ducks (1996)  
138, "Love Bug, The (1969)"  
139, Homeward Bound: The Incredible Journey (1993)  
140, "20,000 Leagues Under the Sea (1954)"  
141, Bedknobs and Broomsticks (1971)  
142, "Sound of Music, The (1965)"  
143, Die Hard (1988)  
144, "Lawnmower Man, The (1992)"  
145, Unhook the Stars (1996)  
146, "Long Kiss Goodnight, The (1996)"  
147, "Ghost and the Darkness, The (1996)"  
148, Jude (1996)  
149, Swingers (1996)  
150, Willy Wonka and the Chocolate Factory (1971)  
151, Sleeper (1973)  
152, "Fish Called Wanda, A (1988)"  
153, Monty Python's Life of Brian (1979)  
154, Dirty Dancing (1987)  
155, Reservoir Dogs (1992)  
156, Platoon (1986)  
157, Weekend at Bernie's (1989)  
158, Basic Instinct (1992)  
159, Glengarry Glen Ross (1992)  
160, Top Gun (1986)  
161, On Golden Pond (1981)  
162, "Return of the Pink Panther, The (1974)"  
163, "Abyss, The (1989)"  
164, Jean de Florette (1986)  
165, Manon of the Spring (Manon des sources) (1986)  
166, Private Benjamin (1980)  
167, Monty Python and the Holy Grail (1974)  
168, "Wrong Trousers, The (1993)"  
169, Cinema Paradiso (1988)  
170, Delicatessen (1991)  
171, "Empire Strikes Back, The (1980)"  
172, "Princess Bride, The (1987)"  
173, Raiders of the Lost Ark (1981)  
174, Brazil (1985)  
175, Aliens (1986)  
176, "Good, The Bad and The Ugly, The (1966)"  
177, 12 Angry Men (1957)  
178, "Clockwork Orange, A (1971)"  
179, Apocalypse Now (1979)  
180, Return of the Jedi (1983)  
181, GoodFellas (1990)  
182, Alien (1979)  
183, Army of Darkness (1993)  
184, Psycho (1960)  
185, "Blues Brothers, The (1980)"  
186, "Godfather: Part II, The (1974)"  
187, Full Metal Jacket (1987)  
188, "Grand Day Out, A (1992)"  
189, Henry V (1989)  
190, Amadeus (1984)  
191, Raging Bull (1980)  
192, "Right Stuff, The (1983)"  
193, "Sting, The (1973)"  
194, "Terminator, The (1984)"  
195, Dead Poets Society (1989)  
196, "Graduate, The (1967)"  
197, Nikita (La Femme Nikita) (1990)  
198, "Bridge on the River Kwai, The (1957)"  
199, "Shining, The (1980)"  
200, Evil Dead II (1987)  
201, Groundhog Day (1993)

202,Unforgiven (1992)  
203,Back to the Future (1985)  
204,Patton (1970)  
205,Akira (1988)  
206,Cyrano de Bergerac (1990)  
207,Young Frankenstein (1974)  
208,This Is Spinal Tap (1984)  
209,Indiana Jones and the Last Crusade (1989)  
210,M\*A\*S\*H (1970)  
211,"Unbearable Lightness of Being, The (1988)"  
212,"Room with a View, A (1986)"  
213,Pink Floyd - The Wall (1982)  
214,Field of Dreams (1989)  
215,When Harry Met Sally... (1989)  
216,Bram Stoker's Dracula (1992)  
217,Cape Fear (1991)  
218,"Nightmare on Elm Street, A (1984)"  
219,"Mirror Has Two Faces, The (1996)"  
220,Breaking the Waves (1996)  
221,Star Trek: First Contact (1996)  
222,Sling Blade (1996)  
223,Ridicule (1996)  
224,101 Dalmatians (1996)  
225,Die Hard 2 (1990)  
226,Star Trek VI: The Undiscovered Country (1991)  
227,Star Trek: The Wrath of Khan (1982)  
228,Star Trek III: The Search for Spock (1984)  
229,Star Trek IV: The Voyage Home (1986)  
230,Batman Returns (1992)  
231,Young Guns (1988)  
232,Under Siege (1992)  
233,Jaws (1975)  
234,Mars Attacks! (1996)  
235,Citizen Ruth (1996)  
236,Jerry Maguire (1996)  
237,Raising Arizona (1987)  
238,Sneakers (1992)  
239,Beavis and Butt-head Do America (1996)  
240,"Last of the Mohicans, The (1992)"  
241,Kolya (1996)  
242,Jungle2Jungle (1997)  
243,Smilla's Sense of Snow (1997)  
244,"Devil's Own, The (1997)"  
245,Chasing Amy (1997)  
246,Turbo: A Power Rangers Movie (1997)  
247,Grosse Pointe Blank (1997)  
248,Austin Powers: International Man of Mystery (1997)  
249,"Fifth Element, The (1997)"  
250,Shall We Dance? (1996)  
251,"Lost World: Jurassic Park, The (1997)"  
252,"Pillow Book, The (1995)"  
253,Batman & Robin (1997)  
254,My Best Friend's Wedding (1997)  
255,When the Cats Away (Chacun cherche son chat) (1996)  
256,Men in Black (1997)  
257,Contact (1997)  
258,George of the Jungle (1997)  
259,Event Horizon (1997)  
260,Air Bud (1997)  
261,In the Company of Men (1997)  
262,Steel (1997)  
263,Mimic (1997)  
264,"Hunt for Red October, The (1990)"  
265,Kull the Conqueror (1997)  
266,unknown  
267,Chasing Amy (1997)  
268,"Full Monty, The (1997)"  
269,Gattaca (1997)  
270,Starship Troopers (1997)  
271,Good Will Hunting (1997)  
272,Heat (1995)  
273,Sabrina (1995)  
274,Sense and Sensibility (1995)  
275,Leaving Las Vegas (1995)

276,Restoration (1995)  
277,Bed of Roses (1996)  
278,Once Upon a Time... When We Were Colored (1995)  
279,Up Close and Personal (1996)  
280,"River Wild, The (1994)"  
281,"Time to Kill, A (1996)"  
282,Emma (1996)  
283,Tin Cup (1996)  
284,Secrets & Lies (1996)  
285,"English Patient, The (1996)"  
286,Marvin's Room (1996)  
287,Scream (1996)  
288,Evita (1996)  
289,Fierce Creatures (1997)  
290,Absolute Power (1997)  
291,Rosewood (1997)  
292,Donnie Brasco (1997)  
293,Liar Liar (1997)  
294,Breakdown (1997)  
295,"Promesse, La (1996)"  
296,Ulee's Gold (1997)  
297,Face/Off (1997)  
298,Hoodlum (1997)  
299,Air Force One (1997)  
300,In & Out (1997)  
301,L.A. Confidential (1997)  
302,Ulee's Gold (1997)  
303,Fly Away Home (1996)  
304,"Ice Storm, The (1997)"  
305,"Mrs. Brown (Her Majesty, Mrs. Brown) (1997)"  
306,"Devil's Advocate, The (1997)"  
307,FairyTale: A True Story (1997)  
308,Deceiver (1997)  
309,"Rainmaker, The (1997)"  
310,"Wings of the Dove, The (1997)"  
311,Midnight in the Garden of Good and Evil (1997)  
312,Titanic (1997)  
313,3 Ninjas: High Noon At Mega Mountain (1998)  
314,Apt Pupil (1998)  
315,As Good As It Gets (1997)  
316,In the Name of the Father (1993)  
317,Schindler's List (1993)  
318,Everyone Says I Love You (1996)  
319,Paradise Lost: The Child Murders at Robin Hood Hills (1996)  
320,Mother (1996)  
321,Murder at 1600 (1997)  
322,Dante's Peak (1997)  
323,Lost Highway (1997)  
324,Crash (1996)  
325,G.I. Jane (1997)  
326,Cop Land (1997)  
327,Conspiracy Theory (1997)  
328,Desperate Measures (1998)  
329,187 (1997)  
330,"Edge, The (1997)"  
331,Kiss the Girls (1997)  
332,"Game, The (1997)"  
333,U Turn (1997)  
334,How to Be a Player (1997)  
335,Playing God (1997)  
336,"House of Yes, The (1997)"  
337,Bean (1997)  
338,Mad City (1997)  
339,Boogie Nights (1997)  
340,Critical Care (1997)  
341,"Man Who Knew Too Little, The (1997)"  
342,Alien: Resurrection (1997)  
343,"Apostle, The (1997)"  
344,Deconstructing Harry (1997)  
345,Jackie Brown (1997)  
346,Wag the Dog (1997)  
347,Desperate Measures (1998)  
348,Hard Rain (1998)  
349,Fallen (1998)

350,"Prophecy II, The (1998)"  
351,Spice World (1997)  
352,Deep Rising (1998)  
353,"Wedding Singer, The (1998)"  
354,Sphere (1998)  
355,"Client, The (1994)"  
356,One Flew Over the Cuckoo's Nest (1975)  
357,Spawn (1997)  
358,"Assignment, The (1997)"  
359,Wonderland (1997)  
360,Incognito (1997)  
361,Blues Brothers 2000 (1998)  
362,Sudden Death (1995)  
363,Ace Ventura: When Nature Calls (1995)  
364,Powder (1995)  
365,Dangerous Minds (1995)  
366,Clueless (1995)  
367,Bio-Dome (1996)  
368,Black Sheep (1996)  
369,Mary Reilly (1996)  
370,"Bridges of Madison County, The (1995)"  
371,Jeffrey (1995)  
372,Judge Dredd (1995)  
373,Mighty Morphin Power Rangers: The Movie (1995)  
374>Showgirls (1995)  
375,Houseguest (1994)  
376,Heavyweights (1994)  
377,Miracle on 34th Street (1994)  
378,Tales From the Crypt Presents: Demon Knight (1995)  
379,Star Trek: Generations (1994)  
380,Muriel's Wedding (1994)  
381,"Adventures of Priscilla, Queen of the Desert, The (1994)"  
382,"Flintstones, The (1994)"  
383,Naked Gun 33 1/3: The Final Insult (1994)  
384,True Lies (1994)  
385,Addams Family Values (1993)  
386,"Age of Innocence, The (1993)"  
387,Beverly Hills Cop III (1994)  
388,Black Beauty (1994)  
389,Fear of a Black Hat (1993)  
390,Last Action Hero (1993)  
391,"Man Without a Face, The (1993)"  
392,Mrs. Doubtfire (1993)  
393,Radioland Murders (1994)  
394,Robin Hood: Men in Tights (1993)  
395,Serial Mom (1994)  
396,Striking Distance (1993)  
397,Super Mario Bros. (1993)  
398,"Three Musketeers, The (1993)"  
399,"Little Rascals, The (1994)"  
400,"Brady Bunch Movie, The (1995)"  
401,Ghost (1990)  
402,Batman (1989)  
403,Pinocchio (1940)  
404,Mission: Impossible (1996)  
405,Thinner (1996)  
406,Spy Hard (1996)  
407,"Close Shave, A (1995)"  
408,Jack (1996)  
409,Kingpin (1996)  
410,"Nutty Professor, The (1996)"  
411,"Very Brady Sequel, A (1996)"  
412,Tales from the Crypt Presents: Bordello of Blood (1996)  
413,My Favorite Year (1982)  
414,"Apple Dumpling Gang, The (1975)"  
415,Old Yeller (1957)  
416,"Parent Trap, The (1961)"  
417,Cinderella (1950)  
418,Mary Poppins (1964)  
419,Alice in Wonderland (1951)  
420,William Shakespeare's Romeo and Juliet (1996)  
421,Aladdin and the King of Thieves (1996)  
422,E.T. the Extra-Terrestrial (1982)  
423,Children of the Corn: The Gathering (1996)

424,Bob Roberts (1992)  
425,"Transformers: The Movie, The (1986)"  
426,To Kill a Mockingbird (1962)  
427,Harold and Maude (1971)  
428,"Day the Earth Stood Still, The (1951)"  
429,Duck Soup (1933)  
430,Highlander (1986)  
431,Fantasia (1940)  
432,Heathers (1989)  
433,Forbidden Planet (1956)  
434,Butch Cassidy and the Sundance Kid (1969)  
435,"American Werewolf in London, An (1981)"  
436,Amityville 1992: It's About Time (1992)  
437,Amityville 3-D (1983)  
438,Amityville: A New Generation (1993)  
439,Amityville II: The Possession (1982)  
440,"Amityville Horror, The (1979)"  
441,"Amityville Curse, The (1990)"  
442,"Birds, The (1963)"  
443,"Blob, The (1958)"  
444,"Body Snatcher, The (1945)"  
445,Burnt Offerings (1976)  
446,Carrie (1976)  
447,"Omen, The (1976)"  
448,Star Trek: The Motion Picture (1979)  
449,Star Trek V: The Final Frontier (1989)  
450,Grease (1978)  
451,Jaws 2 (1978)  
452,Jaws 3-D (1983)  
453,Bastard Out of Carolina (1996)  
454,Jackie Chan's First Strike (1996)  
455,Beverly Hills Ninja (1997)  
456,Free Willy 3: The Rescue (1997)  
457,Nixon (1995)  
458,"Cry, the Beloved Country (1995)"  
459,"Crossing Guard, The (1995)"  
460,Smoke (1995)  
461,Like Water For Chocolate (Como agua para chocolate) (1992)  
462,"Secret of Roan Inish, The (1994)"  
463,Vanya on 42nd Street (1994)  
464,"Jungle Book, The (1994)"  
465,Red Rock West (1992)  
466,"Bronx Tale, A (1993)"  
467,Rudy (1993)  
468,Short Cuts (1993)  
469,Tombstone (1993)  
470,Courage Under Fire (1996)  
471,Dragonheart (1996)  
472,James and the Giant Peach (1996)  
473,Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb (1963)  
474,Trainspotting (1996)  
475,"First Wives Club, The (1996)"  
476,Matilda (1996)  
477,"Philadelphia Story, The (1940)"  
478,Vertigo (1958)  
479,North by Northwest (1959)  
480,"Apartment, The (1960)"  
481,Some Like It Hot (1959)  
482,Casablanca (1942)  
483,"Maltese Falcon, The (1941)"  
484,My Fair Lady (1964)  
485,Sabrina (1954)  
486,Roman Holiday (1953)  
487,Sunset Blvd. (1950)  
488,Notorious (1946)  
489,To Catch a Thief (1955)  
490,"Adventures of Robin Hood, The (1938)"  
491,East of Eden (1955)  
492,"Thin Man, The (1934)"  
493,His Girl Friday (1940)  
494,Around the World in 80 Days (1956)  
495,It's a Wonderful Life (1946)  
496,Bringing Up Baby (1938)  
497,"African Queen, The (1951)"

498,Cat on a Hot Tin Roof (1958)  
499,Fly Away Home (1996)  
500,Dumbo (1941)  
501,Bananas (1971)  
502,"Candidate, The (1972)"  
503,Bonnie and Clyde (1967)  
504,Dial M for Murder (1954)  
505,Rebel Without a Cause (1955)  
506,"Streetcar Named Desire, A (1951)"  
507,"People vs. Larry Flynt, The (1996)"  
508,My Left Foot (1989)  
509,"Magnificent Seven, The (1954)"  
510,Lawrence of Arabia (1962)  
511,Wings of Desire (1987)  
512,"Third Man, The (1949)"  
513,Annie Hall (1977)  
514,"Boot, Das (1981)"  
515,Local Hero (1983)  
516,Manhattan (1979)  
517,Miller's Crossing (1990)  
518,"Treasure of the Sierra Madre, The (1948)"  
519,"Great Escape, The (1963)"  
520,"Deer Hunter, The (1978)"  
521,Down by Law (1986)  
522,Cool Hand Luke (1967)  
523,"Great Dictator, The (1940)"  
524,"Big Sleep, The (1946)"  
525,Ben-Hur (1959)  
526,Gandhi (1982)  
527,"Killing Fields, The (1984)"  
528,My Life as a Dog (Mitt liv som hund) (1985)  
529,"Man Who Would Be King, The (1975)"  
530,Shine (1996)  
531,Kama Sutra: A Tale of Love (1996)  
532,"Daytrippers, The (1996)"  
533,Traveller (1997)  
534,Addicted to Love (1997)  
535,Ponette (1996)  
536,My Own Private Idaho (1991)  
537,Anastasia (1997)  
538,Mouse Hunt (1997)  
539,Money Train (1995)  
540,Mortal Kombat (1995)  
541,Pocahontas (1995)  
542,"Misérables, Les (1995)"  
543,Things to Do in Denver when You're Dead (1995)  
544,Vampire in Brooklyn (1995)  
545,Broken Arrow (1996)  
546,"Young Poisoner's Handbook, The (1995)"  
547,"NeverEnding Story III, The (1994)"  
548,Rob Roy (1995)  
549,Die Hard: With a Vengeance (1995)  
550,Lord of Illusions (1995)  
551,Species (1995)  
552,"Walk in the Clouds, A (1995)"  
553,Waterworld (1995)  
554,White Man's Burden (1995)  
555,Wild Bill (1995)  
556,Farinelli: il castrato (1994)  
557,Heavenly Creatures (1994)  
558,Interview with the Vampire (1994)  
559,"Kid in King Arthur's Court, A (1995)"  
560,Mary Shelley's Frankenstein (1994)  
561,"Quick and the Dead, The (1995)"  
562,Stephen King's The Langoliers (1995)  
563,Tales from the Hood (1995)  
564,Village of the Damned (1995)  
565,Clear and Present Danger (1994)  
566,Wes Craven's New Nightmare (1994)  
567,Speed (1994)  
568,Wolf (1994)  
569,Wyatt Earp (1994)  
570,Another Stakeout (1993)  
571,Blown Away (1994)

572,Body Snatchers (1993)  
573,Boxing Helena (1993)  
574,City Slickers II: The Legend of Curly's Gold (1994)  
575,Cliffhanger (1993)  
576,Coneheads (1993)  
577,Demolition Man (1993)  
578,Fatal Instinct (1993)  
579,"Englishman Who Went Up a Hill, But Came Down a Mountain, The (1995)"  
580,Kalifornia (1993)  
581,"Piano, The (1993)"  
582,Romeo Is Bleeding (1993)  
583,"Secret Garden, The (1993)"  
584,Son in Law (1993)  
585,Terminal Velocity (1994)  
586,"Hour of the Pig, The (1993)"  
587,Beauty and the Beast (1991)  
588,"Wild Bunch, The (1969)"  
589,Hellraiser: Bloodline (1996)  
590,Primal Fear (1996)  
591,True Crime (1995)  
592,Stalingrad (1993)  
593,Heavy (1995)  
594,"Fan, The (1996)"  
595,"Hunchback of Notre Dame, The (1996)"  
596,Eraser (1996)  
597,"Big Squeeze, The (1996)"  
598,Police Story 4: Project S (Chao ji ji hua) (1993)  
599,Daniel Defoe's Robinson Crusoe (1996)  
600,For Whom the Bell Tolls (1943)  
601,"American in Paris, An (1951)"  
602,Rear Window (1954)  
603,It Happened One Night (1934)  
604,Meet Me in St. Louis (1944)  
605,All About Eve (1950)  
606,Rebecca (1940)  
607,Spellbound (1945)  
608,Father of the Bride (1950)  
609,Gigi (1958)  
610,Laura (1944)  
611,Lost Horizon (1937)  
612,My Man Godfrey (1936)  
613,Giant (1956)  
614,"39 Steps, The (1935)"  
615,Night of the Living Dead (1968)  
616,"Blue Angel, The (Blaue Engel, Der) (1930)"  
617,Picnic (1955)  
618,Extreme Measures (1996)  
619,"Chamber, The (1996)"  
620,"Davy Crockett, King of the Wild Frontier (1955)"  
621,Swiss Family Robinson (1960)  
622,Angels in the Outfield (1994)  
623,"Three Caballeros, The (1945)"  
624,"Sword in the Stone, The (1963)"  
625,So Dear to My Heart (1949)  
626,Robin Hood: Prince of Thieves (1991)  
627,Sleepers (1996)  
628,Victor/Victoria (1982)  
629,"Great Race, The (1965)"  
630,"Crying Game, The (1992)"  
631,Sophie's Choice (1982)  
632,"Christmas Carol, A (1938)"  
633,Microcosmos: Le peuple de l'herbe (1996)  
634,"Fog, The (1980)"  
635,Escape from New York (1981)  
636,"Howling, The (1981)"  
637,"Return of Martin Guerre, The (Retour de Martin Guerre, Le) (1982)"  
638,"Tin Drum, The (Blechtrommel, Die) (1979)"  
639,"Cook the Thief His Wife & Her Lover, The (1989)"  
640,Paths of Glory (1957)  
641,"Grifters, The (1990)"  
642,The Innocent (1994)  
643,"Thin Blue Line, The (1988)"  
644,Paris Is Burning (1990)  
645,Once Upon a Time in the West (1969)

646,Ran (1985)  
647,"Quiet Man, The (1952)"  
648,Once Upon a Time in America (1984)  
649,"Seventh Seal, The (Sjunde inseglet, Det) (1957)"  
650,Glory (1989)  
651,Rosencrantz and Guildenstern Are Dead (1990)  
652,Touch of Evil (1958)  
653,Chinatown (1974)  
654,Stand by Me (1986)  
655,M (1931)  
656,"Manchurian Candidate, The (1962)"  
657,Pump Up the Volume (1990)  
658,Arsenic and Old Lace (1944)  
659,Fried Green Tomatoes (1991)  
660,High Noon (1952)  
661,Somewhere in Time (1980)  
662,Being There (1979)  
663,"Paris, Texas (1984)"  
664,Alien 3 (1992)  
665,Blood For Dracula (Andy Warhol's Dracula) (1974)  
666,Audrey Rose (1977)  
667,Blood Beach (1981)  
668,Body Parts (1991)  
669,Body Snatchers (1993)  
670,Bride of Frankenstein (1935)  
671,Candyman (1992)  
672,Cape Fear (1962)  
673,Cat People (1982)  
674,"Nosferatu (Nosferatu, eine Symphonie des Grauens) (1922)"  
675,"Crucible, The (1996)"  
676,Fire on the Mountain (1996)  
677,Volcano (1997)  
678,Conan the Barbarian (1981)  
679,Kull the Conqueror (1997)  
680,Wishmaster (1997)  
681,I Know What You Did Last Summer (1997)  
682,Rocket Man (1997)  
683,In the Line of Fire (1993)  
684,Executive Decision (1996)  
685,"Perfect World, A (1993)"  
686,McHale's Navy (1997)  
687,Leave It to Beaver (1997)  
688,"Jackal, The (1997)"  
689,Seven Years in Tibet (1997)  
690,Dark City (1998)  
691,"American President, The (1995)"  
692,Casino (1995)  
693,Persuasion (1995)  
694,Kicking and Screaming (1995)  
695,City Hall (1996)  
696,"Basketball Diaries, The (1995)"  
697,"Browning Version, The (1994)"  
698,Little Women (1994)  
699,Miami Rhapsody (1995)  
700,"Wonderful, Horrible Life of Leni Riefenstahl, The (1993)"  
701,Barcelona (1994)  
702,Widows' Peak (1994)  
703,"House of the Spirits, The (1993)"  
704,Singin' in the Rain (1952)  
705,Bad Moon (1996)  
706,Enchanted April (1991)  
707,"Sex, Lies, and Videotape (1989)"  
708,Strictly Ballroom (1992)  
709,Better Off Dead... (1985)  
710,"Substance of Fire, The (1996)"  
711,Tin Men (1987)  
712,Othello (1995)  
713,Carrington (1995)  
714,To Die For (1995)  
715,Home for the Holidays (1995)  
716,"Juror, The (1996)"  
717,In the Bleak Midwinter (1995)  
718,Canadian Bacon (1994)  
719,First Knight (1995)



720,Mallrats (1995)  
721,Nine Months (1995)  
722,Boys on the Side (1995)  
723,Circle of Friends (1995)  
724,Exit to Eden (1994)  
725,Fluke (1995)  
726,Immortal Beloved (1994)  
727,Junior (1994)  
728,Nell (1994)  
729,"Queen Margot (Reine Margot, La) (1994)"  
730,"Corrina, Corrina (1994)"  
731,Dave (1993)  
732,Go Fish (1994)  
733,Made in America (1993)  
734,Philadelphia (1993)  
735,Shadowlands (1993)  
736,Sirens (1994)  
737,Threesome (1994)  
738,Pretty Woman (1990)  
739,Jane Eyre (1996)  
740,"Last Supper, The (1995)"  
741,Ransom (1996)  
742,"Crow: City of Angels, The (1996)"  
743,Michael Collins (1996)  
744,"Ruling Class, The (1972)"  
745,Real Genius (1985)  
746,Benny & Joon (1993)  
747,"Saint, The (1997)"  
748,"MatchMaker, The (1997)"  
749,Amistad (1997)  
750,Tomorrow Never Dies (1997)  
751,"Replacement Killers, The (1998)"  
752,Burnt By the Sun (1994)  
753,Red Corner (1997)  
754,Jumanji (1995)  
755,Father of the Bride Part II (1995)  
756,Across the Sea of Time (1995)  
757,Lawnmower Man 2: Beyond Cyberspace (1996)  
758,Fair Game (1995)  
759,Screamers (1995)  
760,Nick of Time (1995)  
761,Beautiful Girls (1996)  
762,Happy Gilmore (1996)  
763,If Lucy Fell (1996)  
764,Boomerang (1992)  
765,Man of the Year (1995)  
766,"Addiction, The (1995)"  
767,Casper (1995)  
768,Congo (1995)  
769,Devil in a Blue Dress (1995)  
770,Johnny Mnemonic (1995)  
771,Kids (1995)  
772,Mute Witness (1994)  
773,"Prophecy, The (1995)"  
774,Something to Talk About (1995)  
775,Three Wishes (1995)  
776,Castle Freak (1995)  
777,Don Juan DeMarco (1995)  
778,Drop Zone (1994)  
779,Dumb & Dumber (1994)  
780,French Kiss (1995)  
781,Little Odessa (1994)  
782,Milk Money (1994)  
783,Beyond Bedlam (1993)  
784,Only You (1994)  
785,"Perez Family, The (1995)"  
786,Roommates (1995)  
787,Relative Fear (1994)  
788,Swimming with Sharks (1995)  
789,Tommy Boy (1995)  
790,"Baby-Sitters Club, The (1995)"  
791,Bullets Over Broadway (1994)  
792,Crooklyn (1994)  
793,It Could Happen to You (1994)

794, Rich (1994)  
795, Speechless (1994)  
796, Timecop (1994)  
797, Bad Company (1995)  
798, Boys Life (1995)  
799, In the Mouth of Madness (1995)  
800, "Air Up There, The (1994)"  
801, Hard Target (1993)  
802, Heaven & Earth (1993)  
803, Jimmy Hollywood (1994)  
804, Manhattan Murder Mystery (1993)  
805, Menace II Society (1993)  
806, Poetic Justice (1993)  
807, "Program, The (1993)"  
808, Rising Sun (1993)  
809, "Shadow, The (1994)"  
810, Thirty-Two Short Films About Glenn Gould (1993)  
811, Andre (1994)  
812, "Celluloid Closet, The (1995)"  
813, "Great Day in Harlem, A (1994)"  
814, One Fine Day (1996)  
815, Candyman: Farewell to the Flesh (1995)  
816, Frisk (1995)  
817, Girl 6 (1996)  
818, Eddie (1996)  
819, Space Jam (1996)  
820, Mrs. Winterbourne (1996)  
821, Faces (1968)  
822, Mulholland Falls (1996)  
823, "Great White Hype, The (1996)"  
824, "Arrival, The (1996)"  
825, "Phantom, The (1996)"  
826, Daylight (1996)  
827, Alaska (1996)  
828, Fled (1996)  
829, Power 98 (1995)  
830, Escape from L.A. (1996)  
831, Bogus (1996)  
832, Bulletproof (1996)  
833, Halloween: The Curse of Michael Myers (1995)  
834, "Gay Divorcee, The (1934)"  
835, Ninotchka (1939)  
836, Meet John Doe (1941)  
837, In the Line of Duty 2 (1987)  
838, Loch Ness (1995)  
839, Last Man Standing (1996)  
840, "Glimmer Man, The (1996)"  
841, Pollyanna (1960)  
842, "Shaggy Dog, The (1959)"  
843, Freeway (1996)  
844, That Thing You Do! (1996)  
845, To Gillian on Her 37th Birthday (1996)  
846, Looking for Richard (1996)  
847, "Murder, My Sweet (1944)"  
848, Days of Thunder (1990)  
849, "Perfect Candidate, A (1996)"  
850, Two or Three Things I Know About Her (1966)  
851, "Bloody Child, The (1996)"  
852, Braindead (1992)  
853, Bad Taste (1987)  
854, Diva (1981)  
855, Night on Earth (1991)  
856, Paris Was a Woman (1995)  
857, Amityville: Dollhouse (1996)  
858, April Fool's Day (1986)  
859, "Believers, The (1987)"  
860, Nosferatu a Venezia (1986)  
861, Jingle All the Way (1996)  
862, "Garden of Finzi-Contini, The (Giardino dei Finzi-Contini, Il) (1970)"  
863, My Fellow Americans (1996)  
864, "Ice Storm, The (1997)"  
865, Michael (1996)  
866, "Whole Wide World, The (1996)"  
867, Hearts and Minds (1996)

868,Fools Rush In (1997)  
869,Touch (1997)  
870,Vegas Vacation (1997)  
871,Love Jones (1997)  
872,Picture Perfect (1997)  
873,Career Girls (1997)  
874,She's So Lovely (1997)  
875,Money Talks (1997)  
876,Excess Baggage (1997)  
877,That Darn Cat! (1997)  
878,"Peacemaker, The (1997)"  
879,Soul Food (1997)  
880,Money Talks (1997)  
881,Washington Square (1997)  
882,Telling Lies in America (1997)  
883,Year of the Horse (1997)  
884,Phantoms (1998)  
885,"Life Less Ordinary, A (1997)"  
886,Eve's Bayou (1997)  
887,One Night Stand (1997)  
888,"Tango Lesson, The (1997)"  
889,Mortal Kombat: Annihilation (1997)  
890,Bent (1997)  
891,Flubber (1997)  
892,For Richer or Poorer (1997)  
893,Home Alone 3 (1997)  
894,Scream 2 (1997)  
895,"Sweet Hereafter, The (1997)"  
896,Time Tracers (1995)  
897,"Postman, The (1997)"  
898,"Winter Guest, The (1997)"  
899,Kundun (1997)  
900,Mr. Magoo (1997)  
901,"Big Lebowski, The (1998)"  
902,Afterglow (1997)  
903,Ma vie en rose (My Life in Pink) (1997)  
904,Great Expectations (1998)  
905,Oscar & Lucinda (1997)  
906,Vermin (1998)  
907,Half Baked (1998)  
908,Dangerous Beauty (1998)  
909,Nil By Mouth (1997)  
910,Twilight (1998)  
911,U.S. Marshalls (1998)  
912,Love and Death on Long Island (1997)  
913,Wild Things (1998)  
914,Primary Colors (1998)  
915,Lost in Space (1998)  
916,Mercury Rising (1998)  
917,City of Angels (1998)  
918,"City of Lost Children, The (1995)"  
919,Two Bits (1995)  
920,Farewell My Concubine (1993)  
921,Dead Man (1995)  
922,Raise the Red Lantern (1991)  
923,White Squall (1996)  
924,Unforgettable (1996)  
925,Down Periscope (1996)  
926,"Flower of My Secret, The (Flor de mi secreto, La) (1995)"  
927,"Craft, The (1996)"  
928,Harriet the Spy (1996)  
929,Chain Reaction (1996)  
930,"Island of Dr. Moreau, The (1996)"  
931,First Kid (1996)  
932,"Funeral, The (1996)"  
933,"Preacher's Wife, The (1996)"  
934,Paradise Road (1997)  
935,Brassed Off (1996)  
936,"Thousand Acres, A (1997)"  
937,"Smile Like Yours, A (1997)"  
938,Murder in the First (1995)  
939,Airheads (1994)  
940,With Honors (1994)  
941,What's Love Got to Do with It (1993)

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942,Killing Zoe (1994)  
943,Renaissance Man (1994)  
944,Charade (1963)  
945,"Fox and the Hound, The (1981)"  
946,"Big Blue, The (Grand bleu, Le) (1988)"  
947,Booty Call (1997)  
948,How to Make an American Quilt (1995)  
949,Georgia (1995)  
950,"Indian in the Cupboard, The (1995)"  
951,Blue in the Face (1995)  
952,Unstrung Heroes (1995)  
953,Unzipped (1995)  
954,Before Sunrise (1995)  
955,Nobody's Fool (1994)  
956,Pushing Hands (1992)  
957,To Live (Huozhe) (1994)  
958,Dazed and Confused (1993)  
959,Naked (1993)  
960,Orlando (1993)  
961,Ruby in Paradise (1993)  
962,Some Folks Call It a Sling Blade (1993)  
963,"Month by the Lake, A (1995)"  
964,Funny Face (1957)  
965,"Affair to Remember, An (1957)"  
966,Little Lord Fauntleroy (1936)  
967,"Inspector General, The (1949)"  
968,Winnie the Pooh and the Blustery Day (1968)  
969,Hear My Song (1991)  
970,Mediterraneo (1991)  
971,Passion Fish (1992)  
972,Grateful Dead (1995)  
973,Eye for an Eye (1996)  
974,Fear (1996)  
975,Solo (1996)  
976,"Substitute, The (1996)"  
977,Heaven's Prisoners (1996)  
978,"Trigger Effect, The (1996)"  
979,Mother Night (1996)  
980,Dangerous Ground (1997)  
981,Maximum Risk (1996)  
982,"Rich Man's Wife, The (1996)"  
983,Shadow Conspiracy (1997)  
984,Blood & Wine (1997)  
985,Turbulence (1997)  
986,Underworld (1997)  
987,"Beautician and the Beast, The (1997)"  
988,Cats Don't Dance (1997)  
989,Anna Karenina (1997)  
990,Keys to Tulsa (1997)  
991,Head Above Water (1996)  
992,Hercules (1997)  
993,"Last Time I Committed Suicide, The (1997)"  
994,"Kiss Me, Guido (1997)"  
995,"Big Green, The (1995)"  
996,Stuart Saves His Family (1995)  
997,Cabin Boy (1994)  
998,Clean Slate (1994)  
999,Lightning Jack (1994)  
1000,"Stupids, The (1996)"  
1001,"Pest, The (1997)"  
1002,That Darn Cat! (1997)  
1003,Geronimo: An American Legend (1993)  
1004,"Double vie de Véronique, La (Double Life of Veronique, The) (1991)"  
1005,Until the End of the World (Bis ans Ende der Welt) (1991)  
1006,Waiting for Guffman (1996)  
1007,I Shot Andy Warhol (1996)  
1008,Stealing Beauty (1996)  
1009,Basquiat (1996)  
1010,2 Days in the Valley (1996)  
1011,Private Parts (1997)  
1012,Anaconda (1997)  
1013,Romy and Michele's High School Reunion (1997)  
1014,Shiloh (1997)  
1015,Con Air (1997)

1016,Trees Lounge (1996)  
1017,Tie Me Up! Tie Me Down! (1990)  
1018,"Die xue shuang xiong (Killer, The) (1989)"  
1019,Gaslight (1944)  
1020,8 1/2 (1963)  
1021,"Fast, Cheap & Out of Control (1997)"  
1022,Fathers' Day (1997)  
1023,Mrs. Dalloway (1997)  
1024,Fire Down Below (1997)  
1025,"Lay of the Land, The (1997)"  
1026,"Shooter, The (1995)"  
1027,Grumpier Old Men (1995)  
1028,Jury Duty (1995)  
1029,"Beverly Hillbillies, The (1993)"  
1030,Lassie (1994)  
1031,Little Big League (1994)  
1032,Homeward Bound II: Lost in San Francisco (1996)  
1033,"Quest, The (1996)"  
1034,Cool Runnings (1993)  
1035,Drop Dead Fred (1991)  
1036,Grease 2 (1982)  
1037,Switchback (1997)  
1038,Hamlet (1996)  
1039,Two if by Sea (1996)  
1040,Forget Paris (1995)  
1041,Just Cause (1995)  
1042,Rent-a-Kid (1995)  
1043,"Paper, The (1994)"  
1044,Fearless (1993)  
1045,Malice (1993)  
1046,Multiplicity (1996)  
1047,She's the One (1996)  
1048,House Arrest (1996)  
1049,"Ghost and Mrs. Muir, The (1947)"  
1050,"Associate, The (1996)"  
1051,Dracula: Dead and Loving It (1995)  
1052,Now and Then (1995)  
1053,Mr. Wrong (1996)  
1054,"Simple Twist of Fate, A (1994)"  
1055,Cronos (1992)  
1056,"Pallbearer, The (1996)"  
1057,"War, The (1994)"  
1058,Don't Be a Menace to South Central While Drinking Your Juice in the Hood (1996)  
1059,"Adventures of Pinocchio, The (1996)"  
1060,"Evening Star, The (1996)"  
1061,Four Days in September (1997)  
1062,"Little Princess, A (1995)"  
1063,Crossfire (1947)  
1064,Koyaanisqatsi (1983)  
1065,Balto (1995)  
1066,Bottle Rocket (1996)  
1067,"Star Maker, The (Uomo delle stelle, L') (1995)"  
1068,Amateur (1994)  
1069,Living in Oblivion (1995)  
1070,Party Girl (1995)  
1071,"Pyromaniac's Love Story, A (1995)"  
1072,Shallow Grave (1994)  
1073,Reality Bites (1994)  
1074,"Man of No Importance, A (1994)"  
1075,"Pagemaster, The (1994)"  
1076,Love and a .45 (1994)  
1077,Oliver & Company (1988)  
1078,Joe's Apartment (1996)  
1079,Celestial Clockwork (1994)  
1080,Curdled (1996)  
1081,Female Perversions (1996)  
1082,Albino Alligator (1996)  
1083,Anne Frank Remembered (1995)  
1084,Carried Away (1996)  
1085,It's My Party (1995)  
1086,Bloodsport 2 (1995)  
1087,Double Team (1997)  
1088,Speed 2: Cruise Control (1997)  
1089,Sliver (1993)

1090,Pete's Dragon (1977)  
1091,Dear God (1996)  
1092,Live Nude Girls (1995)  
1093,"Thin Line Between Love and Hate, A (1996)"  
1094,High School High (1996)  
1095,Commandments (1997)  
1096,"Hate (Haine, La) (1995)"  
1097,Flirting With Disaster (1996)  
1098,"Red Firecracker, Green Firecracker (1994)"  
1099,What Happened Was... (1994)  
1100,Six Degrees of Separation (1993)  
1101,Two Much (1996)  
1102,Trust (1990)  
1103,C'est arrivé près de chez vous (1992)  
1104,Firestorm (1998)  
1105,"Newton Boys, The (1998)"  
1106,Beyond Rangoon (1995)  
1107,Feast of July (1995)  
1108,Death and the Maiden (1994)  
1109,Tank Girl (1995)  
1110,Double Happiness (1994)  
1111,Cobb (1994)  
1112,Mrs. Parker and the Vicious Circle (1994)  
1113,Faithful (1996)  
1114,Twelfth Night (1996)  
1115,"Mark of Zorro, The (1940)"  
1116,Surviving Picasso (1996)  
1117,Up in Smoke (1978)  
1118,Some Kind of Wonderful (1987)  
1119,I'm Not Rappaport (1996)  
1120,"Umbrellas of Cherbourg, The (Parapluies de Cherbourg, Les) (1964)"  
1121,They Made Me a Criminal (1939)  
1122,"Last Time I Saw Paris, The (1954)"  
1123,"Farewell to Arms, A (1932)"  
1124,"Innocents, The (1961)"  
1125,"Old Man and the Sea, The (1958)"  
1126,"Truman Show, The (1998)"  
1127,Heidi Fleiss: Hollywood Madam (1995)  
1128,Chungking Express (1994)  
1129,Jupiter's Wife (1994)  
1130,Safe (1995)  
1131,Feeling Minnesota (1996)  
1132,Escape to Witch Mountain (1975)  
1133,Get on the Bus (1996)  
1134,"Doors, The (1991)"  
1135,Ghosts of Mississippi (1996)  
1136,Beautiful Thing (1996)  
1137,Best Men (1997)  
1138,Hackers (1995)  
1139,"Road to Wellville, The (1994)"  
1140,"War Room, The (1993)"  
1141,When We Were Kings (1996)  
1142,Hard Eight (1996)  
1143,"Quiet Room, The (1996)"  
1144,Blue Chips (1994)  
1145,Calendar Girl (1993)  
1146,My Family (1995)  
1147,Tom & Viv (1994)  
1148,Walkabout (1971)  
1149,Last Dance (1996)  
1150,Original Gangstas (1996)  
1151,In Love and War (1996)  
1152,Backbeat (1993)  
1153,Alphaville (1965)  
1154,"Rendezvous in Paris (Rendez-vous de Paris, Les) (1995)"  
1155,Cyclo (1995)  
1156,"Relic, The (1997)"  
1157,"Fille seule, La (A Single Girl) (1995)"  
1158,Stalker (1979)  
1159,Love! Valour! Compassion! (1997)  
1160,Palookaville (1996)  
1161,Phat Beach (1996)  
1162,"Portrait of a Lady, The (1996)"  
1163,Zeus and Roxanne (1997)

1164,Big Bully (1996)  
1165,Love & Human Remains (1993)  
1166,"Sum of Us, The (1994)"  
1167,Little Buddha (1993)  
1168,Fresh (1994)  
1169,Spanking the Monkey (1994)  
1170,Wild Reeds (1994)  
1171,"Women, The (1939)"  
1172,Bliss (1997)  
1173,Caught (1996)  
1174,Hugo Pool (1997)  
1175>Welcome To Sarajevo (1997)  
1176,Dunston Checks In (1996)  
1177,Major Payne (1994)  
1178,Man of the House (1995)  
1179,I Love Trouble (1994)  
1180,"Low Down Dirty Shame, A (1994)"  
1181,Cops and Robbersons (1994)  
1182,"Cowboy Way, The (1994)"  
1183,"Endless Summer 2, The (1994)"  
1184,In the Army Now (1994)  
1185,"Inkwell, The (1994)"  
1186,Switchblade Sisters (1975)  
1187,Young Guns II (1990)  
1188,Prefontaine (1997)  
1189,That Old Feeling (1997)  
1190,"Letter From Death Row, A (1998)"  
1191,"Boys of St. Vincent, The (1993)"  
1192,Before the Rain (Pred dozhdot) (1994)  
1193,Once Were Warriors (1994)  
1194,Strawberry and Chocolate (Fresa y chocolate) (1993)  
1195,"Savage Nights (Nuits fauves, Les) (1992)"  
1196,"Family Thing, A (1996)"  
1197,Purple Noon (1960)  
1198,Cemetery Man (Dellamorte Dellamore) (1994)  
1199,Kim (1950)  
1200,Marlene Dietrich: Shadow and Light (1996)  
1201,"Maybe, Maybe Not (Bewegte Mann, Der) (1994)"  
1202,Top Hat (1935)  
1203,To Be or Not to Be (1942)  
1204,"Secret Agent, The (1996)"  
1205,Amos & Andrew (1993)  
1206,Jade (1995)  
1207,Kiss of Death (1995)  
1208,Mixed Nuts (1994)  
1209,Virtuosity (1995)  
1210,Blue Sky (1994)  
1211,Flesh and Bone (1993)  
1212,Guilty as Sin (1993)  
1213,In the Realm of the Senses (Ai no corrida) (1976)  
1214,Barb Wire (1996)  
1215,Kissed (1996)  
1216,Assassins (1995)  
1217,Friday (1995)  
1218,"Goofy Movie, A (1995)"  
1219,Higher Learning (1995)  
1220,When a Man Loves a Woman (1994)  
1221,Judgment Night (1993)  
1222,King of the Hill (1993)  
1223,"Scout, The (1994)"  
1224,Angus (1995)  
1225,Night Falls on Manhattan (1997)  
1226,"Awfully Big Adventure, An (1995)"  
1227,Under Siege 2: Dark Territory (1995)  
1228,Poison Ivy II (1995)  
1229,Ready to Wear (Pret-A-Porter) (1994)  
1230,Marked for Death (1990)  
1231,Madonna: Truth or Dare (1991)  
1232,N nette et Boni (1996)  
1233,Chairman of the Board (1998)  
1234,"Big Bang Theory, The (1994)"  
1235,"Other Voices, Other Rooms (1997)"  
1236,Twisted (1996)  
1237,Full Speed (1996)

1238,Cutthroat Island (1995)  
1239,Ghost in the Shell (Kokaku kidotai) (1995)  
1240,"Van, The (1996)"  
1241,"Old Lady Who Walked in the Sea, The (Vieille qui marchait dans la mer, La) (1991)"  
1242,Night Flier (1997)  
1243,Metro (1997)  
1244,Gridlock'd (1997)  
1245,Bushwhacked (1995)  
1246,Bad Girls (1994)  
1247,Blink (1994)  
1248,For Love or Money (1993)  
1249,Best of the Best 3: No Turning Back (1995)  
1250,A Chef in Love (1996)  
1251,"Contempt (Mépris, Le) (1963)"  
1252,"Tie That Binds, The (1995)"  
1253,Gone Fishin' (1997)  
1254,Broken English (1996)  
1255,"Designated Mourner, The (1997)"  
1256,"Designated Mourner, The (1997)"  
1257,Trial and Error (1997)  
1258,Pie in the Sky (1995)  
1259>Total Eclipse (1995)  
1260,"Run of the Country, The (1995)"  
1261,Walking and Talking (1996)  
1262,Foxfire (1996)  
1263,Nothing to Lose (1994)  
1264,Star Maps (1997)  
1265,Bread and Chocolate (Pane e cioccolata) (1973)  
1266,Clockers (1995)  
1267,Bitter Moon (1992)  
1268,Love in the Afternoon (1957)  
1269,Life with Mikey (1993)  
1270,North (1994)  
1271,Talking About Sex (1994)  
1272,Color of Night (1994)  
1273,Robocop 3 (1993)  
1274,Killer (Bulletproof Heart) (1994)  
1275,Sunset Park (1996)  
1276,Set It Off (1996)  
1277,Selena (1997)  
1278,Wild America (1997)  
1279,Gang Related (1997)  
1280,Manny & Lo (1996)  
1281,"Grass Harp, The (1995)"  
1282,Out to Sea (1997)  
1283,Before and After (1996)  
1284,Princess Caraboo (1994)  
1285,Shall We Dance? (1937)  
1286,Ed (1996)  
1287,Denise Calls Up (1995)  
1288,Jack and Sarah (1995)  
1289,Country Life (1994)  
1290,Celtic Pride (1996)  
1291,"Simple Wish, A (1997)"  
1292,Star Kid (1997)  
1293,Ayn Rand: A Sense of Life (1997)  
1294,Kicked in the Head (1997)  
1295,Indian Summer (1996)  
1296,Love Affair (1994)  
1297,"Band Wagon, The (1953)"  
1298,Penny Serenade (1941)  
1299,'Til There Was You (1997)  
1300,Stripes (1981)  
1301,Late Bloomers (1996)  
1302,"Getaway, The (1994)"  
1303,New York Cop (1996)  
1304,National Lampoon's Senior Trip (1995)  
1305,Delta of Venus (1994)  
1306,Carmen Miranda: Bananas Is My Business (1994)  
1307,Babyfever (1994)  
1308,"Very Natural Thing, A (1974)"  
1309,"Walk in the Sun, A (1945)"  
1310,Waiting to Exhale (1995)  
1311,"Pompatus of Love, The (1996)"



1312, Palmetto (1998)  
1313, Surviving the Game (1994)  
1314, Inventing the Abbotts (1997)  
1315, "Horse Whisperer, The (1998)"  
1316, "Journey of August King, The (1995)"  
1317, Catwalk (1995)  
1318, "Neon Bible, The (1995)"  
1319, Homage (1995)  
1320, Open Season (1996)  
1321, Metisse (Café au Lait) (1993)  
1322, "Wooden Man's Bride, The (Wu Kui) (1994)"  
1323, Loaded (1994)  
1324, August (1996)  
1325, Boys (1996)  
1326, Captives (1994)  
1327, Of Love and Shadows (1994)  
1328, "Low Life, The (1994)"  
1329, An Unforgettable Summer (1994)  
1330, "Last Klezmer: Leopold Kozlowski, His Life and Music, The (1995)"  
1331, My Life and Times With Antonin Artaud (En compagnie d'Antonin Artaud) (1993)  
1332, Midnight Dancers (Sibak) (1994)  
1333, Somebody to Love (1994)  
1334, American Buffalo (1996)  
1335, Kazaam (1996)  
1336, Larger Than Life (1996)  
1337, Two Deaths (1995)  
1338, Stefano Quantestorie (1993)  
1339, "Crude Oasis, The (1995)"  
1340, Hedd Wyn (1992)  
1341, "Convent, The (Convento, O) (1995)"  
1342, Lotto Land (1995)  
1343, "Story of Xinghua, The (1993)"  
1344, "Day the Sun Turned Cold, The (Tianguo niezi) (1994)"  
1345, Dingo (1992)  
1346, "Ballad of Narayama, The (Narayama Bushiko) (1958)"  
1347, Every Other Weekend (1990)  
1348, Mille bolle blu (1993)  
1349, Crows and Sparrows (1949)  
1350, Lover's Knot (1996)  
1351, Shadow of Angels (Schatten der Engel) (1976)  
1352, 1-900 (1994)  
1353, Venice/Venice (1992)  
1354, Infinity (1996)  
1355, Ed's Next Move (1996)  
1356, For the Moment (1994)  
1357, The Deadly Cure (1996)  
1358, Boys in Venice (1996)  
1359, "Sexual Life of the Belgians, The (1994)"  
1360, "Search for One-eye Jimmy, The (1996)"  
1361, American Strays (1996)  
1362, "Leopard Son, The (1996)"  
1363, Bird of Prey (1996)  
1364, Johnny 100 Pesos (1993)  
1365, JLG/JLG - autoportrait de décembre (1994)  
1366, Faust (1994)  
1367, Mina Tannenbaum (1994)  
1368, "Forbidden Christ, The (Cristo proibito, Il) (1950)"  
1369, I Can't Sleep (J'ai pas sommeil) (1994)  
1370, "Machine, The (1994)"  
1371, "Stranger, The (1994)"  
1372, Good Morning (1971)  
1373, Falling in Love Again (1980)  
1374, "Cement Garden, The (1993)"  
1375, Meet Wally Sparks (1997)  
1376, Hotel de Love (1996)  
1377, Rhyme & Reason (1997)  
1378, Love and Other Catastrophes (1996)  
1379, Hollow Reed (1996)  
1380, Losing Chase (1996)  
1381, "Bonheur, Le (1965)"  
1382, "Second Jungle Book: Mowgli & Baloo, The (1997)"  
1383, Squeeze (1996)  
1384, Roseanna's Grave (For Roseanna) (1997)  
1385, Tetsuo II: Body Hammer (1992)

1386,Fall (1997)  
1387,Gabbeh (1996)  
1388,Mondo (1996)  
1389,"Innocent Sleep, The (1995)"  
1390,For Ever Mozart (1996)  
1391,"Locusts, The (1997)"  
1392,Stag (1997)  
1393,Swept from the Sea (1997)  
1394,Hurricane Streets (1998)  
1395,Stonewall (1995)  
1396,Of Human Bondage (1934)  
1397,Anna (1996)  
1398,Stranger in the House (1997)  
1399,Picture Bride (1995)  
1400,M. Butterfly (1993)  
1401,"Ciao, Professore! (1993)"  
1402,Caro Diario (Dear Diary) (1994)  
1403,Withnail and I (1987)  
1404,Boy's Life 2 (1997)  
1405,When Night Is Falling (1995)  
1406,"Specialist, The (1994)"  
1407,Gordy (1995)  
1408,"Swan Princess, The (1994)"  
1409,Harlem (1993)  
1410,Barbarella (1968)  
1411,Land Before Time III: The Time of the Great Giving (1995) (V)  
1412,Street Fighter (1994)  
1413,Coldblooded (1995)  
1414,"Next Karate Kid, The (1994)"  
1415,No Escape (1994)  
1416,"Turning, The (1992)"  
1417,"Joy Luck Club, The (1993)"  
1418,Highlander III: The Sorcerer (1994)  
1419,Gilligan's Island: The Movie (1998)  
1420,My Crazy Life (Mi vida loca) (1993)  
1421,Suture (1993)  
1422,"Walking Dead, The (1995)"  
1423,I Like It Like That (1994)  
1424,I'll Do Anything (1994)  
1425,Grace of My Heart (1996)  
1426,Drunks (1995)  
1427,SubUrbia (1997)  
1428,Sliding Doors (1998)  
1429,Ill Gotten Gains (1997)  
1430,Legal Deceit (1997)  
1431,"Mighty, The (1998)"  
1432,Men of Means (1998)  
1433,Shooting Fish (1997)  
1434,"Steal Big, Steal Little (1995)"  
1435,Mr. Jones (1993)  
1436,House Party 3 (1994)  
1437,Panther (1995)  
1438,Jason's Lyric (1994)  
1439,Above the Rim (1994)  
1440,Moonlight and Valentino (1995)  
1441,"Scarlet Letter, The (1995)"  
1442,8 Seconds (1994)  
1443,That Darn Cat! (1965)  
1444,Ladybird Ladybird (1994)  
1445,"Bye Bye, Love (1995)"  
1446,Century (1993)  
1447,My Favorite Season (1993)  
1448,Pather Panchali (1955)  
1449,Golden Earrings (1947)  
1450,Foreign Correspondent (1940)  
1451,Lady of Burlesque (1943)  
1452,Angel on My Shoulder (1946)  
1453,Angel and the Badman (1947)  
1454,"Outlaw, The (1943)"  
1455,Beat the Devil (1954)  
1456,Love Is All There Is (1996)  
1457,"Damsel in Distress, A (1937)"  
1458,Madame Butterfly (1995)  
1459,Sleepover (1995)

1460,Here Comes Cookie (1935)  
1461,"Thieves (Voleurs, Les) (1996)"  
1462,"Boys, Les (1997)"  
1463,"Stars Fell on Henrietta, The (1995)"  
1464,Last Summer in the Hamptons (1995)  
1465,Margaret's Museum (1995)  
1466,"Saint of Fort Washington, The (1993)"  
1467,"Cure, The (1995)"  
1468,Tom and Huck (1995)  
1469,Gumby: The Movie (1995)  
1470,Hideaway (1995)  
1471,"Visitors, The (Visiteurs, Les) (1993)"  
1472,"Little Princess, The (1939)"  
1473,Nina Takes a Lover (1994)  
1474,Bhaji on the Beach (1993)  
1475,Raw Deal (1948)  
1476,Nightwatch (1997)  
1477,Dead Presidents (1995)  
1478,Reckless (1995)  
1479,Herbie Rides Again (1974)  
1480,S.F.W. (1994)  
1481,"Gate of Heavenly Peace, The (1995)"  
1482,"Man in the Iron Mask, The (1998)"  
1483,"Jerky Boys, The (1994)"  
1484,"Colonel Chabert, Le (1994)"  
1485,Girl in the Cadillac (1995)  
1486,Even Cowgirls Get the Blues (1993)  
1487,Germinal (1993)  
1488,Chasers (1994)  
1489,Fausto (1993)  
1490,Tough and Deadly (1995)  
1491,Window to Paris (1994)  
1492,"Modern Affair, A (1995)"  
1493,"Mostro, Il (1994)"  
1494,Flirt (1995)  
1495,Carpool (1996)  
1496,"Line King: Al Hirschfeld, The (1996)"  
1497,Farmer & Chase (1995)  
1498,Grosse Fatigue (1994)  
1499,Santa with Muscles (1996)  
1500,Prisoner of the Mountains (Kavkazsky Plennik) (1996)  
1501,Naked in New York (1994)  
1502,Gold Diggers: The Secret of Bear Mountain (1995)  
1503,"Bewegte Mann, Der (1994)"  
1504,Killer: A Journal of Murder (1995)  
1505,Nelly & Monsieur Arnaud (1995)  
1506,Three Lives and Only One Death (1996)  
1507,"Babysitter, The (1995)"  
1508,Getting Even with Dad (1994)  
1509,Mad Dog Time (1996)  
1510,Children of the Revolution (1996)  
1511,"World of Apu, The (Apu Sansar) (1959)"  
1512,Sprung (1997)  
1513,Dream With the Fishes (1997)  
1514,Wings of Courage (1995)  
1515,"Wedding Gift, The (1994)"  
1516,Race the Sun (1996)  
1517,Losing Isaiah (1995)  
1518,New Jersey Drive (1995)  
1519,"Fear, The (1995)"  
1520,Mr. Wonderful (1993)  
1521,Trial by Jury (1994)  
1522,"Good Man in Africa, A (1994)"  
1523,Kaspar Hauser (1993)  
1524,"Object of My Affection, The (1998)"  
1525,Witness (1985)  
1526,Senseless (1998)  
1527,Nowhere (1997)  
1528,Underground (1995)  
1529,Jefferson in Paris (1995)  
1530,Far From Home: The Adventures of Yellow Dog (1995)  
1531,Foreign Student (1994)  
1532,I Don't Want to Talk About It (De eso no se habla) (1993)  
1533,Twin Town (1997)

1534,"Enfer, L' (1994)"  
1535,Aiqing wansui (1994)  
1536,Cosi (1996)  
1537,All Over Me (1997)  
1538,Being Human (1993)  
1539,"Amazing Panda Adventure, The (1995)"  
1540,"Beans of Egypt, Maine, The (1994)"  
1541,"Scarlet Letter, The (1926)"  
1542,Johns (1996)  
1543,It Takes Two (1995)  
1544,Frankie Starlight (1995)  
1545,Shadows (Cienie) (1988)  
1546,"Show, The (1995)"  
1547,The Courtyard (1995)  
1548,Dream Man (1995)  
1549,Destiny Turns on the Radio (1995)  
1550,"Glass Shield, The (1994)"  
1551,"Hunted, The (1995)"  
1552,"Underneath, The (1995)"  
1553,Safe Passage (1994)  
1554,"Secret Adventures of Tom Thumb, The (1993)"  
1555,Condition Red (1995)  
1556,Yankee Zulu (1994)  
1557,Aparajito (1956)  
1558,Hostile Intentions (1994)  
1559,Clean Slate (Coup de Torchon) (1981)  
1560,Tigrero: A Film That Was Never Made (1994)  
1561,"Eye of Vichy, The (Oeil de Vichy, L') (1993)"  
1562,"Promise, The (Versprechen, Das) (1994)"  
1563,To Cross the Rubicon (1991)  
1564,Daens (1992)  
1565,"Man from Down Under, The (1943)"  
1566,Careful (1992)  
1567,Vermont Is For Lovers (1992)  
1568,"Vie est belle, La (Life is Rosey) (1987)"  
1569,Quartier Mozart (1992)  
1570,Touki Bouki (Journey of the Hyena) (1973)  
1571,Wend Kuuni (God's Gift) (1982)  
1572,Spirits of the Dead (Tre passi nel delirio) (1968)  
1573,Pharaoh's Army (1995)  
1574,"I, Worst of All (Yo, la peor de todas) (1990)"  
1575,"Hungarian Fairy Tale, A (1987)"  
1576,"Death in the Garden (Mort en ce jardin, La) (1956)"  
1577,"Collectionneuse, La (1967)"  
1578,Baton Rouge (1988)  
1579,Liebelei (1933)  
1580,"Woman in Question, The (1950)"  
1581,T-Men (1947)  
1582,"Invitation, The (Zaproszenie) (1986)"  
1583,"Symphonie pastorale, La (1946)"  
1584,American Dream (1990)  
1585,Lashou shentan (1992)  
1586,Terror in a Texas Town (1958)  
1587,Salut cousin! (1996)  
1588,Schizopolis (1996)  
1589,"To Have, or Not (1995)"  
1590,Duoluo tianshi (1995)  
1591,"Magic Hour, The (1998)"  
1592,Death in Brunswick (1991)  
1593,Everest (1998)  
1594,Shopping (1994)  
1595,Nemesis 2: Nebula (1995)  
1596,Romper Stomper (1992)  
1597,City of Industry (1997)  
1598,Someone Else's America (1995)  
1599,Guantanamera (1994)  
1600,Office Killer (1997)  
1601,"Price Above Rubies, A (1998)"  
1602,Angela (1995)  
1603,He Walked by Night (1948)  
1604,Love Serenade (1996)  
1605,Deceiver (1997)  
1606,Hurricane Streets (1998)  
1607,Buddy (1997)

1608,B\*A\*P\*S (1997)  
1609,"Truth or Consequences, N.M. (1997)"  
1610,Intimate Relations (1996)  
1611,"Leading Man, The (1996)"  
1612,Tokyo Fist (1995)  
1613,"Reluctant Debutante, The (1958)"  
1614,Warriors of Virtue (1997)  
1615,Desert Winds (1995)  
1616,Hugo Pool (1997)  
1617,King of New York (1990)  
1618,All Things Fair (1996)  
1619,"Sixth Man, The (1997)"  
1620,Butterfly Kiss (1995)  
1621,"Paris, France (1993)"  
1622,"C  r  monie, La (1995)"  
1623,Hush (1998)  
1624,Nightwatch (1997)  
1625,Nobody Loves Me (Keiner liebt mich) (1994)  
1626,"Wife, The (1995)"  
1627,Lamerica (1994)  
1628,Nico Icon (1995)  
1629,"Silence of the Palace, The (Saint el Qusur) (1994)"  
1630,"Slingshot, The (1993)"  
1631,Land and Freedom (Tierra y libertad) (1995)  
1632,   k  ldum klaka (Cold Fever) (1994)  
1633,Etz Hadomim Tafus (Under the Domin Tree) (1994)  
1634,Two Friends (1986)  
1635,Brothers in Trouble (1995)  
1636,Girls Town (1996)  
1637,Normal Life (1996)  
1638,Bitter Sugar (Azucar Amargo) (1996)  
1639,"Eighth Day, The (1996)"  
1640,Dadetown (1995)  
1641,Some Mother's Son (1996)  
1642,Angel Baby (1995)  
1643,Sudden Manhattan (1996)  
1644,"Butcher Boy, The (1998)"  
1645,Men With Guns (1997)  
1646,Hana-bi (1997)  
1647,"Niagara, Niagara (1997)"  
1648,"Big One, The (1997)"  
1649,"Butcher Boy, The (1998)"  
1650,"Spanish Prisoner, The (1997)"  
1651,Temptress Moon (Feng Yue) (1996)  
1652,Entertaining Angels: The Dorothy Day Story (1996)  
1653,Chairman of the Board (1998)  
1654,"Favor, The (1994)"  
1655,Little City (1998)  
1656,Target (1995)  
1657,"Substance of Fire, The (1996)"  
1658,Getting Away With Murder (1996)  
1659,Small Faces (1995)  
1660,"New Age, The (1994)"  
1661,Rough Magic (1995)  
1662,Nothing Personal (1995)  
1663,8 Heads in a Duffel Bag (1997)  
1664,"Brother's Kiss, A (1997)"  
1665,Ripe (1996)  
1666,"Next Step, The (1995)"  
1667,Wedding Bell Blues (1996)  
1668,MURDER and murder (1996)  
1669,Tainted (1998)  
1670,"Further Gesture, A (1996)"  
1671,Kika (1993)  
1672,Mirage (1995)  
1673,Mamma Roma (1962)  
1674,"Sunchaser, The (1996)"  
1675,"War at Home, The (1996)"  
1676,Sweet Nothing (1995)  
1677,Mat' i syn (1997)  
1678,B. Monkey (1998)  
1679,Sliding Doors (1998)  
1680,You So Crazy (1994)  
1681,Scream of Stone (Schrei aus Stein) (1991)

**File: integration\_test/app\_screenshots\_test.dart**

```
import 'dart:io';

import 'package:flutter/material.dart';
import 'package:flutter_test/flutter_test.dart';
import 'package:integration_test/integration_test.dart';

import 'package:federated_learning_in_mobile/main.dart' as app;

const String _liveServerUrl = String.fromEnvironment(
 'FL_TEST_SERVER_URL',
 defaultValue: '',
);

void main() {
 final IntegrationTestWidgetsFlutterBinding binding =
 IntegrationTestWidgetsFlutterBinding.ensureInitialized();

 setUpAll(() async {
 if (Platform.isAndroid) {
 await binding.convertFlutterSurfaceToImage();
 }
 });

 testWidgets('launches app and captures screenshots across app situations', (
 WidgetTester tester,
) async {
 app.main();
 await tester.pumpAndSettle();

 expect(find.text('Federated Client'), findsOneWidget);
 expect(find.text('SERVER CONFIGURATION'), findsOneWidget);

 expect(
 _liveServerUrl.isNotEmpty,
 isTrue,
 reason:
 'Provide the server URL with --dart-define=FL_TEST_SERVER_URL=http://<HOST_IP>:8080',
);

 await _enterServerUrl(tester, _liveServerUrl);
 await _enterClientId(tester, 'integration_screenshots_client');
 await tester.tap(find.text('CONNECT TO SERVER'));

 final connected = await _pumpUntilFound(
 tester,
 find.textContaining('Status: Connected'),
 timeout: const Duration(seconds: 30),
 throwOnTimeout: false,
);
 expect(connected, isTrue, reason: 'App did not reach Connected state');

 await _assertNoVisibleErrorSnackBar(tester);
 await _takeScreenshot(binding, tester, '01_connected_dashboard');

 await tester.tap(find.byIcon(Icons.movie));
 await tester.pumpAndSettle();
 expect(find.text('Movie Recommendations'), findsOneWidget);
 await _assertNoVisibleErrorSnackBar(tester);
 await _takeScreenshot(binding, tester, '02_recommendations_live');

 await tester.pageBack();
 await tester.pumpAndSettle();
 expect(find.text('Federated Client'), findsOneWidget);

 await tester.scrollUntilVisible(
 find.text('LIVE LOGS'),
 350,
 scrollable: find.byType(Scrollable).first,
);
 await tester.pumpAndSettle();
 await _assertNoVisibleErrorSnackBar(tester);
 await _takeScreenshot(binding, tester, '03_dashboard_logs_connected');
```

```

 });
}

Future<void> _enterServerUrl(WidgetTester tester, String url) async {
 final serverUrlField = find.byType(TextField).first;
 await tester.tap(serverUrlField);
 await tester.pumpAndSettle();
 await tester.enterText(serverUrlField, url);
 await tester.pumpAndSettle();
}

Future<void> _enterClientId(WidgetTester tester, String clientId) async {
 final clientIdField = find.byType(TextField).at(1);
 await tester.tap(clientIdField);
 await tester.pumpAndSettle();
 await tester.enterText(clientIdField, clientId);
 await tester.pumpAndSettle();
}

Future<bool> _pumpUntilFound(
 WidgetTester tester,
 Finder finder, {
 required Duration timeout,
 bool throwOnTimeout = true,
}) async {
 final maxIterations = timeout.inMilliseconds ~/ 250;
 for (int i = 0; i < maxIterations; i++) {
 await tester.pump(const Duration(milliseconds: 250));
 if (finder.evaluate().isNotEmpty) {
 return true;
 }
 }
 if (throwOnTimeout) {
 throw TestFailure('Timed out waiting for $finder');
 }
 return false;
}

Future<void> _assertNoVisibleErrorSnackBar(WidgetTester tester) async {
 await tester.pump();
 expect(find.byType(SnackBar), findsNothing);
}

Future<void> _takeScreenshot(
 IntegrationTestWidgetsFlutterBinding binding,
 WidgetTester tester,
 String name,
) async {
 await tester.pump(const Duration(milliseconds: 250));
 await tester.pumpAndSettle();
 await binding.takeScreenshot(name);
}

```

### File: ios/Flutter/flutter\_export\_environment.sh

```

#!/bin/sh
This is a generated file; do not edit or check into version control.
export "FLUTTER_ROOT=/opt/homebrew/share/flutter"
export "FLUTTER_APPLICATION_PATH=/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile"
export "COCOPODS_PARALLEL_CODE_SIGN=true"
export "FLUTTER_TARGET=integration_test/app_screenshots_test.dart"
export "FLUTTER_BUILD_DIR=build"
export "FLUTTER_BUILD_NAME=1.0.0"
export "FLUTTER_BUILD_NUMBER=1"
export "DART_DEFINES=RkxfVEVTVF9TRVJWRVJfVJVJMPWh0dHA6Ly8xOTIuMTY4LjI5LjY3OjgwODA=,RkxVVFRFul9WRVJTSU90PTMuNDEuNQ==,RkxVVFRFul9DSEFOTkVMPXN0YWJsZQ==,RkxVVFRFul9HSVRfVJVJMPWh0dHBz0i8vZ2l0aHVhLmNvbS9mbHV0dGVyL2ZsdXR0ZXIuZ2l0,RkxVVFRFul9GUkFN RVdPUktfUkVWSVNJT049MmM5ZWlyMdczOQ==,RkxVVFRFul9FTkdJTkVfUkVWSVNJT049MDUyZjMxZDExNQ==,RkxVVFRFul9EQVJUX1ZFUlNJT049My4xMS4z"
export "DART_OBFUSCATION=false"
export "TRACK_WIDGET_CREATION=true"
export "TREE_SHAKE_ICONS=false"
export "PACKAGE_CONFIG=/Users/jonat/Documents/GitHub/master_thesis/federated_learning_in_mobile/.dart_tool/

```

package\_config.json"

#### File: ios/Flutter/ephemeral/flutter\_lldb\_helper.py

```
#
Generated file, do not edit.
#

import lldb

def handle_new_rx_page(frame: lldb.SBFrame, bp_loc, extra_args, intern_dict):
 """Intercept NOTIFY_DEBUGGER_ABOUT_RX_PAGES and touch the pages."""
 base = frame.register["x0"].GetValueAsAddress()
 page_len = frame.register["x1"].GetValueAsUnsigned()

 # Note: NOTIFY_DEBUGGER_ABOUT_RX_PAGES will check contents of the
 # first page to see if handled it correctly. This makes diagnosing
 # misconfiguration (e.g. missing breakpoint) easier.
 data = bytearray(page_len)
 data[0:8] = b'IHELPED!'

 error = lldb.SBError()
 frame.GetThread().GetProcess().WriteMemory(base, data, error)
 if not error.Success():
 print(f'Failed to write into {base}[+{page_len}]', error)
 return

def __lldb_init_module(debugger: lldb.SBDebugger, _):
 target = debugger.GetDummyTarget()
 # Caveat: must use BreakpointCreateByRegEx here and not
 # BreakpointCreateByName. For some reasons callback function does not
 # get carried over from dummy target for the later.
 bp = target.BreakpointCreateByRegEx("^NOTIFY_DEBUGGER_ABOUT_RX_PAGES$")
 bp.SetScriptCallbackFunction('{}handle_new_rx_page'.format(__name__))
 bp.SetAutoContinue(True)
 print("-- LLDB integration loaded --")
```

#### File: ios/Pods/Local Podspecs/Flutter.podspec.json

```
{
 "name": "Flutter",
 "version": "1.0.0",
 "summary": "A UI toolkit for beautiful and fast apps.",
 "homepage": "https://flutter.dev",
 "license": {
 "type": "BSD"
 },
 "authors": {
 "Flutter Dev Team": "flutter-dev@googlegroups.com"
 },
 "source": {
 "git": "https://github.com/flutter/engine",
 "tag": "1.0.0"
 },
 "platforms": {
 "ios": "13.0"
 },
 "vendored_frameworks": "path/to/nothing"
}
```

#### File: ios/Pods/Local Podspecs/battery\_plus.podspec.json

```
{
 "name": "battery_plus",
 "version": "1.0.0",
 "summary": "Flutter plugin for accessing information about the battery_plus.",
 "description": "A Flutter plugin to access various information about the battery of the device the app is running on.\nDownloaded by pub (not CocoaPods).",
 "homepage": "https://plus.fluttercommunity.dev/",
 "license": {
 "type": "BSD",
 "file": "../LICENSE"
 },
 "authors": {
```



```

 "Flutter Community Team": "authors@fluttercommunity.dev"
 },
 "source": {
 "http": "https://github.com/fluttercommunity/plus_plugins/tree/main/packages/battery_plus"
 },
 "documentation_url": "https://pub.dev/packages/battery_plus",
 "source_files": "battery_plus/Sources/battery_plus/**/*.{h,m}",
 "public_header_files": "battery_plus/Sources/battery_plus/include/**/*.h",
 "dependencies": {
 "Flutter": []
 },
 "platforms": {
 "ios": "12.0"
 },
 "pod_target_xcconfig": {
 "DEFINES_MODULE": "YES"
 },
 "resource_bundles": {
 "battery_plus_privacy": [
 "battery_plus/Sources/battery_plus/PrivacyInfo.xcprivacy"
]
 }
}

```

#### File: ios/Pods/Local Podspecs/device\_info\_plus.podspec.json

```

{
 "name": "device_info_plus",
 "version": "0.0.1",
 "summary": "Flutter Device Info Plus",
 "description": "Get current device information from within the Flutter application.\nDownloaded by pub (not CocoaPods).",
 "homepage": "https://github.com/fluttercommunity/plus_plugins",
 "license": {
 "type": "BSD",
 "file": "../LICENSE"
 },
 "authors": {
 "Flutter Community Team": "authors@fluttercommunity.dev"
 },
 "source": {
 "http": "https://github.com/fluttercommunity/plus_plugins/tree/main/packages/device_info_plus"
 },
 "documentation_url": "https://pub.dev/packages/device_info_plus",
 "source_files": "Classes/**/*",
 "public_header_files": "Classes/**/*.h",
 "dependencies": {
 "Flutter": []
 },
 "platforms": {
 "ios": "12.0"
 },
 "pod_target_xcconfig": {
 "DEFINES_MODULE": "YES"
 },
 "resource_bundles": {
 "device_info_plus_privacy": [
 "PrivacyInfo.xcprivacy"
]
 }
}

```

#### File: ios/Pods/Local Podspecs/integration\_test.podspec.json

```

{
 "name": "integration_test",
 "version": "0.0.1",
 "summary": "Adapter for integration tests.",
 "description": "Runs tests that use the flutter_test API as integration tests.",
 "homepage": "https://github.com/flutter/flutter/tree/main/packages/integration_test",
 "license": {
 "type": "BSD",
 "text": "Use of this source code is governed by a BSD-style license that can be found in the LICENSE file.\n"
 }
}

```

```

},
"authors": {
 "Flutter Team": "flutter-dev@googlegroups.com"
},
"source": {
 "http": "https://github.com/flutter/flutter/tree/main/packages/integration_test"
},
"source_files": "integration_test/Sources/integration_test/**/*.{h,m}",
"public_header_files": "integration_test/Sources/integration_test/**/*.h",
"dependencies": {
 "Flutter": []
},
"ios": {
 "frameworks": "UIKit"
},
"platforms": {
 "ios": "12.0"
},
"pod_target_xcconfig": {
 "DEFINES_MODULE": "YES",
 "EXCLUDED_ARCHS[sdk=iphonesimulator*]": "i386"
}
}

```

#### File: ios/Pods/Local Podspecs/open\_filex.podspec.json

```

{
 "name": "open_filex",
 "version": "0.0.2",
 "summary": "A new Flutter project.",
 "description": "A new Flutter project.",
 "homepage": "http://example.com",
 "license": {
 "file": "../LICENSE"
 },
 "authors": {
 "Your Company": "email@example.com"
 },
 "source": {
 "path": "."
 },
 "source_files": "Classes/**/*",
 "public_header_files": "Classes/**/*.h",
 "dependencies": {
 "Flutter": []
 },
 "platforms": {
 "ios": "8.0"
 }
}

```

#### File: ios/Pods/Local Podspecs/path\_provider\_foundation.podspec.json

```

{
 "name": "path_provider_foundation",
 "version": "0.0.1",
 "summary": "An iOS and macOS implementation of the path_provider plugin.",
 "description": "An iOS and macOS implementation of the Flutter plugin for getting commonly used locations on the filesystem.",
 "homepage": "https://github.com/flutter/packages/tree/main/packages/path_provider/path_provider_foundation",
 "license": {
 "type": "BSD",
 "file": "../LICENSE"
 },
 "authors": {
 "Flutter Dev Team": "flutter-dev@googlegroups.com"
 },
 "source": {
 "http": "https://github.com/flutter/packages/tree/main/packages/path_provider/path_provider_foundation"
 },
 "source_files": "path_provider_foundation/Sources/path_provider_foundation/**/*.swift",
 "ios": {
 "dependencies": {

```

```

 "Flutter": []
 },
 "xcconfig": {
 "LIBRARY_SEARCH_PATHS": "$(TOOLCHAIN_DIR)/usr/lib/swift/${PLATFORM_NAME}/ $(SDKROOT)/usr/lib/swift",
 "LD_RUNPATH_SEARCH_PATHS": "/usr/lib/swift"
 }
},
"osx": {
 "dependencies": {
 "FlutterMacOS": []
 }
},
"platforms": {
 "ios": "13.0",
 "osx": "10.15"
},
"swift_versions": "5.0",
"resource_bundles": {
 "path_provider_foundation_privacy": [
 "path_provider_foundation/Sources/path_provider_foundation/Resources/PrivacyInfo.xcprivacy"
]
},
"swift_version": "5.0"
}

```

#### File: ios/Pods/Local Podspecs/share\_plus.podspec.json

```

{
 "name": "share_plus",
 "version": "0.0.1",
 "summary": "Flutter Share",
 "description": "A Flutter plugin to share content from your Flutter app via the platform's share dialog.\nDownloaded by pub (not CocoaPods).",
 "homepage": "https://github.com/fluttercommunity/plus_plugins",
 "license": {
 "type": "BSD",
 "file": "../LICENSE"
 },
 "authors": {
 "Flutter Community Team": "authors@fluttercommunity.dev"
 },
 "source": {
 "http": "https://github.com/fluttercommunity/plus_plugins/tree/main/packages/share_plus/share_plus"
 },
 "documentation_url": "https://pub.dev/packages/share_plus",
 "source_files": "Classes/**/*",
 "public_header_files": "Classes/**/*.h",
 "dependencies": {
 "Flutter": []
 },
 "ios": {
 "weak_frameworks": "LinkPresentation"
 },
 "platforms": {
 "ios": "11.0"
 },
 "pod_target_xcconfig": {
 "DEFINES_MODULE": "YES"
 }
}

```

#### File: ios/Pods/Target Support Files/Pods-Runner/Pods-Runner-frameworks.sh

```

#!/bin/sh
set -e
set -u
set -o pipefail

function on_error {
 echo "$(realpath -mq "${0}"):$1: error: Unexpected failure"
}
trap 'on_error $LINENO' ERR

if [-z ${FRAMEWORKS_FOLDER_PATH+x}]; then

```

```

If FRAMEWORKS_FOLDER_PATH is not set, then there's nowhere for us to copy
frameworks to, so exit 0 (signalling the script phase was successful).
exit 0
fi

echo "mkdir -p ${CONFIGURATION_BUILD_DIR}/${FRAMEWORKS_FOLDER_PATH}"
mkdir -p "${CONFIGURATION_BUILD_DIR}/${FRAMEWORKS_FOLDER_PATH}"

COCOAPODS_PARALLEL_CODE_SIGN="${COCOAPODS_PARALLEL_CODE_SIGN:-false}"
SWIFT_STDLIB_PATH="${TOOLCHAIN_DIR}/usr/lib/swift/${PLATFORM_NAME}"
BCSYMBOLMAP_DIR="BCSymbolMaps"

This protects against multiple targets copying the same framework dependency at the same time. The solution
was originally proposed here: https://lists.samba.org/archive/rsync/2008-February/020158.html
RSYNC_PROTECT_TMP_FILES=(--filter "P *.??????")

Copies and strips a vendored framework
install_framework()
{
 if [-r "${BUILT_PRODUCTS_DIR}/$1"]; then
 local source="${BUILT_PRODUCTS_DIR}/$1"
 elif [-r "${BUILT_PRODUCTS_DIR}/$(basename "$1")"]; then
 local source="${BUILT_PRODUCTS_DIR}/$(basename "$1")"
 elif [-r "$1"]; then
 local source="$1"
 fi

 local destination="${TARGET_BUILD_DIR}/${FRAMEWORKS_FOLDER_PATH}"

 if [-L "${source}"]; then
 echo "Symlinked..."
 source="$(readlink -f "${source}")"
 fi

 if [-d "${source}/${BCSYMBOLMAP_DIR}"]; then
 # Locate and install any .bcsymbolmaps if present, and remove them from the .framework before the frame
 work is copied
 find "${source}/${BCSYMBOLMAP_DIR}" -name "*.bcsymbolmap"|while read f; do
 echo "Installing $f"
 install_bcsymbolmap "$f" "$destination"
 rm "$f"
 done
 rmdir "${source}/${BCSYMBOLMAP_DIR}"
 fi

 # Use filter instead of exclude so missing patterns don't throw errors.
 echo "rsync --delete -av "${RSYNC_PROTECT_TMP_FILES[@]}" --links --filter \"- CVS/\" --filter \"- .svn/\"
 --filter \"- .git/\" --filter \"- .hg/\" --filter \"- Headers/\" --filter \"- PrivateHeaders/\" --filter \"-
 Modules/\" \"${source}\" \"${destination}\""
 rsync --delete -av "${RSYNC_PROTECT_TMP_FILES[@]}" --links --filter "- CVS/" --filter "- .svn/" --filter
 "- .git/" --filter "- .hg/" --filter "- Headers" --filter "- PrivateHeaders" --filter "- Modules" "${source}
)" "${destination}"

 local basename
 basename="$(basename -s .framework "$1")"
 binary="${destination}/${basename}.framework/${basename}"

 if ! [-r "$binary"]; then
 binary="${destination}/${basename}"
 elif [-L "${binary}"]; then
 echo "Destination binary is symlinked..."
 dirname="$(dirname "${binary}")"
 binary="${dirname}/${readlink "${binary}"}"
 fi

 # Strip invalid architectures so "fat" simulator / device frameworks work on device
 if [["$(file "$binary")" == *"dynamically linked shared library"*]]; then
 strip_invalid_archs "$binary"
 fi

 # Resign the code if required by the build settings to avoid unstable apps
 code_sign_if_enabled "${destination}/${basename}"

```

```

Embed linked Swift runtime libraries. No longer necessary as of Xcode 7.
if ["${XCODE_VERSION_MAJOR}" -lt 7]; then
 local swift_runtime_libs
 swift_runtime_libs=$(xcrun otool -LX "$binary" | grep --color=never @rpath/libswift | sed -E s/@rpath\\
/(\.dylib\).*\/\1/g | uniq -u)
 for lib in $swift_runtime_libs; do
 echo "rsync -auv \"${SWIFT_STDLIB_PATH}/${lib}\" \"${destination}\""
 rsync -auv "${SWIFT_STDLIB_PATH}/${lib}" "${destination}"
 code_sign_if_enabled "${destination}/${lib}"
 done
fi
}
Copies and strips a vendored dSYM
install_dsym() {
 local source="$1"
 warn_missing_arch=${2:-true}
 if [-r "$source"]; then
 # Copy the dSYM into the targets temp dir.
 echo "rsync --delete -av "${RSYNC_PROTECT_TMP_FILES[@]}" --filter \"- CVS/\" --filter \"- .svn/\" --filter \"- .git/\" --filter \"- .hg/\" --filter \"- Headers\" --filter \"- PrivateHeaders\" --filter \"- Modules\" \"${source}\" \"${DERIVED_FILES_DIR}\""
 rsync --delete -av "${RSYNC_PROTECT_TMP_FILES[@]}" --filter "- CVS/" --filter "- .svn/" --filter "- .git/" --filter "- .hg/" --filter "- Headers" --filter "- PrivateHeaders" --filter "- Modules" "${source}" "${DERIVED_FILES_DIR}"

 local basename
 basename=$(basename -s .dSYM "$source")
 binary_name=$(ls "$source/Contents/Resources/DWARF")
 binary="${DERIVED_FILES_DIR}/${basename}.dSYM/Contents/Resources/DWARF/${binary_name}"

 # Strip invalid architectures from the dSYM.
 if [["$(file "$binary")" == *"Mach-O *"dSYM companion"*]]; then
 strip_invalid_archs "$binary" "$warn_missing_arch"
 fi
 if [[$STRIP_BINARY_RETVAL == 0]]; then
 # Move the stripped file into its final destination.
 echo "rsync --delete -av "${RSYNC_PROTECT_TMP_FILES[@]}" --links --filter \"- CVS/\" --filter \"- .svn/\" --filter \"- .git/\" --filter \"- .hg/\" --filter \"- Headers\" --filter \"- PrivateHeaders\" --filter \"- Modules\" \"${DERIVED_FILES_DIR}/${basename}.framework.dSYM\" \"${DWARF_DSYM_FOLDER_PATH}\""
 rsync --delete -av "${RSYNC_PROTECT_TMP_FILES[@]}" --links --filter "- CVS/" --filter "- .svn/" --filter "- .git/" --filter "- .hg/" --filter "- Headers" --filter "- PrivateHeaders" --filter "- Modules" "${DERIVED_FILES_DIR}/${basename}.dSYM" "${DWARF_DSYM_FOLDER_PATH}"
 else
 # The dSYM was not stripped at all, in this case touch a fake folder so the input/output paths from Xcode do not reexecute this script because the file is missing.
 mkdir -p "${DWARF_DSYM_FOLDER_PATH}"
 touch "${DWARF_DSYM_FOLDER_PATH}/${basename}.dSYM"
 fi
 fi
}

Used as a return value for each invocation of `strip_invalid_archs` function.
STRIP_BINARY_RETVAL=0

Strip invalid architectures
strip_invalid_archs() {
 binary="$1"
 warn_missing_arch=${2:-true}
 # Get architectures for current target binary
 binary_archs="$(lipo -info "$binary" | rev | cut -d ':' -f1 | awk '{ $1=$1; print }' | rev)"
 # Intersect them with the architectures we are building for
 intersected_archs="$(echo ${ARCHS[@]} $binary_archs | tr ' ' '\n' | sort | uniq -d)"
 # If there are no archs supported by this binary then warn the user
 if [[-z "$intersected_archs"]]; then
 if [["$warn_missing_arch" == "true"]]; then
 echo "warning: [CP] Vendored binary '$binary' contains architectures ($binary_archs) none of which match the current build architectures ($ARCHS)."
 fi
 STRIP_BINARY_RETVAL=1
 return
 fi
 stripped=""
 for arch in $binary_archs; do

```

```

 if ! [["${ARCHS}" == *"$sarch"*]]; then
 # Strip non-valid architectures in-place
 lipo -remove "$sarch" -output "$binary" "$binary"
 stripped="$stripped $sarch"
 fi
done
if [["$stripped"]]; then
 echo "Stripped $binary of architectures:$stripped"
fi
STRIP_BINARY_RETVAL=0
}

Copies the bcsymbolmap files of a vendored framework
install_bcsymbolmap() {
 local bcsymbolmap_path="$1"
 local destination="${BUILT_PRODUCTS_DIR}"
 echo "rsync --delete -av "${RSYNC_PROTECT_TMP_FILES[@]}" --filter "- CVS/" --filter "- .svn/" --filter "- .git/" --filter "- .hg/" --filter "- Headers" --filter "- PrivateHeaders" --filter "- Modules" "${bcsymbolmap_path}" "${destination}"
 rsync --delete -av "${RSYNC_PROTECT_TMP_FILES[@]}" --filter "- CVS/" --filter "- .svn/" --filter "- .git/" --filter "- .hg/" --filter "- Headers" --filter "- PrivateHeaders" --filter "- Modules" "${bcsymbolmap_path}" "${destination}"
}

Signs a framework with the provided identity
code_sign_if_enabled() {
 if [-n "${EXPANDED_CODE_SIGN_IDENTITY:-}" -a "${CODE_SIGNING_REQUIRED:-}" != "NO" -a "${CODE_SIGNING_ALLOWED}" != "NO"]; then
 # Use the current code_sign_identity
 echo "Code Signing $1 with Identity ${EXPANDED_CODE_SIGN_IDENTITY_NAME}"
 local code_sign_cmd="/usr/bin/codesign --force --sign ${EXPANDED_CODE_SIGN_IDENTITY} ${OTHER_CODE_SIGN_FLAGS:-} --preserve-metadata=identifier,entitlements '$1'"

 if ["${COCOPODS_PARALLEL_CODE_SIGN}" == "true"]; then
 code_sign_cmd="$code_sign_cmd &"
 fi
 echo "$code_sign_cmd"
 eval "$code_sign_cmd"
 fi
}

if [["$CONFIGURATION" == "Debug"]]; then
 install_framework "${BUILT_PRODUCTS_DIR}/battery_plus/battery_plus.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/device_info_plus/device_info_plus.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/integration_test/integration_test.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/open_filex/open_filex.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/path_provider_foundation/path_provider_foundation.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/share_plus/share_plus.framework"
fi
if [["$CONFIGURATION" == "Profile"]]; then
 install_framework "${BUILT_PRODUCTS_DIR}/battery_plus/battery_plus.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/device_info_plus/device_info_plus.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/integration_test/integration_test.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/open_filex/open_filex.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/path_provider_foundation/path_provider_foundation.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/share_plus/share_plus.framework"
fi
if [["$CONFIGURATION" == "Release"]]; then
 install_framework "${BUILT_PRODUCTS_DIR}/battery_plus/battery_plus.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/device_info_plus/device_info_plus.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/integration_test/integration_test.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/open_filex/open_filex.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/path_provider_foundation/path_provider_foundation.framework"
 install_framework "${BUILT_PRODUCTS_DIR}/share_plus/share_plus.framework"
fi
if ["${COCOPODS_PARALLEL_CODE_SIGN}" == "true"]; then
 wait
fi

```

**File: ios/Pods/Target Support Files/Pods-Runner/Pods-Runner-umbrella.h**

```

#ifdef __OBJC__
#import <UIKit/UIKit.h>
#else

```

```
#ifndef FOUNDATION_EXPORT
#if defined(__cplusplus)
#define FOUNDATION_EXPORT extern "C"
#else
#define FOUNDATION_EXPORT extern
#endif
#endif

FOUNDATION_EXPORT double Pods_RunnerVersionNumber;
FOUNDATION_EXPORT const unsigned char Pods_RunnerVersionString[];
```

**File: ios/Pods/Target Support Files/Pods-RunnerTests/Pods-RunnerTests-umbrella.h**

```

#ifdef __OBJC__
#import <UIKit/UIKit.h>
#else
#ifndef FOUNDATION_EXPORT
#if defined(__cplusplus)
#define FOUNDATION_EXPORT extern "C"
#else
#define FOUNDATION_EXPORT extern
#endif
#endif
#endif

FOUNDATION_EXPORT double Pods_RunnerTestsVersionNumber;
FOUNDATION_EXPORT const unsigned char Pods_RunnerTestsVersionString[];
```

**File: ios/Pods/Target Support Files/battery\_plus/battery\_plus-umbrella.h**

```

#ifdef __OBJC__
#import <UIKit/UIKit.h>
#else
#ifndef FOUNDATION_EXPORT
#if defined(__cplusplus)
#define FOUNDATION_EXPORT extern "C"
#else
#define FOUNDATION_EXPORT extern
#endif
#endif
#endif

#import "FPPBatteryPlusPlugin.h"

FOUNDATION_EXPORT double battery_plusVersionNumber;
FOUNDATION_EXPORT const unsigned char battery_plusVersionString[];
```

**File: ios/Pods/Target Support Files/device\_info\_plus/device\_info\_plus-umbrella.h**

```

#ifdef __OBJC__
#import <UIKit/UIKit.h>
#else
#ifndef FOUNDATION_EXPORT
#if defined(__cplusplus)
#define FOUNDATION_EXPORT extern "C"
#else
#define FOUNDATION_EXPORT extern
#endif
#endif
#endif

#import "FPPDeviceInfoPlusPlugin.h"

FOUNDATION_EXPORT double device_info_plusVersionNumber;
FOUNDATION_EXPORT const unsigned char device_info_plusVersionString[];
```

**File: ios/Pods/Target Support Files/integration\_test/integration\_test-umbrella.h**

```

#ifdef __OBJC__
#import <UIKit/UIKit.h>
#else
#ifdef FOUNDATION_EXPORT
#if defined(__cplusplus)
#define FOUNDATION_EXPORT extern "C"
#else
#define FOUNDATION_EXPORT extern
#endif
#endif
#endif

#import "FLTIntegrationTestRunner.h"
#import "IntegrationTestIosTest.h"
#import "IntegrationTestPlugin.h"

FOUNDATION_EXPORT double integration_testVersionNumber;
FOUNDATION_EXPORT const unsigned char integration_testVersionString[];
```

**File: ios/Pods/Target Support Files/open\_filex/open\_filex-umbrella.h**

```

#ifdef __OBJC__
#import <UIKit/UIKit.h>
#else
#ifdef FOUNDATION_EXPORT
#if defined(__cplusplus)
#define FOUNDATION_EXPORT extern "C"
#else
#define FOUNDATION_EXPORT extern
#endif
#endif
#endif

#import "OpenFilePlugin.h"

FOUNDATION_EXPORT double open_filexVersionNumber;
FOUNDATION_EXPORT const unsigned char open_filexVersionString[];
```

**File: ios/Pods/Target Support Files/path\_provider\_foundation/path\_provider\_foundation-umbrella.h**

```

#ifdef __OBJC__
#import <UIKit/UIKit.h>
#else
#ifdef FOUNDATION_EXPORT
#if defined(__cplusplus)
#define FOUNDATION_EXPORT extern "C"
#else
#define FOUNDATION_EXPORT extern
#endif
#endif
#endif

FOUNDATION_EXPORT double path_provider_foundationVersionNumber;
FOUNDATION_EXPORT const unsigned char path_provider_foundationVersionString[];
```

**File: ios/Pods/Target Support Files/share\_plus/share\_plus-umbrella.h**

```

#ifdef __OBJC__
#import <UIKit/UIKit.h>
#else
#ifdef FOUNDATION_EXPORT
#if defined(__cplusplus)
#define FOUNDATION_EXPORT extern "C"
#else
#define FOUNDATION_EXPORT extern
#endif
#endif
#endif
```



```
#import "FPPSharePlusPlugin.h"

FOUNDATION_EXPORT double share_plusVersionNumber;
FOUNDATION_EXPORT const unsigned char share_plusVersionString[];
```

**File: ios/Runner/GeneratedPluginRegistrant.h**

```

//
// Generated file. Do not edit.
//

// clang-format off

#ifdef GeneratedPluginRegistrant_h
#define GeneratedPluginRegistrant_h

#import <Flutter/Flutter.h>

NS_ASSUME_NONNULL_BEGIN

@interface GeneratedPluginRegistrant : NSObject
+ (void)registerWithRegistry:(NSObject<FlutterPluginRegistry>*)registry;
@end

NS_ASSUME_NONNULL_END
#endif /* GeneratedPluginRegistrant_h */
```

**File: ios/Runner/Runner-Bridging-Header.h**

```

#import "GeneratedPluginRegistrant.h"
```

**File: ios/Runner/Assets.xcassets/AppIcon.appiconset/Contents.json**

```

{
 "images" : [
 {
 "size" : "20x20",
 "idiom" : "iphone",
 "filename" : "Icon-App-20x20@2x.png",
 "scale" : "2x"
 },
 {
 "size" : "20x20",
 "idiom" : "iphone",
 "filename" : "Icon-App-20x20@3x.png",
 "scale" : "3x"
 },
 {
 "size" : "29x29",
 "idiom" : "iphone",
 "filename" : "Icon-App-29x29@1x.png",
 "scale" : "1x"
 },
 {
 "size" : "29x29",
 "idiom" : "iphone",
 "filename" : "Icon-App-29x29@2x.png",
 "scale" : "2x"
 },
 {
 "size" : "29x29",
 "idiom" : "iphone",
 "filename" : "Icon-App-29x29@3x.png",
 "scale" : "3x"
 },
 {
 "size" : "40x40",
 "idiom" : "iphone",
 "filename" : "Icon-App-40x40@2x.png",
 "scale" : "2x"
 },
 {

```

```
"size" : "40x40",
"idiom" : "iphone",
"filename" : "Icon-App-40x40@3x.png",
"scale" : "3x"
},
{
 "size" : "60x60",
 "idiom" : "iphone",
 "filename" : "Icon-App-60x60@2x.png",
 "scale" : "2x"
},
{
 "size" : "60x60",
 "idiom" : "iphone",
 "filename" : "Icon-App-60x60@3x.png",
 "scale" : "3x"
},
{
 "size" : "20x20",
 "idiom" : "ipad",
 "filename" : "Icon-App-20x20@1x.png",
 "scale" : "1x"
},
{
 "size" : "20x20",
 "idiom" : "ipad",
 "filename" : "Icon-App-20x20@2x.png",
 "scale" : "2x"
},
{
 "size" : "29x29",
 "idiom" : "ipad",
 "filename" : "Icon-App-29x29@1x.png",
 "scale" : "1x"
},
{
 "size" : "29x29",
 "idiom" : "ipad",
 "filename" : "Icon-App-29x29@2x.png",
 "scale" : "2x"
},
{
 "size" : "40x40",
 "idiom" : "ipad",
 "filename" : "Icon-App-40x40@1x.png",
 "scale" : "1x"
},
{
 "size" : "40x40",
 "idiom" : "ipad",
 "filename" : "Icon-App-40x40@2x.png",
 "scale" : "2x"
},
{
 "size" : "76x76",
 "idiom" : "ipad",
 "filename" : "Icon-App-76x76@1x.png",
 "scale" : "1x"
},
{
 "size" : "76x76",
 "idiom" : "ipad",
 "filename" : "Icon-App-76x76@2x.png",
 "scale" : "2x"
},
{
 "size" : "83.5x83.5",
 "idiom" : "ipad",
 "filename" : "Icon-App-83.5x83.5@2x.png",
 "scale" : "2x"
},
{
 "size" : "1024x1024",
 "idiom" : "ios-marketing",
```

```

 "filename" : "Icon-App-1024x1024@1x.png",
 "scale" : "1x"
 }
],
 "info" : {
 "version" : 1,
 "author" : "xcode"
 }
}

```

#### File: ios/Runner/Assets.xcassets/LaunchImage.imageset/Contents.json

```

{
 "images" : [
 {
 "idiom" : "universal",
 "filename" : "LaunchImage.png",
 "scale" : "1x"
 },
 {
 "idiom" : "universal",
 "filename" : "LaunchImage@2x.png",
 "scale" : "2x"
 },
 {
 "idiom" : "universal",
 "filename" : "LaunchImage@3x.png",
 "scale" : "3x"
 }
],
 "info" : {
 "version" : 1,
 "author" : "xcode"
 }
}

```

#### File: ios/Runner/Assets.xcassets/LaunchImage.imageset/README.md

# Launch Screen Assets

You can customize the launch screen with your own desired assets by replacing the image files in this directory.

You can also do it by opening your Flutter project's Xcode project with `open ios/Runner.xcworkspace`, selecting `Runner/Assets.xcassets` in the Project Navigator and dropping in the desired images.

#### File: lib/main.dart

```

import 'package:flutter/material.dart';
import 'services/fl_client.dart';
import 'services/api_client.dart';
import 'utils/metrics_collector.dart';
import 'screens/recommendations_page.dart';
import 'package:device_info_plus/device_info_plus.dart';
import 'package:share_plus/share_plus.dart';

void main() {
 runApp(const FederatedLearningApp());
}

class FederatedLearningApp extends StatelessWidget {
 const FederatedLearningApp({super.key});

 @override
 Widget build(BuildContext context) {
 return MaterialApp(
 title: 'Federated Learning Client',
 theme: ThemeData(
 colorScheme: ColorScheme.dark(
 surface: const Color(0xFF212D3F),
 primary: const Color(0xFF00BCD4),
 secondary: const Color(0xFF00BCD4),
),
 scaffoldBackgroundColor: const Color(0xFF1A2332),
),
);
 }
}

```

```

 cardTheme: const CardThemeData(
 color: Color(0xFF212D3F),
 elevation: 0,
 shape: RoundedRectangleBorder(
 borderRadius: BorderRadius.all(Radius.circular(12)),
),
),
 useMaterial3: true,
),
 home: const FederatedLearningHomePage(),
);
 }
}

class FederatedLearningHomePage extends StatefulWidget {
 const FederatedLearningHomePage({super.key});

 @override
 State<FederatedLearningHomePage> createState() =>
 _FederatedLearningHomePageState();
}

class _FederatedLearningHomePageState extends State<FederatedLearningHomePage> {
 final TextEditingController _serverUrlController = TextEditingController(
 text:
 '', // User must enter their PC's IP address (e.g., http://192.168.x.x:8000)
);
 final TextEditingController _clientIdController = TextEditingController(
 text: 'android_client_${DateTime.now().millisecondsSinceEpoch % 10000}',
);

 FederatedLearningClient? _client;
 bool _isConnected = false;
 bool _isTraining = false;
 String _status = 'Ready';
 final List<String> _logs = [];
 MetricsCollector? _metricsCollector;
 int _trainingRound = 0;
 String? _deviceId;

 @override
 void dispose() {
 _serverUrlController.dispose();
 _clientIdController.dispose();
 super.dispose();
 }

 void _addLog(String message) {
 final timestamp = DateTime.now().toString().substring(11, 19);
 final logMessage = '[$timestamp] $message';
 // Debug print to console
 debugPrint('[FL_CLIENT_LOG] $logMessage');
 print(
 '[FL_CLIENT_LOG] $logMessage',
); // Also use print for better visibility
 setState(() {
 _logs.add(logMessage);
 if (_logs.length > 100) {
 _logs.removeAt(0);
 }
 });
 }

 Future<void> _connectToServer() async {
 final serverUrl = _serverUrlController.text.trim();

 if (serverUrl.isEmpty) {
 _showError(
 'Please enter server URL (e.g., http://192.168.1.100:8000)\n\nOn your PC, run: ipconfig\nto find your IP address',
);
 return;
 }
 }
}

```

```

// Validate URL format
if (!serverUrl.startsWith('http://') && !serverUrl.startsWith('https://')) {
 _showError('Server URL must start with http:// or https://');
 return;
}

// Parse and validate the host address
try {
 final uri = Uri.parse(serverUrl);
 final host = uri.host.toLowerCase();

 // Check for invalid addresses
 if (host == 'localhost' ||
 host == '192.168.29.147' ||
 host == '127.0.0.1' ||
 host == '0.0.0.0' ||
 host.isEmpty) {
 String errorMsg = 'Invalid server address: $host\n\n';
 if (host == '0.0.0.0') {
 errorMsg += '0.0.0.0 is not a valid address to connect to!\n\n';
 } else {
 errorMsg +=
 'localhost/127.0.0.1 will not work from mobile device!\n\n';
 }
 errorMsg += 'Use your PC\'s actual IP address instead:\n';
 errorMsg += '• Windows: Run "ipconfig" and look for "IPv4 Address"\n';
 errorMsg += '• Mac/Linux: Run "ifconfig" or "ip addr"\n';
 errorMsg += '• Example: http://192.168.1.100:8000';
 _showError(errorMsg);
 return;
 }

 // Warn if port is missing
 if (uri.port == 0) {
 _showError('Server URL must include a port number (e.g., :8000)');
 return;
 }

 _addLog('NETWORK: DNS lookup resolved for coordinator.');
```

```

} catch (e) {
 _showError(
 'Invalid URL format: $e\n\nPlease use format: http://IP_ADDRESS:PORT\nExample: http://192.168.1.100
:8000',
);
 return;
}

setState(() {
 _status = 'Connecting...';
 _isTraining = true;
});

try {
 final apiClient = ApiClient(serverUrl: serverUrl);
 await apiClient.healthCheck();

 _addLog('INFO: Handshake protocol version 3.2.0.');
```

```

// Create client
_client = FederatedLearningClient(
 clientId: _clientIdController.text,
 serverUrl: serverUrl,
 numUsers: 943, // Will be updated after fetching model
 numItems: 1682, // Will be updated after fetching model
 embeddingDim: 16,
);

await _client!.initialize();

// Get device info for metrics collection
try {
 final deviceInfo = DeviceInfoPlugin();
 if (Theme.of(context).platform == TargetPlatform.android) {
 final androidInfo = await deviceInfo.androidInfo;
```

```

 deviceId = '${androidInfo.brand}${androidInfo.model}'.replaceAll(
 '\s',
 '_');
 } else {
 deviceId = 'device${DateTime.now().millisecondsSinceEpoch}';
 }
} catch (e) {
 deviceId = 'device${DateTime.now().millisecondsSinceEpoch}';
 _addLog('Warning: Could not get device info: $e');
}

// Initialize metrics collector (non-blocking - continue even if it fails)
String? experimentId;
try {
 experimentId = createExperimentId(
 embeddingDim: _client!.embeddingDim,
 numClients: 1,
 deviceId: _deviceId,
);

 _metricsCollector = MetricsCollector(experimentId: experimentId);
 await _metricsCollector!.initialize();
 _addLog('Metrics collector initialized');

 // Set experiment config
 _metricsCollector!.setConfig({
 'num_users': _client!.numUsers,
 'num_items': _client!.numItems,
 'embedding_dim': _client!.embeddingDim,
 'client_id': _clientIdController.text,
 'device_id': _deviceId,
 'server_url': serverUrl,
 });
} catch (e) {
 _addLog('Warning: Metrics collector initialization failed: $e');
 _addLog('Training will continue but results won\'t be saved');
 // Continue anyway - don't block connection
}

setState(() {
 _isConnected = true;
 _status = 'Connected';
 _isTraining = false;
});

_addLog('SYSTEM: Client initialized successfully.');
```

```

if (experimentId != null) {
 _addLog('Experiment ID: $experimentId');
 _addLog('Metrics will be saved after training');
}

// Try to fetch initial model (optional - model might not be initialized yet)
try {
 await _client!.fetchGlobalModel();
 _addLog('SYSTEM: Local model weight buffer allocated (512MB).');
} catch (e) {
 _addLog('WARN: Model not initialized on server yet.');
```

```

 _addLog('INFO: You can fetch model later when starting training.');
```

```

} catch (e, stackTrace) {
 setState(() {
 _status = 'Connection failed';
 _isTraining = false;
 });
 debugPrint('[FL_CLIENT_ERROR] Connection error: $e');
 debugPrint('[FL_CLIENT_ERROR] Stack trace: $stackTrace');
 print('[FL_CLIENT_ERROR] Connection error: $e');
 print('[FL_CLIENT_ERROR] Stack trace: $stackTrace');

 // Provide helpful error message based on error type
 String errorMessage = 'Failed to connect to server.\n\n';

 if (e.toString().contains('Connection refused') ||

```

```

 e.toString().contains('SocketException')) {
 errorMessage += 'Connection was refused. This usually means:\n';
 errorMessage += '1. Server is not running\n';
 errorMessage += '2. Wrong IP address or port\n';
 errorMessage += '3. Firewall blocking connection\n';
 errorMessage += '4. Device not on same network\n\n';
 errorMessage += 'Troubleshooting:\n';
 errorMessage += '• Verify server is running on your PC\n';
 errorMessage +=
 '• Check IP address: Run "ipconfig" (Windows) or "ifconfig" (Mac/Linux)\n';
 errorMessage += '• Ensure both devices are on the same Wi-Fi network\n';
 errorMessage += '• Try disabling firewall temporarily\n';
 errorMessage += '• Check server logs for errors';
 } else if (e.toString().contains('timeout') ||
 e.toString().contains('TimeoutException')) {
 errorMessage += 'Connection timed out. Check:\n';
 errorMessage += '• Server is running and accessible\n';
 errorMessage += '• Network connection is stable\n';
 errorMessage += '• Firewall is not blocking the connection';
 } else {
 errorMessage += 'Error: $e';
 }
 }
 _addLog('Error: $e');
 _showError(errorMessage);
}

Future<void> _runTrainingRound() async {
 if (_client == null || !_isConnected) {
 _showError('Please connect to server first');
 return;
 }

 setState(() {
 _isTraining = true;
 _status = 'Training Round ${_trainingRound + 1}...';
 });

 try {
 // First, make sure we have the global model (it updates numUsers and numItems)
 if (_trainingRound == 0) {
 _addLog('Fetching global model...');
 try {
 await _client!.fetchGlobalModel();
 _addLog(
 'Global model fetched. Model: ${_client!.numUsers} users, ${_client!.numItems} items',
);
 } catch (e) {
 _addLog('Warning: Could not fetch model, using current dimensions');
 }
 }

 // Load some sample data (in real app, this would come from device)
 // For demo, create minimal sample data
 if (_client!.localData.isEmpty) {
 // Generate sample data based on ACTUAL model dimensions (updated after fetchGlobalModel)
 final sampleData = <List<dynamic>>[];
 final random = DateTime.now().millisecondsSinceEpoch;
 final numUsers = _client!.numUsers;
 final numItems = _client!.numItems;

 _addLog(
 'Generating sample data for $numUsers users, $numItems items...',
);

 for (int i = 0; i < 10; i++) {
 final userId = (random + i) % numUsers;
 final itemId = (random + i * 2) % numItems;

 // Validate IDs are in range (just to be safe)
 if (userId >= 0 &&
 userId < numUsers &&
 itemId >= 0 &&
 itemId < numItems) {

```

```

 sampleData.add([
 userId,
 itemId,
 1.0, // Binarized rating
]);
 } else {
 _addLog(
 'Warning: Skipping invalid sample: user=$userId (max=$numUsers), item=$itemId (max=$numItems)',
);
 }
}
_client!.loadLocalData(sampleData);
_addLog('Loaded ${sampleData.length} sample interactions');
}
}

_addLog('Starting training round ${_trainingRound + 1}...');

// Run training round
final metrics = await _client!.runTrainingRound();

_trainingRound++;

// Collect metrics
if (_metricsCollector != null) {
 final trainMetrics = metrics['train'] as Map<String, dynamic>? ?? {};
 final testMetrics = metrics['test'] as Map<String, dynamic>? ?? {};
 final uploadMetrics = metrics['upload'] as Map<String, dynamic>? ?? {};
 final resourceMetrics =
 uploadMetrics['resource_metrics'] as Map<String, dynamic>? ?? {};

 _metricsCollector!.addRoundMetrics(
 roundNum: _trainingRound - 1, // Use round number before increment
 trainLoss: (trainMetrics['loss'] as num?)?.toDouble() ?? 0.0,
 testMetrics: testMetrics,
 aggregationInfo: {
 'num_clients': 1,
 'total_samples': trainMetrics['samples'] ?? 0,
 },
 clientMetrics: [
 {
 'client_id': _clientIdController.text,
 'loss': trainMetrics['loss'] ?? 0.0,
 'samples': trainMetrics['samples'] ?? 0,
 'epochs': trainMetrics['epochs'] ?? 0,
 },
],
 resourceMetrics: resourceMetrics,
);

 // Save results after each round (both locally and to server)
 try {
 // Save locally first
 final jsonPath = await _metricsCollector!.saveJson();
 final csvPath = await _metricsCollector!.saveCsvSummary();

 _addLog('Results saved locally:');
 _addLog(' JSON: ${jsonPath.split('/').last}');
 _addLog(' CSV: ${csvPath.split('/').last}');

 // Also upload to server (saves to PC automatically!)
 await _uploadResultsToServer();
 } catch (e) {
 _addLog('Warning: Failed to save results: $e');
 }
}

setState(() {
 _status = 'Round ${_trainingRound - 1} complete';
 _isTraining = false;
});

_addLog('Training round ${_trainingRound - 1} complete');

```



```

 _addLog(
 'Loss: ${metrics['train']?['loss']?.toStringAsFixed(4) ?? "N/A"}',
);
 _addLog(
 'Accuracy: ${metrics['test']?['accuracy']?.toStringAsFixed(4) ?? "N/A"}',
);

 // Show success message after 10 rounds
 if (_trainingRound > 10) {
 _addLog('■ All 10 rounds complete!');
 _addLog('Results have been uploaded to your PC automatically.');
```

// Calculate and add final metrics
 \_calculateFinalMetrics();

 // Show dialog after final round
 Future.delayed(const Duration(seconds: 1), () {
 \_uploadResultsToServer(showSuccessDialog: true);
 });
 }
 } catch (e, stackTrace) {
 setState(() {
 \_status = 'Training failed';
 \_isTraining = false;
 });
 debugPrint('[FL\_CLIENT\_ERROR] Training error: \$e');
 debugPrint('[FL\_CLIENT\_ERROR] Stack trace: \$stackTrace');
 print('[FL\_CLIENT\_ERROR] Training error: \$e');
 print('[FL\_CLIENT\_ERROR] Stack trace: \$stackTrace');
 \_addLog('Training error: \$e');
 \_showError('Training failed: \$e');
 }
}

void \_ensureSampleDataLoaded() {
 if (\_client == null || \_client!.localData.isEmpty) {
 return;
 }

 final sampleData = <List<dynamic>>[];
 final seed = DateTime.now().millisecondsSinceEpoch;
 final numUsers = \_client!.numUsers;
 final numItems = \_client!.numItems;
 final sampleCount = numItems > 0 ? 15 : 0;

 for (int i = 0; i < sampleCount; i++) {
 final userId = (seed + i) % numUsers;
 final itemId = (seed + i \* 3) % numItems;
 sampleData.add([userId, itemId, 1.0]);
 }

 \_client!.loadLocalData(sampleData);
 \_addLog('Demo data loaded: \${sampleData.length} interactions');
}

Future<void> \_runDemoMode() async {
 if (\_client == null || !\_isConnected) {
 \_showError('Please connect to server first');
 return;
 }

 if (\_isTraining) {
 \_showError('Training is already in progress');
 return;
 }

 setState(() {
 \_status = 'Demo mode: preparing model...';
 });

 try {
 await \_client!.fetchGlobalModel();
 \_ensureSampleDataLoaded();
 \_addLog('Demo mode: running one training round...');
 }
}

```

 await _runTrainingRound();

 if (!mounted) {
 return;
 }

 if (_client!.model != null) {
 _addLog('Demo mode: opening recommendations screen.');
```

\_openRecommendationsScreen();
 }
 } catch (e) {
 \_showError('Demo mode failed: \$e');
 }
}

/// Run 10 rounds automatically
Future<void> \_runAll10Rounds() async {
 if (\_client == null || !\_isConnected) {
 \_showError('Please connect to server first');
 return;
 }

 if (\_isTraining) {
 \_showError('Training already in progress');
 return;
 }

 // Reset round counter
 \_trainingRound = 0;

 // Confirm action
 final confirmed = await showDialog<bool>(
 context: context,
 builder: (context) => AlertDialog(
 title: const Text('Run 10 Training Rounds'),
 content: const Text(
 'This will automatically run 10 training rounds.\n\n'
 'The training will continue until all 10 rounds are complete.\n'
 'You can stop by closing the app if needed.\n\n'
 'Proceed?',
 ),
 actions: [
 TextButton(
 onPressed: () => Navigator.of(context).pop(false),
 child: const Text('Cancel'),
 ),
 ElevatedButton(
 onPressed: () => Navigator.of(context).pop(true),
 child: const Text('Start'),
 ),
 ],
 ),
 );

 if (confirmed != true) {
 return;
 }

 setState(() {
 \_status = 'Running 10 rounds...';
 });

 \_addLog('Starting automatic 10-round training...');

 // Run 10 rounds sequentially
 for (int round = 1; round <= 10; round++) {
 if (!\_isConnected || \_client == null) {
 \_addLog('Training stopped: connection lost');
 break;
 }

 \_addLog('=== Round \$round/10 ===');
 await \_runTrainingRound();
 }
}

```

 // Small delay between rounds
 await Future.delayed(const Duration(milliseconds: 500));
 }

 if (_trainingRound > 10) {
 setState(() {
 _status = 'All 10 rounds complete!';
 });
 }
}

/// Calculate final metrics after all rounds are complete
void _calculateFinalMetrics() {
 if (_metricsCollector == null) return;

 final rounds = _metricsCollector!.experimentData['rounds'] as List;
 if (rounds.isEmpty) return;

 // Get metrics from last round
 final lastRound = rounds.last as Map<String, dynamic>;
 final lastTestMetrics =
 lastRound['test_metrics'] as Map<String, dynamic>? ?? {};

 // Calculate averages across all rounds
 double avgAccuracy = 0.0;
 double avgLoss = 0.0;
 double avgHit10 = 0.0;
 double avgNDCG10 = 0.0;
 int count = 0;

 for (var round in rounds) {
 final roundMap = round as Map<String, dynamic>;
 final testMetrics =
 roundMap['test_metrics'] as Map<String, dynamic>? ?? {};

 if (testMetrics['accuracy'] != null) {
 avgAccuracy += (testMetrics['accuracy'] as num).toDouble();
 avgLoss += (roundMap['train_loss'] as num?)?.toDouble() ?? 0.0;
 avgHit10 += (testMetrics['Hit@10'] as num?)?.toDouble() ?? 0.0;
 avgNDCG10 += (testMetrics['NDCG@10'] as num?)?.toDouble() ?? 0.0;
 count++;
 }
 }

 if (count > 0) {
 avgAccuracy /= count;
 avgLoss /= count;
 avgHit10 /= count;
 avgNDCG10 /= count;
 }

 // Set final metrics
 _metricsCollector!.addFinalMetrics({
 'accuracy': lastTestMetrics['accuracy'] ?? avgAccuracy,
 'mse': lastTestMetrics['mse'] ?? 0.0,
 'mae': lastTestMetrics['mae'] ?? 0.0,
 'Hit@10': lastTestMetrics['Hit@10'] ?? avgHit10,
 'NDCG@10': lastTestMetrics['NDCG@10'] ?? avgNDCG10,
 'Precision@10': lastTestMetrics['Precision@10'] ?? 0.0,
 'Recall@10': lastTestMetrics['Recall@10'] ?? 0.0,
 'final_train_loss': avgLoss,
 'avg_accuracy': avgAccuracy,
 'avg_Hit@10': avgHit10,
 'avg_NDCG@10': avgNDCG10,
 });

 // Add client metrics summary
 _metricsCollector!.addClientMetrics(_clientIdController.text, {
 'total_rounds': rounds.length,
 'total_samples':
 (rounds.last as Map)['aggregation']?['total_samples'] ?? 0,
 'final_loss': avgLoss,
 'final_accuracy': avgAccuracy,
 });
}

```

```

// Add mobile metrics summary
double totalBatteryDrain = 0.0;
int totalTimeMs = 0;
for (var round in rounds) {
 final roundMap = round as Map<String, dynamic>;
 final resourceMetrics =
 roundMap['resource_metrics'] as Map<String, dynamic> ?? {};
 totalBatteryDrain +=
 (resourceMetrics['battery_drain'] as num?)?.toDouble() ?? 0.0;
 totalTimeMs +=
 (resourceMetrics['training_time_ms'] as num?)?.toInt() ?? 0;
}

_metricsCollector!.addMobileMetrics(_deviceId ?? 'unknown', {
 'total_rounds': rounds.length,
 'total_battery_drain': totalBatteryDrain,
 'total_training_time_ms': totalTimeMs,
 'avg_battery_drain_per_round': rounds.isNotEmpty
 ? totalBatteryDrain / rounds.length
 : 0.0,
 'avg_training_time_ms_per_round': rounds.isNotEmpty
 ? totalTimeMs / rounds.length
 : 0.0,
});

_addLog('Final metrics calculated and saved');
}

void _showError(String message) {
 ScaffoldMessenger.of(context).showSnackBar(
 SnackBar(content: Text(message), backgroundColor: Colors.red),
);
}

Future<void> _showResultsLocation() async {
 if (_metricsCollector == null) return;

 try {
 await _metricsCollector!.initialize();
 final resultsPath = _metricsCollector!.resultsPath;
 final resultFiles = await _metricsCollector!.getResultFiles();

 showDialog(
 context: context,
 builder: (context) => AlertDialog(
 title: const Text('Results Location'),
 content: SingleChildScrollView(
 child: Column(
 crossAxisAlignment: CrossAxisAlignment.start,
 mainAxisAlignment: MainAxisAlignment.min,
 children: [
 const Text(
 'Results are saved to Downloads folder:',
 style: TextStyle(fontWeight: FontWeight.bold),
),
 const SizedBox(height: 8),
 SelectableText(
 resultsPath,
 style: const TextStyle(fontFamily: 'monospace', fontSize: 12),
),
 const SizedBox(height: 16),
 const Text(
 'Files:',
 style: TextStyle(fontWeight: FontWeight.bold),
),
 const SizedBox(height: 8),
 if (resultFiles.isEmpty)
 const Text(
 'No files saved yet',
 style: TextStyle(fontStyle: FontStyle.italic),
)
 else
 ...resultFiles.map(

```

```

 (f) => Text(
 '• ${f.path.split('/').last}',
 style: const TextStyle(fontSize: 12),
),
),
 const SizedBox(height: 16),
 Text(
 'Training rounds: $_trainingRound',
 style: const TextStyle(fontWeight: FontWeight.bold),
),
 const SizedBox(height: 16),
 const Text(
 'How to access files:',
 style: TextStyle(fontWeight: FontWeight.bold),
),
 const SizedBox(height: 8),
 const Text(
 '1. Open Android File Manager\n'
 '2. Go to Downloads folder\n'
 '3. Look for "FL_Results" folder\n'
 '4. Your JSON and CSV files are inside\n\n'
 'Or use ADB:\n'
 'adb pull /storage/emulated/0/Download/FL_Results/',
 style: TextStyle(fontSize: 12),
),
],
),
),
actions: [
 if (resultFiles.isNotEmpty)
 TextButton.icon(
 onPressed: () => _shareResults(),
 icon: const Icon(Icons.share),
 label: const Text('Share Files'),
),
 TextButton(
 onPressed: () => Navigator.of(context).pop(),
 child: const Text('Close'),
),
],
),
);
} catch (e) {
 _showError('Failed to get results location: $e');
}
}

Future<void> _shareResults() async {
 if (_metricsCollector == null) return;

 try {
 final resultFiles = await _metricsCollector!.getResultFiles();
 if (resultFiles.isEmpty) {
 _showError('No result files to share');
 return;
 }

 // Share the most recent JSON file
 final jsonFiles = resultFiles
 .where((f) => f.path.endsWith('.json'))
 .toList();
 if (jsonFiles.isEmpty) {
 _showError('No JSON files to share');
 return;
 }

 // Sort by modification time (newest first)
 jsonFiles.sort(
 (a, b) => b.lastModifiedSync().compareTo(a.lastModifiedSync()),
);
 final latestFile = jsonFiles.first;

 final xFile = XFile(latestFile.path, mimeType: 'application/json');
 await Share.shareXFiles(

```

```

 [xFile],
 text: 'Federated Learning Experiment Results',
 subject: 'FL Results: ${_metricsCollector!.experimentId}',
);

 _addLog('Shared file: ${latestFile.path.split('/').last}');
} catch (e) {
 _showError('Failed to share files: $e');
}
}

/// Upload results to server (saves directly to PC)
Future<void> _uploadResultsToServer({bool showSuccessDialog = false}) async {
 if (_client == null || !_isConnected) {
 _showError('Please connect to server first');
 return;
 }

 if (_metricsCollector == null) {
 _showError('No results to upload. Run training rounds first.');
```

return;

```
 }

 setState(() {
 _status = 'Uploading results to PC...';
 });

 try {
 final experimentData = _metricsCollector!.getExperimentData();

 _addLog('Uploading results to server (PC)...');
 _addLog('Experiment ID: ${_metricsCollector!.experimentId}');

 final uploadResponse = await _client!.apiClient.uploadMobileResults(
 experimentId: _metricsCollector!.experimentId,
 experimentData: experimentData,
);

 final jsonPath =
 uploadResponse['json_file'] as String? ??
 'mobile_results/${_metricsCollector!.experimentId}.json';
 final csvPath = uploadResponse['csv_file'] as String?;
 final message =
 uploadResponse['message'] as String? ?? 'Results saved to PC';

 _addLog('✓ Results uploaded successfully!');
 _addLog(' Saved on PC: $message');
 _addLog(' JSON: $jsonPath');
 if (csvPath != null) {
 _addLog(' CSV: $csvPath');
 }

 setState(() {
 _status = 'Results uploaded to PC';
 });

 if (showSuccessDialog) {
 showDialog(
 context: context,
 builder: (context) => AlertDialog(
 title: const Text('✓ Results Uploaded to PC'),
 content: SingleChildScrollView(
 child: Column(
 crossAxisAlignment: CrossAxisAlignment.start,
 mainAxisAlignment: MainAxisAlignment.min,
 children: [
 const Text(
 'Your results have been saved directly to your PC!',
 style: TextStyle(fontWeight: FontWeight.bold),
),
 const SizedBox(height: 16),
 const Text(
 'Location on your PC:',
 style: TextStyle(fontWeight: FontWeight.bold),
),
],
),
),
),
);
 }
 } catch (e) {
 _showError('Failed to upload results: $e');
 }
}


```

```

),
 const SizedBox(height: 8),
 SelectableText(
 jsonPath,
 style: const TextStyle(
 fontFamily: 'monospace',
 fontSize: 12,
),
),
),
 if (csvPath != null) ...[
 const SizedBox(height: 8),
 SelectableText(
 csvPath,
 style: const TextStyle(
 fontFamily: 'monospace',
 fontSize: 12,
),
),
],
 const SizedBox(height: 16),
 const Text(
 'How to access:',
 style: TextStyle(fontWeight: FontWeight.bold),
),
 const SizedBox(height: 8),
 const Text(
 '1. Go to your project folder\n'
 '2. Open the "mobile_results" folder\n'
 '3. Find your JSON and CSV files\n\n'
 'Or check your server terminal - it shows the full path!',
 style: TextStyle(fontSize: 12),
),
],
),
),
actions: [
 TextButton(
 onPressed: () => Navigator.of(context).pop(),
 child: const Text('OK'),
),
],
),
);
}
} catch (uploadError) {
 _addLog('X Upload failed: $uploadError');
 setState(() {
 _status = 'Upload failed';
 });

 String errorMsg = 'Failed to upload results to PC.\n\n';
 if (uploadError.toString().contains('Connection refused') ||
 uploadError.toString().contains('SocketException')) {
 errorMsg += 'Cannot connect to server.\n';
 errorMsg += '• Make sure server is running\n';
 errorMsg += '• Check server URL is correct\n';
 errorMsg += '• Verify both devices are on same network';
 } else {
 errorMsg += 'Error: $uploadError';
 }

 _showError(errorMsg);
}
}

void _openRecommendationsScreen() {
 if (_client == null || !_isConnected) {
 _showError('Please connect to server first.');
```

```

}

Navigator.of(context).push(
 MaterialPageRoute(builder: (_) => RecommendationsPage(client: _client!)),
);
}

@override
Widget build(BuildContext context) {
 return Scaffold(
 backgroundColor: const Color(0xFF1A2332),
 body: SafeArea(
 child: SingleChildScrollView(
 padding: const EdgeInsets.all(20.0),
 child: Column(
 crossAxisAlignment: CrossAxisAlignment.stretch,
 children: [
 // Header with title and settings icon
 Row(
 mainAxisAlignment: MainAxisAlignment.spaceBetween,
 crossAxisAlignment: CrossAxisAlignment.start,
 children: [
 Expanded(
 child: Column(
 crossAxisAlignment: CrossAxisAlignment.start,
 children: [
 const Text(
 'Federated Client',
 maxLines: 1,
 overflow: TextOverflow.ellipsis,
 style: TextStyle(
 fontSize: 32,
 fontWeight: FontWeight.bold,
 color: Colors.white,
),
),
 const SizedBox(height: 4),
 Text(
 'Researcher Edition v2.4',
 maxLines: 1,
 overflow: TextOverflow.ellipsis,
 style: TextStyle(fontSize: 16, color: Colors.grey[400]),
),
],
),
),
 const SizedBox(width: 12),
 Container(
 decoration: BoxDecoration(
 color: const Color(0xFF212D3F),
 borderRadius: BorderRadius.circular(12),
 border: Border.all(
 color: const Color(0xFF00BCD4).withValues(alpha: 0.3),
 width: 1,
),
),
),
 child: Row(
 mainAxisAlignment: MainAxisAlignment.min,
 children: [
 IconButton(
 tooltip: 'Open recommendations',
 icon: const Icon(
 Icons.movie,
 color: Color(0xFF00BCD4),
),
 onPressed: (_isConnected && _client?.model != null)
 ? _openRecommendationsScreen
 : null,
),
 IconButton(
 icon: const Icon(
 Icons.settings,
 color: Color(0xFF00BCD4),
),

```



```

 onPressed: () {
 // Settings action - can be implemented later
 ScaffoldMessenger.of(context).showSnackBar(
 const SnackBar(
 content: Text('Settings coming soon'),
),
);
 },
],
),
],
),
),
],
),
const SizedBox(height: 24),

// Connection Section
Card(
 child: Padding(
 padding: const EdgeInsets.all(20.0),
 child: Column(
 crossAxisAlignment: CrossAxisAlignment.stretch,
 children: [
 Row(
 children: [
 Icon(
 Icons.dns,
 color: const Color(0xFF00BCD4),
 size: 20,
),
 const SizedBox(width: 8),
 const Text(
 'SERVER CONFIGURATION',
 style: TextStyle(
 fontSize: 13,
 fontWeight: FontWeight.w600,
 letterSpacing: 0.5,
 color: Colors.grey,
),
),
],
),
 const SizedBox(height: 20),
 Text(
 'SERVER URL',
 style: TextStyle(
 fontSize: 12,
 fontWeight: FontWeight.w600,
 color: const Color(0xFF00BCD4),
 letterSpacing: 0.5,
),
),
 const SizedBox(height: 8),
 TextField(
 controller: _serverUrlController,
 decoration: InputDecoration(
 hintText: 'wss://fl-coordinator.research.net',
 hintStyle: TextStyle(color: Colors.grey[600]),
 filled: true,
 fillColor: const Color(0xFF1A2332),
 border: OutlineInputBorder(
 borderRadius: BorderRadius.circular(8),
 borderSide: BorderSide.none,
),
 contentPadding: const EdgeInsets.symmetric(
 horizontal: 16,
 vertical: 14,
),
),
 style: const TextStyle(
 color: Colors.white70,
 fontSize: 15,
),
),
],
),
),
),

```

```

 enabled: !_isConnected,
 keyboardType: TextInputType.url,
),
 const SizedBox(height: 16),
 Text(
 'CLIENT ID',
 style: TextStyle(
 fontSize: 12,
 fontWeight: FontWeight.w600,
 color: Colors.grey[400],
 letterSpacing: 0.5,
),
),
),
 const SizedBox(height: 8),
 TextField(
 controller: _clientIdController,
 decoration: InputDecoration(
 hintText: 'android_client_6149',
 hintStyle: TextStyle(color: Colors.grey[600]),
 filled: true,
 fillColor: const Color(0xFF1A2332),
 border: OutlineInputBorder(
 borderRadius: BorderRadius.circular(8),
 borderSide: BorderSide.none,
),
),
 contentPadding: const EdgeInsets.symmetric(
 horizontal: 16,
 vertical: 14,
),
),
 style: const TextStyle(
 color: Colors.white70,
 fontSize: 15,
),
 enabled: !_isConnected,
),
 const SizedBox(height: 20),
 SizedBox(
 height: 54,
 child: ElevatedButton(
 onPressed: _isConnected ? null : _connectToServer,
 style: ElevatedButton.styleFrom(
 backgroundColor: const Color(0xFF00BCD4),
 foregroundColor: Colors.black87,
 disabledBackgroundColor: Colors.grey[700],
 shape: RoundedRectangleBorder(
 borderRadius: BorderRadius.circular(8),
),
),
 elevation: 0,
),
 child: Row(
 mainAxisAlignment: MainAxisAlignment.center,
 children: [
 const Icon(Icons.flash_on, size: 20),
 const SizedBox(width: 8),
 const Text(
 'CONNECT TO SERVER',
 style: TextStyle(
 fontSize: 15,
 fontWeight: FontWeight.bold,
 letterSpacing: 0.5,
),
),
],
),
),
),
),
),
),
),
),
),
),
),
),
const SizedBox(height: 16),

```

```

// Status Section
Card(
 child: Padding(
 padding: const EdgeInsets.all(20.0),
 child: Column(
 crossAxisAlignment: CrossAxisAlignment.stretch,
 children: [
 Row(
 mainAxisAlignment: MainAxisAlignment.spaceBetween,
 children: [
 Expanded(
 child: Row(
 children: [
 Container(
 width: 12,
 height: 12,
 decoration: BoxDecoration(
 shape: BoxShape.circle,
 color: _isConnected
 ? const Color(0xFF4CAF50)
 : Colors.grey,
),
),
 const SizedBox(width: 10),
 Expanded(
 child: Text(
 'Status: $_status',
 maxLines: 1,
 overflow: TextOverflow.ellipsis,
 style: const TextStyle(
 fontSize: 18,
 fontWeight: FontWeight.w600,
 color: Colors.white,
),
),
),
],
),
),
],
),
 const SizedBox(width: 8),
 Container(
 padding: const EdgeInsets.symmetric(
 horizontal: 12,
 vertical: 6,
),
 decoration: BoxDecoration(
 color: const Color(0xFF1A2332),
 borderRadius: BorderRadius.circular(6),
),
 child: Text(
 _isTraining ? 'TRAINING' : 'IDLE',
 style: const TextStyle(
 fontSize: 12,
 fontWeight: FontWeight.bold,
 color: Color(0xFF4CAF50),
 letterSpacing: 0.5,
),
),
),
],
),
 const SizedBox(height: 20),
 Row(
 children: [
 Expanded(
 child: SizedBox(
 height: 100,
 child: ElevatedButton(
 onPressed: (_isConnected && !_isTraining)
 ? _runTrainingRound
 : null,
 style: ElevatedButton.styleFrom(
 backgroundColor: const Color(0xFF2A3A4F),
 foregroundColor: Colors.white70,
),
),
),
),
],
),
),
)

```

```

 disabledBackgroundColor: const Color(
 0xFF1E2936,
),
 shape: RoundedRectangleBorder(
 borderRadius: BorderRadius.circular(12),
),
 elevation: 0,
),
 child: const Column(
 mainAxisAlignment: MainAxisAlignment.center,
 children: [
 Icon(
 Icons.play_arrow,
 size: 32,
 color: Color(0xFF00BCD4),
),
 SizedBox(height: 8),
 Text(
 'Run 1 Round',
 style: TextStyle(
 fontSize: 16,
 fontWeight: FontWeight.w600,
),
),
],
),
),
),
),
const SizedBox(width: 12),
Expanded(
 child: SizedBox(
 height: 100,
 child: ElevatedButton(
 onPressed: (_isConnected && !_isTraining)
 ? _runAll10Rounds
 : null,
 style: ElevatedButton.styleFrom(
 backgroundColor: const Color(0xFF2A3A4F),
 foregroundColor: Colors.white70,
 disabledBackgroundColor: const Color(
 0xFF1E2936,
),
 shape: RoundedRectangleBorder(
 borderRadius: BorderRadius.circular(12),
),
 elevation: 0,
),
),
 child: const Column(
 mainAxisAlignment: MainAxisAlignment.center,
 children: [
 Icon(
 Icons.fast_forward,
 size: 32,
 color: Color(0xFF00BCD4),
),
 SizedBox(height: 8),
 Text(
 'Run 10 Rounds',
 style: TextStyle(
 fontSize: 16,
 fontWeight: FontWeight.w600,
),
),
],
),
),
),
const SizedBox(height: 12),
SizedBox(
 height: 52,

```

```

child: ElevatedButton.icon(
 onPressed: (_isConnected && !_isTraining)
 ? _runDemoMode
 : null,
 icon: const Icon(
 Icons.present_to_all,
 color: Color(0xFF00BCD4),
),
 label: const Text(
 'Demo Mode: Train and Show Recommendations',
 style: TextStyle(fontWeight: FontWeight.w600),
),
 style: ElevatedButton.styleFrom(
 backgroundColor: const Color(0xFF243447),
 foregroundColor: Colors.white,
 disabledBackgroundColor: const Color(0xFF1E2936),
 shape: RoundedRectangleBorder(
 borderRadius: BorderRadius.circular(10),
),
 elevation: 0,
),
),
),
if (_isTraining) ...[
 const SizedBox(height: 8),
 const Row(
 mainAxisAlignment: MainAxisAlignment.center,
 children: [
 SizedBox(
 width: 16,
 height: 16,
 child: CircularProgressIndicator(strokeWidth: 2),
),
 SizedBox(width: 8),
 Text('Training...'),
],
),
],
if (_metricsCollector != null && _trainingRound > 0) ...[
 const SizedBox(height: 12),
 Row(
 children: [
 Expanded(
 child: OutlinedButton.icon(
 onPressed: (_isConnected && !_isTraining)
 ? _uploadResultsToServer
 : null,
 icon: const Icon(Icons.cloud_upload, size: 18),
 label: const Text(
 'Upload to PC',
 style: TextStyle(fontSize: 13),
),
 style: OutlinedButton.styleFrom(
 foregroundColor: const Color(0xFF00BCD4),
 side: const BorderSide(
 color: Color(0xFF00BCD4),
),
 padding: const EdgeInsets.symmetric(
 vertical: 12,
),
),
),
),
 const SizedBox(width: 8),
 Expanded(
 child: OutlinedButton.icon(
 onPressed: _showResultsLocation,
 icon: const Icon(Icons.folder, size: 18),
 label: const Text(
 'Location',
 style: TextStyle(fontSize: 13),
),
 style: OutlinedButton.styleFrom(
 foregroundColor: Colors.grey[400],

```

```

 side: BorderSide(color: Colors.grey[700]!),
 padding: const EdgeInsets.symmetric(
 vertical: 12,
),
),
),
],
),
],
),
],
),
),
), // End of Status Section Card
// Logs Section
const SizedBox(height: 16),
Card(
 child: Padding(
 padding: const EdgeInsets.all(20.0),
 child: Column(
 crossAxisAlignment: CrossAxisAlignment.stretch,
 children: [
 Row(
 mainAxisAlignment: MainAxisAlignment.spaceBetween,
 children: [
 Row(
 children: [
 Icon(
 Icons.live_tv,
 color: Colors.grey[400],
 size: 20,
),
 const SizedBox(width: 8),
 const Text(
 'LIVE LOGS',
 style: TextStyle(
 fontSize: 13,
 fontWeight: FontWeight.w600,
 letterSpacing: 0.5,
 color: Colors.grey,
),
),
],
),
 TextButton(
 onPressed: () {
 setState(() {
 _logs.clear();
 });
 },
 style: TextButton.styleFrom(
 foregroundColor: Colors.grey[500],
 padding: const EdgeInsets.symmetric(
 horizontal: 12,
 vertical: 8,
),
),
 child: const Text(
 'CLEAR CONSOLE',
 style: TextStyle(
 fontSize: 11,
 fontWeight: FontWeight.w600,
 letterSpacing: 0.5,
),
),
),
],
),
 const SizedBox(height: 12),
 Container(
 height: 250,
 padding: const EdgeInsets.all(12),
 decoration: BoxDecoration(
 color: const Color(0xFF0D1117),

```

```

 borderRadius: BorderRadius.circular(8),
),
 child: _logs.isEmpty
 ? Center(
 child: Text(
 'No logs yet. Connect to server to see activity.',
 style: TextStyle(
 color: Colors.grey[600],
 fontSize: 13,
 fontStyle: FontStyle.italic,
),
),
),
 : ListView.builder(
 reverse: false,
 itemCount: _logs.length,
 itemBuilder: (context, index) {
 final log = _logs[index];
 Color logColor = const Color(
 0xFF58A6FF,
); // Default blue

 // Color code logs based on type
 if (log.contains('SYSTEM:')) {
 logColor = const Color(0xFF4CAF50); // Green
 } else if (log.contains('NETWORK:')) {
 logColor = const Color(0xFF00BCD4); // Cyan
 } else if (log.contains('INFO:')) {
 logColor = const Color(0xFF58A6FF); // Blue
 } else if (log.contains('WARN:')) {
 logColor = const Color(
 0xFFFFA726,
); // Orange
 } else if (log.contains('ERROR:')) {
 logColor = const Color(0xFFEF5350); // Red
 }

 return Padding(
 padding: const EdgeInsets.symmetric(
 vertical: 2.0,
),
 child: Text(
 log,
 style: TextStyle(
 color: logColor,
 fontFamily: 'Courier',
 fontSize: 12,
 height: 1.4,
),
),
);
 },
),
), // End of Container
],
),
),
),
);
}
}

```

#### File: lib/models/matrix\_factorization.dart

```

/// Matrix Factorization model for movie recommendations
/// Pure Dart implementation compatible with PyTorch server model
library;

class MatrixFactorization {
 final int numUsers;

```

```

final int numItems;
final int embeddingDim;

// Embedding matrices: [num_users x embedding_dim], [num_items x embedding_dim]
late List<List<double>> userEmbeddings;
late List<List<double>> itemEmbeddings;

MatrixFactorization({
 required this.numUsers,
 required this.numItems,
 required this.embeddingDim,
}) {
 // Initialize embeddings with small random values
 userEmbeddings = _initializeEmbeddings(numUsers, embeddingDim);
 itemEmbeddings = _initializeEmbeddings(numItems, embeddingDim);
}

/// Initialize embedding matrix with small random values (similar to PyTorch)
List<List<double>> _initializeEmbeddings(int size, int dim) {
 final embeddings = <List<double>>[];
 final random = DateTime.now().millisecondsSinceEpoch; // Simple seed

 for (int i = 0; i < size; i++) {
 final row = <double>[];
 for (int j = 0; j < dim; j++) {
 // Generate random value between -0.01 and 0.01
 final value = ((random + i * dim + j) % 2000 - 1000) / 100000.0;
 row.add(value);
 }
 embeddings.add(row);
 }
 return embeddings;
}

/// Forward pass: predict rating for user-item pair
double predict(int userId, int itemId) {
 if (userId >= numUsers || itemId >= numItems || userId < 0 || itemId < 0) {
 throw ArgumentError('Invalid user_id or item_id');
 }

 final userEmb = userEmbeddings[userId];
 final itemEmb = itemEmbeddings[itemId];

 // Dot product: sum(user_emb[i] * item_emb[i])
 double result = 0.0;
 for (int i = 0; i < embeddingDim; i++) {
 result += userEmb[i] * itemEmb[i];
 }

 return result;
}

/// Predict ratings for multiple user-item pairs (batch prediction)
List<double> predictBatch(List<int> userIds, List<int> itemIds) {
 if (userIds.length != itemIds.length) {
 throw ArgumentError('userIds and itemIds must have same length');
 }

 final predictions = <double>[];
 for (int i = 0; i < userIds.length; i++) {
 predictions.add(predict(userIds[i], itemIds[i]));
 }
 return predictions;
}

/// Get model state dictionary (for serialization)
Map<String, dynamic> getStateDict() {
 return {
 'user_embedding.weight': userEmbeddings,
 'item_embedding.weight': itemEmbeddings,
 };
}

/// Load model state from dictionary (for deserialization)

```



```

void loadStateDict(Map<String, dynamic> stateDict) {
 userEmbeddings = List<List<double>>.from(
 stateDict['user_embedding.weight'].map((row) =>
 List<double>.from(row.map((val) => val.toDouble())))
);
 itemEmbeddings = List<List<double>>.from(
 stateDict['item_embedding.weight'].map((row) =>
 List<double>.from(row.map((val) => val.toDouble())))
);
}

/// Get all trainable parameters (flattened for gradient computation)
List<List<List<double>>> getParameters() {
 return [userEmbeddings, itemEmbeddings];
}

/// Update parameters (used during training)
void updateParameters(List<List<double>> userGrad, List<List<double>> itemGrad, double learningRate) {
 // Gradient descent update
 for (int i = 0; i < numUsers; i++) {
 for (int j = 0; j < embeddingDim; j++) {
 userEmbeddings[i][j] -= learningRate * userGrad[i][j];
 }
 }

 for (int i = 0; i < numItems; i++) {
 for (int j = 0; j < embeddingDim; j++) {
 itemEmbeddings[i][j] -= learningRate * itemGrad[i][j];
 }
 }
}
}

```

### File: lib/screens/recommendations\_page.dart

```

import 'package:flutter/material.dart';

import '../services/fl_client.dart';
import '../services/movie_title_service.dart';
import '../services/recommendation_service.dart';

class RecommendationsPage extends StatefulWidget {
 final FederatedLearningClient client;

 const RecommendationsPage({super.key, required this.client});

 @override
 State<RecommendationsPage> createState() => _RecommendationsPageState();
}

class _RecommendationsPageState extends State<RecommendationsPage> {
 final RecommendationService _recommendationService =
 const RecommendationService();
 final MovieTitleService _movieTitleService = const MovieTitleService();

 List<int> _candidateUsers = const <int>[];
 List<MovieRecommendation> _recommendations = const <MovieRecommendation>[];
 Map<int, String> _movieTitles = const <int, String>{};
 int? _selectedUserId;
 int _topN = 10;
 bool _excludeSeenItems = true;
 bool _isLoadingTitles = true;
 final Set<int> _submittingSeenItems = <int>{};
 String? _error;

 @override
 void initState() {
 super.initState();
 _initialize();
 }
}

```

```

Future<void> _initialize() async {
 try {
 final titles = await _movieTitleService.loadTitles();
 if (!mounted) {
 return;
 }
 _movieTitles = titles;
 } catch (_) {
 // Fallback is handled in RecommendationService with Movie #id labels.
 _movieTitles = const <int, String>{};
 }

 _candidateUsers = _recommendationService.getCandidateUsers(widget.client);
 if (_candidateUsers.isNotEmpty) {
 _selectedUserId = _candidateUsers.first;
 _refreshRecommendations();
 } else {
 _error =
 'No users available. Connect and run at least one training round first.';
 }

 if (mounted) {
 setState(() {
 _isLoadingTitles = false;
 });
 }
}

void _refreshRecommendations() {
 if (_selectedUserId == null) {
 return;
 }

 try {
 final results = _recommendationService.topNRecommendations(
 client: widget.client,
 userId: _selectedUserId!,
 topN: _topN,
 movieTitles: _movieTitles,
 excludeSeenItems: _excludeSeenItems,
);

 setState(() {
 _error = null;
 _recommendations = results;
 });
 } catch (e) {
 setState(() {
 _recommendations = const <MovieRecommendation>[];
 _error = 'Failed to generate recommendations: $e';
 });
 }
}

Future<void> _markAsSeen(MovieRecommendation rec, int index) async {
 final userId = _selectedUserId;
 if (userId == null ||
 rec.isSeen ||
 _submittingSeenItems.contains(rec.itemId)) {
 return;
 }

 setState(() {
 _submittingSeenItems.add(rec.itemId);
 _recommendations[index] = rec.copyWith(
 isSeen: true,
 explanation:
 'Predicted score ${rec.score.toStringAsFixed(4)} - Seen before',
);
 });

 try {
 await widget.client.markRecommendationSeen(
 userId: userId,

```

```

 itemId: rec.itemId,
 rating: 1.0,
);

 if (!mounted) {
 return;
 }
 ScaffoldMessenger.of(
 context,
).showSnackBar(SnackBar(content: Text('Marked "${rec.title}" as seen')));
} catch (e) {
 if (!mounted) {
 return;
 }
 setState(() {
 _recommendations[index] = rec;
 });
 ScaffoldMessenger.of(context).showSnackBar(
 SnackBar(
 content: Text('Failed to send seen event: $e'),
 backgroundColor: Colors.redAccent,
),
);
} finally {
 if (mounted) {
 setState(() {
 _submittingSeenItems.remove(rec.itemId);
 });
 }
}
}

@override
Widget build(BuildContext context) {
 return Scaffold(
 appBar: AppBar(
 title: const Text('Movie Recommendations'),
 actions: [
 IconButton(
 tooltip: 'Refresh',
 onPressed: _refreshRecommendations,
 icon: const Icon(Icons.refresh),
),
],
),
 body: Padding(
 padding: const EdgeInsets.all(16),
 child: Column(
 crossAxisAlignment: CrossAxisAlignment.stretch,
 children: [
 _buildControls(),
 const SizedBox(height: 12),
 if (_isLoadingTitles)
 const Text('Loading movie titles...')
 else if (_error != null)
 Text(_error!, style: const TextStyle(color: Colors.redAccent))
 else
 Text(
 'Showing ${_recommendations.length} recommendations for user $_selectedUserId',
 style: Theme.of(context).textTheme.bodyMedium,
),
 const SizedBox(height: 12),
 Expanded(
 child: _recommendations.isEmpty
 ? const Center(child: Text('No recommendations yet'))
 : ListView.separated(
 itemCount: _recommendations.length,
 separatorBuilder: (_, __) => const Divider(height: 1),
 itemBuilder: (context, index) {
 final rec = _recommendations[index];
 final isSubmitting = _submittingSeenItems.contains(
 rec.itemId,
);
 return Padding(

```

```

padding: const EdgeInsets.symmetric(
 horizontal: 12,
 vertical: 10,
),
child: Row(
 crossAxisAlignment: CrossAxisAlignment.start,
 children: [
 CircleAvatar(child: Text('${index + 1}')),
 const SizedBox(width: 12),
 Expanded(
 child: Column(
 crossAxisAlignment: CrossAxisAlignment.start,
 mainAxisAlignment: MainAxisAlignment.min,
 children: [
 Text(
 rec.title,
 style: Theme.of(context)
 .textTheme
 .titleMedium,
),
 const SizedBox(height: 4),
 Text(rec.explanation),
],
),
),
],
const SizedBox(width: 10),
Column(
 mainAxisAlignment: MainAxisAlignment.min,
 crossAxisAlignment: CrossAxisAlignment.end,
 children: [
 Container(
 padding: const EdgeInsets.symmetric(
 horizontal: 8,
 vertical: 2,
),
 decoration: BoxDecoration(
 color: rec.isSeen
 ? Colors.orange.withValues(alpha: 0.15)
 : Colors.green.withValues(alpha: 0.15),
 borderRadius: BorderRadius.circular(8),
 border: Border.all(
 color: rec.isSeen
 ? Colors.orange
 : Colors.green,
),
),
),
 child: Text(
 rec.isSeen ? 'Seen' : 'Unseen',
 style: TextStyle(
 color: rec.isSeen
 ? Colors.orange
 : Colors.green,
 fontSize: 11,
 fontWeight: FontWeight.w600,
),
),
],
const SizedBox(height: 6),
FilledButton.tonal(
 style: FilledButton.styleFrom(
 minimumSize: const Size(0, 28),
 tapTargetSize:
 MaterialTapTargetSize.shrinkWrap,
 visualDensity: VisualDensity.compact,
 padding: const EdgeInsets.symmetric(
 horizontal: 10,
 vertical: 6,
),
),
),
onPressed: (rec.isSeen || isSubmitting)
 ? null
 : () => _markAsSeen(rec, index),
child: isSubmitting
 ? const SizedBox(

```

```

 height: 12,
 width: 12,
 child: CircularProgressIndicator(
 strokeWidth: 2,
),
),
 : Text(rec.isSeen ? 'Seen' : 'Mark seen'),
],
],
),
),
),
);
}

Widget _buildControls() {
 final dropdownItems = _candidateUsers
 .map(
 (userId) =>
 DropdownMenuItem<int>(value: userId, child: Text('User $userId')),
)
 .toList();

 return Card(
 child: Padding(
 padding: const EdgeInsets.all(12),
 child: Column(
 children: [
 Row(
 children: [
 const Text('User:'),
 const SizedBox(width: 12),
 Expanded(
 child: DropdownButton<int>(
 value: _selectedUserId,
 isExpanded: true,
 items: dropdownItems,
 onChanged: (value) {
 if (value == null) {
 return;
 }
 setState(() {
 _selectedUserId = value;
 });
 _refreshRecommendations();
 },
),
),
],
),
 const SizedBox(height: 8),
 Row(
 children: [
 const Text('Top N:'),
 Expanded(
 child: Slider(
 min: 5,
 max: 30,
 divisions: 5,
 value: _topN.toDouble(),
 label: _topN.toString(),
 onChanged: (value) {
 setState(() {
 _topN = value.round();
 });
 },
),
),
],
),
],
),
),
);
}

void _onChangeEnd(_) => _refreshRecommendations(),

```

```

),
),
 Text(_topN.toString()),
],
),
SwitchListTile(
 contentPadding: EdgeInsets.zero,
 title: const Text('Exclude seen items'),
 value: _excludeSeenItems,
 onChanged: (value) {
 setState(() {
 _excludeSeenItems = value;
 });
 _refreshRecommendations();
 },
),
),
],
),
),
);
}
}

```

### File: lib/services/api\_client.dart

```

import 'dart:convert';
import 'dart:io';
import 'package:http/http.dart' as http;

/// API client for communicating with the federated learning server
class ApiClient {
 final String _baseUrl;

 ApiClient({required String serverUrl})
 : _baseUrl = serverUrl.endsWith('/')
 ? serverUrl.substring(0, serverUrl.length - 1)
 : serverUrl;

 /// Get normalized base URL (without trailing slash)
 String get baseUrl => _baseUrl;

 /// Check if server is available
 Future<Map<String, dynamic>> healthCheck() async {
 final url = Uri.parse('$_baseUrl/healthz');
 print('[API_CLIENT] Health check: GET $url');
 print('[API_CLIENT] Parsed host: ${url.host}, port: ${url.port}');

 // Validate host before attempting connection
 if (url.host.isEmpty || url.host == '0.0.0.0') {
 throw Exception(
 'Invalid server address: ${url.host}. Use your PC\'s actual IP address (e.g., 192.168.1.100)',
);
 }

 try {
 final response = await http
 .get(url)
 .timeout(
 const Duration(seconds: 10),
 onTimeout: () {
 throw Exception(
 'Connection timeout: Server did not respond within 10 seconds. Check if server is running and accessible.',
);
 },
);
 print('[API_CLIENT] Health check response: ${response.statusCode}');
 print('[API_CLIENT] Health check headers: ${response.headers}');
 print('[API_CLIENT] Health check body (raw): ${response.body}');
 print('[API_CLIENT] Health check body length: ${response.body.length}');

 if (response.statusCode == 200) {
 try {
 final decoded = jsonDecode(response.body) as Map<String, dynamic>;

```

```

 print('[API_CLIENT] Health check parsed successfully: $decoded');
 return decoded;
 } catch (jsonError) {
 print('[API_CLIENT_ERROR] JSON decode failed: $jsonError');
 print(
 '[API_CLIENT_ERROR] Response body that failed to parse: ${response.body}',
);
 throw Exception(
 'Server returned invalid JSON. Expected JSON but got: ${response.body.substring(0, 100)}...',
);
 }
} else {
 throw Exception(
 'Server health check failed: ${response.statusCode} - ${response.body}',
);
}
} on SocketException catch (e) {
 print('[API_CLIENT_ERROR] Socket error: $e');
 throw Exception(
 'Cannot connect to server at ${url.host}:${url.port}. '
 'Check that:\n'
 '1. Server is running\n'
 '2. IP address is correct\n'
 '3. Port is correct\n'
 '4. Both devices are on the same network',
);
} catch (e, stackTrace) {
 print('[API_CLIENT_ERROR] Health check failed: $e');
 print('[API_CLIENT_ERROR] Stack trace: $stackTrace');
 rethrow;
}
}

/// Register client with server
Future<Map<String, dynamic>> register(String clientId) async {
 final url = Uri.parse('${_baseUrl}/register');
 print('[API_CLIENT] Register: POST $url');
 print('[API_CLIENT] Register payload: client_id=$clientId');
 try {
 final response = await http.post(
 url,
 headers: {'Content-Type': 'application/json'},
 body: jsonEncode({'client_id': clientId}),
);
 print('[API_CLIENT] Register response: ${response.statusCode}');
 print('[API_CLIENT] Register body: ${response.body}');

 if (response.statusCode == 200) {
 return jsonDecode(response.body) as Map<String, dynamic>;
 } else {
 throw Exception(
 'Registration failed: ${response.statusCode} - ${response.body}',
);
 }
 } catch (e, stackTrace) {
 print('[API_CLIENT_ERROR] Registration failed: $e');
 print('[API_CLIENT_ERROR] Stack trace: $stackTrace');
 rethrow;
 }
}

/// Fetch global model parameters from server
Future<Map<String, dynamic>> fetchGlobalParams() async {
 final url = Uri.parse('${_baseUrl}/global-params-json');
 print('[API_CLIENT] Fetch global params: GET $url');
 try {
 final response = await http.get(url);
 print(
 '[API_CLIENT] Fetch global params response: ${response.statusCode}',
);
 print('[API_CLIENT] Response body length: ${response.body.length}');

 if (response.statusCode == 200) {
 final data = jsonDecode(response.body) as Map<String, dynamic>;

```

```

 print('[API_CLIENT] Model config: ${data['model_config']}');
 print(
 '[API_CLIENT] Params count: ${data['params'] as List?}?.length ?? 'N/A'}',
);
 return data;
 } else if (response.statusCode == 404) {
 // Model not initialized on server
 print('[API_CLIENT_ERROR] Model not initialized (404)');
 throw Exception(
 'Model not initialized on server. Please initialize the model first:\n\n'
 'Run on your PC: curl -X POST "$_baseUrl/init-model?num_users=943&num_items=1682&embedding_dim=16'
 "\n\n"
 'Or use the API docs at: $_baseUrl/docs',
);
 } else {
 throw Exception(
 'Failed to fetch global params: ${response.statusCode} - ${response.body}',
);
 }
} catch (e, stackTrace) {
 print('[API_CLIENT_ERROR] Fetch global params failed: $e');
 print('[API_CLIENT_ERROR] Stack trace: $stackTrace');
 rethrow;
}

/// Upload client model parameters to server
Future<Map<String, dynamic>> uploadParams({
 required String clientId,
 required List<Map<String, dynamic>> params,
 required int sampleCount,
}) async {
 final url = Uri.parse('$_baseUrl/upload-params-json');
 final payload = {
 'client_id': clientId,
 'params': params,
 'sample_count': sampleCount,
 };
 print('[API_CLIENT] Upload params: POST $url');
 print(
 '[API_CLIENT] Upload payload: client_id=$clientId, sample_count=$sampleCount, params_count=${params.l
 ength}',
);
 try {
 final response = await http.post(
 url,
 headers: {'Content-Type': 'application/json'},
 body: jsonEncode(payload),
);
 print('[API_CLIENT] Upload response: ${response.statusCode}');
 print('[API_CLIENT] Upload body: ${response.body}');

 if (response.statusCode == 200) {
 return jsonDecode(response.body) as Map<String, dynamic>;
 } else {
 throw Exception(
 'Upload failed: ${response.statusCode} - ${response.body}',
);
 }
 } catch (e, stackTrace) {
 print('[API_CLIENT_ERROR] Upload failed: $e');
 print('[API_CLIENT_ERROR] Stack trace: $stackTrace');
 rethrow;
 }
}

/// Upload resource metrics (CPU, memory, battery, etc.)
Future<void> uploadResourceMetrics({
 required String clientId,
 required Map<String, dynamic> metrics,
}) async {
 // This endpoint should be added to server.py
 // For now, we'll include it in upload-params or create a separate endpoint
 // Implementation depends on server capability

```



```

}

/// Upload mobile experiment results to server (saves to PC)
Future<Map<String, dynamic>> uploadMobileResults({
 required String experimentId,
 required Map<String, dynamic> experimentData,
}) async {
 final url = Uri.parse('${_baseUrl}/upload-mobile-results');
 final payload = {
 'experiment_id': experimentId,
 'experiment_data': experimentData,
 };

 print('[API_CLIENT] Upload mobile results: POST $url');
 print('[API_CLIENT] Experiment ID: $experimentId');

 try {
 final response = await http.post(
 url,
 headers: {'Content-Type': 'application/json'},
 body: jsonEncode(payload),
);

 print('[API_CLIENT] Upload results response: ${response.statusCode}');

 if (response.statusCode == 200) {
 return jsonDecode(response.body) as Map<String, dynamic>;
 } else {
 throw Exception(
 'Upload failed: ${response.statusCode} - ${response.body}',
);
 }
 } catch (e, stackTrace) {
 print('[API_CLIENT_ERROR] Upload results failed: $e');
 print('[API_CLIENT_ERROR] Stack trace: $stackTrace');
 rethrow;
 }
}

/// Send recommendation feedback (e.g., mark item as seen) to server.
Future<Map<String, dynamic>> sendRecommendationSeenEvent({
 required String clientId,
 required int userId,
 required int itemId,
 double rating = 1.0,
}) async {
 final url = Uri.parse('${_baseUrl}/recommendation-events');
 final payload = {
 'client_id': clientId,
 'user_id': userId,
 'item_id': itemId,
 'rating': rating,
 };

 print('[API_CLIENT] Recommendation event: POST $url');
 print('[API_CLIENT] Recommendation payload: $payload');

 try {
 final response = await http
 .post(
 url,
 headers: {'Content-Type': 'application/json'},
 body: jsonEncode(payload),
)
 .timeout(const Duration(seconds: 10));

 print(
 '[API_CLIENT] Recommendation event response: ${response.statusCode}',
);
 print('[API_CLIENT] Recommendation event body: ${response.body}');

 if (response.statusCode == 200) {
 return jsonDecode(response.body) as Map<String, dynamic>;
 }
 }
}

```

```

 throw Exception(
 'Recommendation event failed: ${response.statusCode} - ${response.body}',
);
 } catch (e, stackTrace) {
 print('[API_CLIENT_ERROR] Recommendation event failed: $e');
 print('[API_CLIENT_ERROR] Stack trace: $stackTrace');
 rethrow;
 }
 }
 }
}

```

#### File: lib/services/fl\_client.dart

```

import '../models/matrix_factorization.dart';
import '../utils/model_encoder.dart';
import '../utils/resource_monitor.dart';
import 'api_client.dart';

/// Federated Learning Client for mobile devices
class FederatedLearningClient {
 final ApiClient apiClient;
 final ResourceMonitor resourceMonitor;

 MatrixFactorization? model;
 String clientId;
 String serverUrl;

 int numUsers;
 int numItems;
 int embeddingDim;

 // Training configuration
 double learningRate = 0.01;
 int localEpochs = 1;
 int batchSize = 32;

 // Local training data: List of (user_id, item_id, rating) tuples
 List<List<dynamic>> localData = [];

 FederatedLearningClient({
 required this.clientId,
 required this.serverUrl,
 required this.numUsers,
 required this.numItems,
 this.embeddingDim = 16,
 }) : apiClient = ApiClient(serverUrl: serverUrl),
 resourceMonitor = ResourceMonitor();

 /// Initialize client (check server, register, initialize monitoring)
 Future<void> initialize() async {
 print('[FL_CLIENT] Initializing client: $clientId');
 try {
 // Check server health
 print('[FL_CLIENT] Checking server health...');
 await apiClient.healthCheck();
 print('[FL_CLIENT] Server health check passed');

 // Register with server
 print('[FL_CLIENT] Registering client...');
 await apiClient.register(clientId);
 print('[FL_CLIENT] Client registered successfully');

 // Initialize resource monitoring
 print('[FL_CLIENT] Initializing resource monitoring...');
 try {
 await resourceMonitor.initialize();
 print('[FL_CLIENT] Resource monitoring initialized');
 } catch (e) {
 print('[FL_CLIENT_WARN] Resource monitoring unavailable: $e');
 }
 } catch (e, stackTrace) {
 print('[FL_CLIENT_ERROR] Initialization failed: $e');
 print('[FL_CLIENT_ERROR] Stack trace: $stackTrace');
 rethrow;
 }
 }
}

```

```

 }
}

/// Load local training data
void loadLocalData(List<List<dynamic>> data) {
 localData = data; // data: [[user_id, item_id, rating], ...]
}

/// Check if local data already contains a user-item interaction.
bool hasInteraction({required int userId, required int itemId}) {
 for (final sample in localData) {
 if (sample.length < 2) {
 continue;
 }
 final sampleUserId = sample[0];
 final sampleItemId = sample[1];
 if (sampleUserId == userId && sampleItemId == itemId) {
 return true;
 }
 }
 return false;
}

/// Add a seen interaction to local data if it does not exist yet.
bool addSeenInteraction({
 required int userId,
 required int itemId,
 double rating = 1.0,
}) {
 if (userId < 0 || userId >= numUsers || itemId < 0 || itemId >= numItems) {
 return false;
 }
 if (hasInteraction(userId: userId, itemId: itemId)) {
 return false;
 }
 localData.add([userId, itemId, rating]);
 return true;
}

/// Mark recommendation as seen and upload feedback event to server.
Future<Map<String, dynamic>> markRecommendationSeen({
 required int userId,
 required int itemId,
 double rating = 1.0,
}) async {
 addSeenInteraction(userId: userId, itemId: itemId, rating: rating);
 return apiClient.sendRecommendationSeenEvent(
 clientId: clientId,
 userId: userId,
 itemId: itemId,
 rating: rating,
);
}

/// Fetch global model from server
Future<void> fetchGlobalModel() async {
 print('[FL_CLIENT] Fetching global model...');
 try {
 final response = await apiClient.fetchGlobalParams();

 // Extract model config
 final modelConfig = response['model_config'] as Map<String, dynamic>;
 numUsers = modelConfig['num_users'] as int;
 numItems = modelConfig['num_items'] as int;
 embeddingDim = modelConfig['embedding_dim'] as int;
 print(
 '[FL_CLIENT] Model config: numUsers=$numUsers, numItems=$numItems, embeddingDim=$embeddingDim',
);

 // Decode parameters
 final paramsJson = response['params'] as List;
 print('[FL_CLIENT] Decoding ${paramsJson.length} parameter tensors...');
 final stateDict = ModelEncoder.jsonParamsToModel(
 paramsJson.map((p) => p as Map<String, dynamic>).toList(),

```

```
);
print('[FL_CLIENT] Parameters decoded successfully');

// Create and load model
print('[FL_CLIENT] Creating model instance...');
model = MatrixFactorization(
 numUsers: numUsers,
 numItems: numItems,
 embeddingDim: embeddingDim,
);
model!.loadStateDict(stateDict);
print('[FL_CLIENT] Model loaded successfully');
} catch (e, stackTrace) {
 print('[FL_CLIENT_ERROR] Fetch global model failed: $e');
 print('[FL_CLIENT_ERROR] Stack trace: $stackTrace');
 rethrow;
}
}

/// Train model locally using local data
Future<Map<String, dynamic>> trainLocal() async {
 print('[FL_CLIENT] Starting local training...');
 if (model == null) {
 print('[FL_CLIENT_ERROR] Model not initialized');
 throw StateError('Model not initialized. Call fetchGlobalModel() first.');
```

```
 }

 if (localData.isEmpty) {
 print('[FL_CLIENT] No local data available, returning empty metrics');
 return {'loss': 0.0, 'samples': 0, 'epochs': 0};
 }

 print(
 '[FL_CLIENT] Training with ${localData.length} samples, $localEpochs epochs, batch size $batchSize',
);

 final startTime = DateTime.now();
 double totalLoss = 0.0;
 int numBatches = 0;

 // Simple SGD training (one epoch for mobile efficiency)
 for (int epoch = 0; epoch < localEpochs; epoch++) {
 // Shuffle data
 localData.shuffle();

 // Process in batches
 for (int i = 0; i < localData.length; i += batchSize) {
 final batchEnd = (i + batchSize < localData.length)
 ? i + batchSize
 : localData.length;
 final batch = localData.sublist(i, batchEnd);

 // Compute gradients (simplified - full implementation would use proper backprop)
 double batchLoss = 0.0;

 // Initialize gradients
 final userGrad = List.generate(
 numUsers,
 (_) => List.filled(embeddingDim, 0.0),
);
 final itemGrad = List.generate(
 numItems,
 (_) => List.filled(embeddingDim, 0.0),
);

 // Compute gradients for batch
 for (var sample in batch) {
 final userId = sample[0] as int;
 final itemId = sample[1] as int;
 final rating = (sample[2] as num).toDouble();

 // Validate IDs are in range before accessing embeddings
 if (userId < 0 ||
 userId >= numUsers ||
```

```

 itemId < 0 ||
 itemId >= numItems) {
 print(
 '[FL_CLIENT_ERROR] Invalid IDs in sample: user=$userId (valid: 0-${numUsers - 1}), item=$itemId (valid: 0-${numItems - 1})',
);
 continue; // Skip invalid samples
}

// Forward pass
final prediction = model!.predict(userId, itemId);
final error = prediction - rating;
batchLoss += error * error;

// Backward pass (simplified gradient)
final userEmb = model!.userEmbeddings[userId];
final itemEmb = model!.itemEmbeddings[itemId];

for (int j = 0; j < embeddingDim; j++) {
 userGrad[userId][j] += error * itemEmb[j];
 itemGrad[itemId][j] += error * userEmb[j];
}

// Average gradients
final actualBatchSize = batch.length;
for (int u = 0; u < numUsers; u++) {
 for (int j = 0; j < embeddingDim; j++) {
 userGrad[u][j] /= actualBatchSize;
 }
}
for (int i = 0; i < numItems; i++) {
 for (int j = 0; j < embeddingDim; j++) {
 itemGrad[i][j] /= actualBatchSize;
 }
}

// Update parameters
model!.updateParameters(userGrad, itemGrad, learningRate);

totalLoss += batchLoss / actualBatchSize;
numBatches++;
}

final trainingTime = DateTime.now().difference(startTime).inMilliseconds;
final avgLoss = numBatches > 0 ? totalLoss / numBatches : 0.0;

print(
 '[FL_CLIENT] Training complete: loss=$avgLoss, samples=${localData.length}, time=${trainingTime}ms',
);

return {
 'loss': avgLoss,
 'samples': localData.length,
 'epochs': localEpochs,
 'training_time_ms': trainingTime,
};
}

/// Upload local model parameters to server
Future<Map<String, dynamic>> uploadParams() async {
 print('[FL_CLIENT] Uploading parameters...');
 if (model == null) {
 print('[FL_CLIENT_ERROR] Model not initialized');
 throw StateError('Model not initialized');
 }

 print('[FL_CLIENT] Getting model state dict...');
 final stateDict = model!.getStateDict();
 print('[FL_CLIENT] Encoding parameters to JSON...');
 final paramsJson = ModelEncoder.modelToJsonParams(stateDict);
 print('[FL_CLIENT] Encoded ${paramsJson.length} parameter tensors');
}

```

```
// Get resource metrics
print('[FL_CLIENT] Getting resource metrics...');
final metrics = await resourceMonitor.getMetrics();
print('[FL_CLIENT] Resource metrics: $metrics');

print('[FL_CLIENT] Uploading to server...');
final response = await apiClient.uploadParams(
 clientId: clientId,
 params: paramsJson,
 sampleCount: localData.length,
);
print('[FL_CLIENT] Upload successful');

return {...response, 'resource_metrics': metrics};
}

/// Evaluate model on local data (used as test set proxy)
Map<String, dynamic> evaluateModel() {
 print('[FL_CLIENT] Evaluating model on local data...');
 if (model == null) {
 print('[FL_CLIENT_ERROR] Model not initialized');
 return {
 'accuracy': 0.0,
 'mse': 0.0,
 'mae': 0.0,
 'Hit@10': 0.0,
 'NDCG@10': 0.0,
 'Precision@10': 0.0,
 'Recall@10': 0.0,
 };
 }

 if (localData.isEmpty) {
 print('[FL_CLIENT] No local data for evaluation');
 return {
 'accuracy': 0.0,
 'mse': 0.0,
 'mae': 0.0,
 'Hit@10': 0.0,
 'NDCG@10': 0.0,
 'Precision@10': 0.0,
 'Recall@10': 0.0,
 };
 }

 double totalLoss = 0.0;
 double totalMae = 0.0;
 int correctPredictions = 0;
 int totalSamples = 0;

 // Evaluate on local data
 for (var sample in localData) {
 final userId = sample[0] as int;
 final itemId = sample[1] as int;
 final rating = (sample[2] as num).toDouble();

 // Validate IDs
 if (userId < 0 ||
 userId >= numUsers ||
 itemId < 0 ||
 itemId >= numItems) {
 continue;
 }

 try {
 final prediction = model!.predict(userId, itemId);
 final error = prediction - rating;

 totalLoss += error * error; // MSE
 totalMae += error.abs();

 // Binary classification accuracy (threshold at 0.5)
 final predBinary = prediction > 0.5 ? 1.0 : 0.0;
 if (predBinary == rating) {
```

```

 correctPredictions++;
 }
 totalSamples++;
} catch (e) {
 print(
 '[FL_CLIENT_ERROR] Evaluation error for user=$userId, item=$itemId: $e',
);
}
}

if (totalSamples == 0) {
 return {
 'accuracy': 0.0,
 'mse': 0.0,
 'mae': 0.0,
 'Hit@10': 0.0,
 'NDCG@10': 0.0,
 'Precision@10': 0.0,
 'Recall@10': 0.0,
 };
}

final mse = totalLoss / totalSamples;
final mae = totalMae / totalSamples;
final accuracy = correctPredictions / totalSamples;

// For recommendation metrics, we'll use simplified versions
// Since we don't have a full test set, we compute basic metrics
final hitAt10 = accuracy; // Simplified: using accuracy as proxy
final ndcgAt10 = accuracy; // Simplified
final precisionAt10 = accuracy;
final recallAt10 = accuracy;

print(
 '[FL_CLIENT] Evaluation complete: accuracy=$accuracy, mse=$mse, mae=$mae',
);

return {
 'accuracy': accuracy,
 'mse': mse,
 'mae': mae,
 'Hit@10': hitAt10,
 'NDCG@10': ndcgAt10,
 'Precision@10': precisionAt10,
 'Recall@10': recallAt10,
 'samples': totalSamples,
};
}

/// Execute one complete federated learning round
Future<Map<String, dynamic>> runTrainingRound() async {
 print('[FL_CLIENT] ===== Starting training round =====');
 try {
 // 1. Fetch global model (will throw if not initialized)
 print('[FL_CLIENT] Step 1/3: Fetching global model...');
 await fetchGlobalModel();

 // 2. Train locally
 print('[FL_CLIENT] Step 2/3: Training locally...');
 final trainMetrics = await trainLocal();

 // 2.5. Evaluate model after training (before uploading)
 print('[FL_CLIENT] Step 2.5/3: Evaluating model...');
 final testMetrics = evaluateModel();

 // 3. Upload parameters
 print('[FL_CLIENT] Step 3/3: Uploading parameters...');
 final uploadMetrics = await uploadParams();

 print('[FL_CLIENT] ===== Training round complete =====');
 return {
 'train': trainMetrics,
 'test': testMetrics,
 'upload': uploadMetrics,
 };
 } catch (e) {
 print('[FL_CLIENT] ===== Training round failed =====');
 return {
 'train': null,
 'test': null,
 'upload': null,
 };
 }
}

```

```

 };
 } catch (e, stackTrace) {
 print('[FL_CLIENT_ERROR] Training round failed: $e');
 print('[FL_CLIENT_ERROR] Stack trace: $stackTrace');
 rethrow;
 }
}
}
}

```

#### File: lib/services/movie\_title\_service.dart

```

import 'package:flutter/services.dart' show rootBundle;

class MovieTitleService {
 const MovieTitleService();

 static const String _assetPath = 'assets/movielens_titles.csv';

 Future<Map<int, String>> loadTitles() async {
 final csvData = await rootBundle.loadString(_assetPath);
 final titles = <int, String>{};

 for (final rawLine in csvData.split('\n')) {
 final line = rawLine.trim();
 if (line.isEmpty || line.startsWith('item_id')) {
 continue;
 }

 final commaIndex = line.indexOf(',');
 if (commaIndex <= 0) {
 continue;
 }

 final idText = line.substring(0, commaIndex).trim();
 final itemId = int.tryParse(idText);
 if (itemId == null) {
 continue;
 }

 var title = line.substring(commaIndex + 1).trim();
 if (title.startsWith('"') && title.endsWith('"') && title.length >= 2) {
 title = title.substring(1, title.length - 1).replaceAll('"', '');
 }

 titles[itemId] = title;
 }

 return titles;
 }
}

```

#### File: lib/services/recommendation\_service.dart

```

import 'dart:math';

import 'fl_client.dart';

class MovieRecommendation {
 final int itemId;
 final String title;
 final double score;
 final bool isSeen;
 final String explanation;

 const MovieRecommendation({
 required this.itemId,
 required this.title,
 required this.score,
 required this.isSeen,
 required this.explanation,
 });

 MovieRecommendation copyWith({
 int? itemId,

```



```
String? title,
double? score,
bool? isSeen,
String? explanation,
}) {
 return MovieRecommendation(
 itemId: itemId ?? this.itemId,
 title: title ?? this.title,
 score: score ?? this.score,
 isSeen: isSeen ?? this.isSeen,
 explanation: explanation ?? this.explanation,
);
}
}

class RecommendationService {
 const RecommendationService();

 List<int> getCandidateUsers(FederatedLearningClient client) {
 final ids =
 client.localData
 .map((sample) => sample[0])
 .whereType<int>()
 .where((id) => id >= 0 && id < client.numUsers)
 .toSet()
 .toList()
 ..sort();

 if (ids.isNotEmpty) {
 return ids;
 }

 if (client.numUsers > 0) {
 return [0];
 }

 return <int>[];
 }

 List<MovieRecommendation> topNRecommendations({
 required FederatedLearningClient client,
 required int userId,
 required int topN,
 Map<int, String> movieTitles = const <int, String>{},
 bool excludeSeenItems = true,
 }) {
 if (client.model == null) {
 throw StateError('Model is not loaded.');
```

```

 continue;
 }

 final score = client.model!.predict(userId, itemId);
 final movieTitle = movieTitles[itemId] ?? 'Movie #$itemId';
 final seenLabel = isSeen ? 'Seen before' : 'New for this user';
 final explanation =
 'Predicted score ${score.toStringAsFixed(4)} - $seenLabel';

 recommendations.add(
 MovieRecommendation(
 itemId: itemId,
 title: movieTitle,
 score: score,
 isSeen: isSeen,
 explanation: explanation,
),
);
};
}

recommendations.sort((a, b) => b.score.compareTo(a.score));
return recommendations.take(max(0, topN)).toList(growable: false);
}
}

```

#### File: lib/utils/metrics\_collector.dart

```

import 'dart:convert';
import 'dart:io';
import 'package:path_provider/path_provider.dart';
import 'package:csv/csv.dart';
import 'package:flutter/foundation.dart';
import 'package:path/path.dart' as path;

/// Metrics Collector for Federated Learning Experiments (Mobile)
/// Collects and saves all experimental metrics to JSON/CSV
class MetricsCollector {
 final String experimentId;
 late Directory resultsDir;

 // Store all collected metrics
 final Map<String, dynamic> experimentData;

 MetricsCollector({
 required this.experimentId,
 }) : experimentData = {
 'experiment_id': experimentId,
 'timestamp': DateTime.now().toIso8601String(),
 'config': <String, dynamic>{},
 'rounds': <Map<String, dynamic>>[],
 'final_metrics': <String, dynamic>{},
 'client_metrics': <Map<String, dynamic>>[],
 'mobile_metrics': <Map<String, dynamic>>[],
 };

 /// Initialize the results directory - saves to Downloads folder for easy access
 Future<void> initialize() async {
 // Check if running on web (not supported)
 try {
 if (kIsWeb) {
 throw UnsupportedError('Metrics collection not supported on web');
 }
 } catch (e) {
 // kIsWeb might not be available, continue anyway
 }

 // Try to use external storage Downloads folder (most accessible)
 try {
 final externalDir = await getExternalStorageDirectory();
 if (externalDir != null) {
 // Navigate to Downloads folder
 // Android external storage structure: /storage/emulated/0/Download
 final downloadsPath = Platform.isAndroid
 ? '/storage/emulated/0/Download/FL_Results'

```

```

 : '${externalDir.path}/Downloads/FL_Results';

 // Try the standard Downloads path first
 final downloadsDir = Directory(downloadsPath);

 // If that doesn't exist, try alternative paths
 if (!await downloadsDir.exists() && Platform.isAndroid) {
 // Alternative: Use external storage root + Download
 final altPath = path.join(externalDir.parent.path, 'Download', 'FL_Results');
 resultsDir = Directory(altPath);
 } else {
 resultsDir = downloadsDir;
 }

 if (!await resultsDir.exists()) {
 await resultsDir.create(recursive: true);
 }

 print('[METRICS_COLLECTOR] Using Downloads folder: ${resultsDir.path}');
 return;
}
} catch (e) {
 print('[METRICS_COLLECTOR] Warning: Downloads folder failed: $e');
}

// Fallback: Try external storage directory
try {
 final externalDir = await getExternalStorageDirectory();
 if (externalDir != null) {
 resultsDir = Directory('${externalDir.path}/FL_Results');
 if (!await resultsDir.exists()) {
 await resultsDir.create(recursive: true);
 }
 print('[METRICS_COLLECTOR] Using external storage: ${resultsDir.path}');
 return;
 }
} catch (e) {
 print('[METRICS_COLLECTOR] Warning: External storage failed: $e');
}

// Fallback: Application documents directory
try {
 final appDocDir = await getApplicationDocumentsDirectory();
 resultsDir = Directory('${appDocDir.path}/FL_Results');

 if (!await resultsDir.exists()) {
 await resultsDir.create(recursive: true);
 }

 print('[METRICS_COLLECTOR] Using app documents: ${resultsDir.path}');
 return;
} catch (e) {
 print('[METRICS_COLLECTOR] Warning: Application documents directory failed: $e');
}

// Last resort: temporary directory
try {
 final tempDir = await getTemporaryDirectory();
 resultsDir = Directory('${tempDir.path}/FL_Results');
 if (!await resultsDir.exists()) {
 await resultsDir.create(recursive: true);
 }
 print('[METRICS_COLLECTOR] Using temporary directory: ${resultsDir.path}');
} catch (e) {
 throw Exception('Could not initialize any storage directory: $e');
}

}

/// Get the results directory path (for display)
String get resultsPath => resultsDir.path;

/// Get the full experiment data as a Map (for uploading to server)
Map<String, dynamic> getExperimentData() {
 return Map<String, dynamic>.from(experimentData);
}

```

```

}

/// Get list of all result files
Future<List<File>> getResultFiles() async {
 try {
 if (!await resultsDir.exists()) {
 return [];
 }

 final files = resultsDir.listSync()
 .whereType<File>()
 .where((f) => f.path.endsWith('.json') || f.path.endsWith('.csv'))
 .toList();

 return files;
 } catch (e) {
 print('[METRICS_COLLECTOR] Error listing files: $e');
 return [];
 }
}

/// Set experiment configuration
void setConfig(Map<String, dynamic> config) {
 experimentData['config'] = config;
 print('[METRICS_COLLECTOR] Set experiment config');
}

/// Add metrics for a training round
void addRoundMetrics({
 required int roundNum,
 required double trainLoss,
 Map<String, dynamic>? testMetrics,
 Map<String, dynamic>? aggregationInfo,
 List<Map<String, dynamic>>? clientMetrics,
 Map<String, dynamic>? resourceMetrics,
}) {
 final roundData = <String, dynamic>{
 'round': roundNum,
 'train_loss': trainLoss,
 'test_metrics': testMetrics ?? <String, dynamic>{},
 'aggregation': aggregationInfo ?? <String, dynamic>{},
 'client_metrics': clientMetrics ?? <Map<String, dynamic>>[],
 'resource_metrics': resourceMetrics ?? <String, dynamic>{},
 'timestamp': DateTime.now().toIso8601String(),
 };

 (experimentData['rounds'] as List).add(roundData);
 print('[METRICS_COLLECTOR] Added metrics for round $roundNum');
}

/// Add final evaluation metrics
void addFinalMetrics(Map<String, dynamic> metrics) {
 experimentData['final_metrics'] = metrics;
 print('[METRICS_COLLECTOR] Added final metrics');
}

/// Add per-client metrics
void addClientMetrics(String clientId, Map<String, dynamic> metrics) {
 final clientData = <String, dynamic>{
 'client_id': clientId,
 ...metrics,
 'timestamp': DateTime.now().toIso8601String(),
 };

 (experimentData['client_metrics'] as List).add(clientData);
}

/// Add mobile device resource metrics
void addMobileMetrics(String deviceId, Map<String, dynamic> metrics) {
 final mobileData = <String, dynamic>{
 'device_id': deviceId,
 ...metrics,
 'timestamp': DateTime.now().toIso8601String(),
 };
}

```

```

 (experimentData['mobile_metrics'] as List).add(mobileData);
}

/// Save metrics to JSON file
Future<String> saveJson([String? filename]) async {
 await initialize();

 filename ??= '$experimentId.json';

 final file = File('${resultsDir.path}/$filename');

 // Convert to JSON with proper formatting
 final jsonString = JsonEncoder.withIndent(' ').convert(experimentData);
 await file.writeAsString(jsonString);

 print('[METRICS_COLLECTOR] Saved JSON to: ${file.path}');
 return file.path;
}

/// Save summary to CSV file
Future<String> saveCsvSummary([String? filename]) async {
 await initialize();

 filename ??= '${experimentId}_summary.csv';

 final file = File('${resultsDir.path}/$filename');

 // Create CSV rows
 final List<List<dynamic>> csvData = [];

 // Header
 csvData.add([
 'Round',
 'Train_Loss',
 'NDCG@10',
 'Hit@10',
 'Precision@10',
 'Recall@10',
 'MSE',
 'MAE',
 'Accuracy',
 'Num_Clients',
 'Total_Samples',
 'Training_Time_MS',
 'Battery_Drain',
]);

 // Round data
 final rounds = experimentData['rounds'] as List;
 for (final roundData in rounds) {
 final testMetrics = roundData['test_metrics'] as Map<String, dynamic>? ?? {};
 final agg = roundData['aggregation'] as Map<String, dynamic>? ?? {};
 final resourceMetrics = roundData['resource_metrics'] as Map<String, dynamic>? ?? {};

 csvData.add([
 roundData['round'],
 roundData['train_loss']?.toString() ?? '',
 testMetrics['NDCG@10']?.toString() ?? '',
 testMetrics['Hit@10']?.toString() ?? '',
 testMetrics['Precision@10']?.toString() ?? '',
 testMetrics['Recall@10']?.toString() ?? '',
 testMetrics['mse']?.toString() ?? '',
 testMetrics['mae']?.toString() ?? '',
 testMetrics['accuracy']?.toString() ?? '',
 agg['num_clients']?.toString() ?? '',
 agg['total_samples']?.toString() ?? '',
 resourceMetrics['training_time_ms']?.toString() ?? '',
 resourceMetrics['battery_drain']?.toString() ?? '',
]);
 }

 // Final metrics row
 final finalMetrics = experimentData['final_metrics'] as Map<String, dynamic>? ?? {};

```

```

csvData.add([
 'FINAL',
 '',
 finalMetrics['NDCG@10']?.toString() ?? '',
 finalMetrics['Hit@10']?.toString() ?? '',
 finalMetrics['Precision@10']?.toString() ?? '',
 finalMetrics['Recall@10']?.toString() ?? '',
 finalMetrics['mse']?.toString() ?? '',
 finalMetrics['mae']?.toString() ?? '',
 finalMetrics['accuracy']?.toString() ?? '',
 '',
 '',
 '',
 '',
 ''
]);

// Write CSV
final csvString = const ListToCsvConverter().convert(csvData);
await file.writeAsString(csvString);

print('[METRICS_COLLECTOR] Saved CSV summary to: ${file.path}');
return file.path;
}

/// Get summary statistics
Map<String, dynamic> getSummary() {
 final rounds = experimentData['rounds'] as List;

 if (rounds.isEmpty) {
 return {'message': 'No rounds collected yet'};
 }

 // Extract metrics across rounds
 final trainLosses = rounds
 .map((r) => (r as Map)['train_loss'] as double?)
 .where((loss) => loss != null)
 .cast<double>()
 .toList();

 final ndcgScores = rounds
 .map((r) {
 final testMetrics = (r as Map)['test_metrics'] as Map?;
 return testMetrics?['NDCG@10'] as double?;
 })
 .where((score) => score != null)
 .cast<double>()
 .toList();

 final hitScores = rounds
 .map((r) {
 final testMetrics = (r as Map)['test_metrics'] as Map?;
 return testMetrics?['Hit@10'] as double?;
 })
 .where((score) => score != null)
 .cast<double>()
 .toList();

 return {
 'experiment_id': experimentId,
 'num_rounds': rounds.length,
 'final_train_loss': trainLosses.isNotEmpty ? trainLosses.last : null,
 'final_ndcg@10': ndcgScores.isNotEmpty ? ndcgScores.last : null,
 'final_hit@10': hitScores.isNotEmpty ? hitScores.last : null,
 'best_ndcg@10': ndcgScores.isNotEmpty ? ndcgScores.reduce((a, b) => a > b ? a : b) : null,
 'best_hit@10': hitScores.isNotEmpty ? hitScores.reduce((a, b) => a > b ? a : b) : null,
 'config': experimentData['config'],
 };
}

}

/// Create standardized experiment ID
String createExperimentId({
 double? dpEpsilon,
 double? alpha,

```

```

 int embeddingDim = 16,
 int numClients = 3,
 int? seed,
 String? deviceId,
 }) {
 final parts = <String>[];

 if (dpEpsilon == null) {
 parts.add('dp_inf');
 } else {
 parts.add('dp_$dpEpsilon');
 }

 if (alpha != null) {
 parts.add('alpha_$alpha');
 }

 parts.add('dim_$embeddingDim');
 parts.add('clients_$numClients');

 if (seed != null) {
 parts.add('seed_$seed');
 }

 if (deviceId != null) {
 // Clean device ID for filename
 final cleanId = deviceId.replaceAll(RegExp(r'^\w-'), '_');
 parts.add('device_$cleanId');
 }

 // Add timestamp for uniqueness
 final timestamp = DateTime.now().millisecondsSinceEpoch;
 parts.add('t$timestamp');

 return parts.join('_');
 }

```

#### File: lib/utils/model\_encoder.dart

```

import 'dart:convert';
import 'package:flutter/foundation.dart';

/// Utilities for encoding/decoding model parameters to/from JSON format
/// Compatible with Python server's portable JSON format

class ModelEncoder {
 /// Convert model parameters to portable JSON format (compatible with server)
 /// Returns list of parameter dictionaries with base64-encoded data
 static List<Map<String, dynamic>> modelToJsonParams(
 Map<String, dynamic> stateDict,
) {
 print('[MODEL_ENCODER] Encoding model parameters to JSON...');
 print('[MODEL_ENCODER] State dict keys: ${stateDict.keys.toList()}');
 final params = <Map<String, dynamic>>[];

 stateDict.forEach((name, value) {
 print('[MODEL_ENCODER] Processing parameter: $name');
 if (value is List<List<double>>) {
 // Convert 2D list to flat array and then to base64
 final flatList = <double>[];
 var shape = <int>[];

 if (value.isNotEmpty) {
 shape = [value.length, value[0].length];
 print('[MODEL_ENCODER] Parameter $name shape: $shape');
 for (var row in value) {
 flatList.addAll(row);
 }
 } else {
 print('[MODEL_ENCODER] Warning: Parameter $name is empty');
 }
 }

 // Convert to Float32 bytes (4 bytes per float)
 });
 }

```

```

 final bytes = <int>[];
 for (var val in flatList) {
 // Convert double to Float32 bytes (simplified - actual implementation
 // should use proper Float32 encoding)
 final bytes32 = _doubleToFloat32Bytes(val);
 bytes.addAll(bytes32);
 }

 print('[MODEL_ENCODER] Parameter $name: ${flatList.length} values, ${bytes.length} bytes');

 // Encode to base64
 final base64Data = base64Encode(bytes);
 print('[MODEL_ENCODER] Parameter $name base64 length: ${base64Data.length}');

 params.add({
 'name': name,
 'shape': shape,
 'dtype': 'float32',
 'data': base64Data,
 });
} else {
 print('[MODEL_ENCODER] Warning: Parameter $name has unsupported type: ${value.runtimeType}');
}
});

print('[MODEL_ENCODER] Encoded ${params.length} parameters');
return params;
}

/// Convert Float32 bytes (4 bytes) from double
static List<int> _doubleToFloat32Bytes(double value) {
 // Simplified: convert double to 4-byte representation
 // For production, use proper Float32 encoding
 final bytes = ByteData(4);
 bytes.setFloat32(0, value.toDouble(), Endian.little);
 return bytes.buffer.asUint8List().toList();
}

/// Decode JSON parameters into model state dictionary
static Map<String, dynamic> jsonParamsToModel(List<Map<String, dynamic>> params) {
 print('[MODEL_ENCODER] Decoding ${params.length} parameters from JSON...');
 final stateDict = <String, dynamic>{};

 for (var param in params) {
 try {
 final name = param['name'] as String;
 final shape = List<int>.from(param['shape'] as List);
 final dtype = param['dtype'] as String;
 final base64Data = param['data'] as String;

 print('[MODEL_ENCODER] Decoding parameter: $name, shape: $shape, dtype: $dtype');
 print('[MODEL_ENCODER] Base64 data length: ${base64Data.length}');

 if (dtype == 'float32') {
 // Decode base64
 final bytes = base64Decode(base64Data);
 print('[MODEL_ENCODER] Decoded bytes length: ${bytes.length}');

 // Convert bytes to Float32 values
 final values = <double>[];
 for (int i = 0; i < bytes.length; i += 4) {
 if (i + 4 <= bytes.length) {
 final byteData = ByteData.view(
 bytes.buffer,
 bytes.offsetInBytes + i,
 4,
);
 values.add(byteData.getFloat32(0, Endian.little));
 }
 }

 print('[MODEL_ENCODER] Decoded ${values.length} float32 values');

 // Reshape to 2D list

```



```

 if (shape.length == 2) {
 final matrix = <List<double>>[];
 for (int i = 0; i < shape[0]; i++) {
 final row = <double>[];
 for (int j = 0; j < shape[1]; j++) {
 final idx = i * shape[1] + j;
 if (idx < values.length) {
 row.add(values[idx]);
 } else {
 print('[MODEL_ENCODER] Warning: Index out of bounds for $name at [$i, $j]');
 }
 }
 matrix.add(row);
 }
 stateDict[name] = matrix;
 print('[MODEL_ENCODER] Successfully decoded parameter $name: ${matrix.length}x${matrix.isNotEmpty ? matrix[0].length : 0}');
 } else {
 print('[MODEL_ENCODER] Warning: Parameter $name has unsupported shape length: ${shape.length}')
 }
 }
 } else {
 print('[MODEL_ENCODER] Warning: Parameter $name has unsupported dtype: $dtype');
 }
} catch (e, stackTrace) {
 print('[MODEL_ENCODER_ERROR] Failed to decode parameter: $e');
 print('[MODEL_ENCODER_ERROR] Stack trace: $stackTrace');
 rethrow;
}
}

print('[MODEL_ENCODER] Decoded ${stateDict.length} parameters');
return stateDict;
}
}

```

#### File: lib/Utils/resource\_monitor.dart

```

import 'dart:io';
import 'package:battery_plus/battery_plus.dart';
import 'package:device_info_plus/device_info_plus.dart';

/// Resource monitoring utilities for mobile devices
/// Tracks CPU, memory, battery, and network usage

class ResourceMonitor {
 final Battery _battery = Battery();
 final DeviceInfoPlugin _deviceInfo = DeviceInfoPlugin();

 int? _initialBatteryLevel;
 DateTime? _startTime;

 /// Initialize monitoring (record initial state)
 Future<void> initialize() async {
 print('[RESOURCE_MONITOR] Initializing resource monitoring...');
 try {
 _startTime = DateTime.now();
 _initialBatteryLevel = await _battery.batteryLevel;
 print('[RESOURCE_MONITOR] Initial battery level: $_initialBatteryLevel%');
 print('[RESOURCE_MONITOR] Start time: $_startTime');
 } catch (e, stackTrace) {
 print('[RESOURCE_MONITOR_ERROR] Initialization failed: $e');
 print('[RESOURCE_MONITOR_ERROR] Stack trace: $stackTrace');
 rethrow;
 }
 }

 /// Get current battery level
 Future<int> getBatteryLevel() async {
 return await _battery.batteryLevel;
 }

 /// Calculate battery drain since initialization

```

```

Future<double> getBatteryDrain() async {
 if (_initialBatteryLevel == null) {
 await initialize();
 }

 final currentLevel = await _battery.batteryLevel;
 return (_initialBatteryLevel! - currentLevel).toDouble();
}

/// Get device information
Future<Map<String, dynamic>> getDeviceInfo() async {
 if (Platform.isAndroid) {
 final androidInfo = await _deviceInfo.androidInfo;
 return {
 'platform': 'Android',
 'model': androidInfo.model,
 'manufacturer': androidInfo.manufacturer,
 'version': androidInfo.version.release,
 'sdkInt': androidInfo.version.sdkInt,
 };
 } else if (Platform.isIOS) {
 final iosInfo = await _deviceInfo.iosInfo;
 return {
 'platform': 'iOS',
 'model': iosInfo.model,
 'name': iosInfo.name,
 'systemVersion': iosInfo.systemVersion,
 };
 }
 return {'platform': 'Unknown'};
}

/// Get resource metrics (for reporting)
Future<Map<String, dynamic>> getMetrics() async {
 print('[RESOURCE_MONITOR] Getting resource metrics...');
 try {
 final batteryLevel = await getBatteryLevel();
 final batteryDrain = await getBatteryDrain();
 final deviceInfo = await getDeviceInfo();

 final elapsedSeconds = _startTime != null
 ? DateTime.now().difference(_startTime!).inSeconds
 : 0;

 final metrics = {
 'battery_level': batteryLevel,
 'battery_drain': batteryDrain,
 'elapsed_seconds': elapsedSeconds,
 'device_info': deviceInfo,
 'timestamp': DateTime.now().toIso8601String(),
 // Note: CPU and memory monitoring requires platform channels
 // For thesis, you can add estimates or use platform-specific packages
 'cpu_percent': _estimateCpuUsage(), // Placeholder
 'memory_mb': _estimateMemoryUsage(), // Placeholder
 };

 print(
 '[RESOURCE_MONITOR] Metrics collected: battery=$batteryLevel%, drain=$batteryDrain%, elapsed=${elapsedSeconds}s',
);
 return metrics;
 } catch (e, stackTrace) {
 print('[RESOURCE_MONITOR_ERROR] Failed to get metrics: $e');
 print('[RESOURCE_MONITOR_ERROR] Stack trace: $stackTrace');
 rethrow;
 }
}

/// Estimate CPU usage (placeholder - requires platform channels for real measurement)
double _estimateCpuUsage() {
 // For thesis, you can implement platform channels or use profiling tools
 // This is a placeholder
 return 15.0; // Estimated percentage
}

```

```

/// Estimate memory usage (placeholder)
double _estimateMemoryUsage() {
 // Requires platform channels for accurate measurement
 // For thesis, you can add actual implementation using dart:ffi or plugins
 return 50.0; // Estimated MB
}

/// Measure network bandwidth for data transfer
static int measureNetworkBytes(
 int bytesTransferred,
 DateTime start,
 DateTime end,
) {
 final duration = end.difference(start);
 final bytesPerSecond = duration.inMilliseconds > 0
 ? (bytesTransferred / (duration.inMilliseconds / 1000)).round()
 : 0;
 return bytesPerSecond;
}

```

#### File: test/recommendation\_service\_test.dart

---

```

import 'package:flutter_test/flutter_test.dart';

import 'package:federated_learning_in_mobile/models/matrix_factorization.dart';
import 'package:federated_learning_in_mobile/services/fl_client.dart';
import 'package:federated_learning_in_mobile/services/recommendation_service.dart';

void main() {
 test('RecommendationService returns titles and explanation text', () {
 final client = FederatedLearningClient(
 clientId: 'test-client',
 serverUrl: 'http://localhost:8000',
 numUsers: 3,
 numItems: 5,
 embeddingDim: 4,
);

 client.model = MatrixFactorization(
 numUsers: client.numUsers,
 numItems: client.numItems,
 embeddingDim: client.embeddingDim,
);

 client.loadLocalData([
 [0, 1, 1.0],
 [0, 3, 1.0],
]);

 const service = RecommendationService();
 final results = service.topNRecommendations(
 client: client,
 userId: 0,
 topN: 3,
 movieTitles: const {0: 'Toy Story (1995)', 2: 'Four Rooms (1995)'},
 excludeSeenItems: false,
);

 expect(results.length, 3);
 expect(results.first.title.isNotEmpty, isTrue);
 expect(results.first.explanation.contains('Predicted score'), isTrue);
 });
}

```

#### File: test/widget\_test.dart

---

```

// This is a basic Flutter widget test.
//
// To perform an interaction with a widget in your test, use the WidgetTester
// utility in the flutter_test package. For example, you can send tap and scroll
// gestures. You can also use WidgetTester to find child widgets in the widget
// tree, read text, and verify that the values of widget properties are correct.

```

```
import 'package:flutter/material.dart';
import 'package:flutter_test/flutter_test.dart';

import 'package:federated_learning_in_mobile/main.dart';

void main() {
 testWidgets('App launches and shows dashboard controls', (
 WidgetTester tester,
) async {
 // Build our app and trigger a frame.
 await tester.pumpWidget(const FederatedLearningApp());

 // Verify that the app shows the server URL field
 expect(find.text('Federated Client'), findsOneWidget);
 expect(find.text('SERVER URL'), findsOneWidget);
 expect(find.text('CONNECT TO SERVER'), findsOneWidget);
 expect(
 find.text('Demo Mode: Train and Show Recommendations'),
 findsOneWidget,
);
 expect(find.byIcon(Icons.movie), findsOneWidget);
 });
}
```

#### File: test\_driver/integration\_test.dart

---

```
import 'dart:io';

import 'package:integration_test/integration_test_driver_extended.dart';

const String _outputDirectoryPath = 'artifacts/integration_screenshots';

Future<void> main() async {
 final outputDirectory = Directory(_outputDirectoryPath);
 await outputDirectory.create(recursive: true);

 await integrationDriver(onScreenshot: _saveScreenshot);
}

Future<bool> _saveScreenshot(
 String screenshotName,
 List<int> screenshotBytes, [
 Map<String, Object?>? args,
]) async {
 final screenshotFile = File('${_outputDirectoryPath}/${screenshotName}.png');
 await screenshotFile.parent.create(recursive: true);
 await screenshotFile.writeAsBytes(screenshotBytes);
 stdout.writeln('Saved screenshot to ${screenshotFile.path}');
 return true;
}
```

#### File: tooling/build\_movie\_titles\_asset.py

---

```
import csv
from pathlib import Path

ROOT = Path(__file__).resolve().parents[2]
source = ROOT / "ml-100k" / "u.item"
out_dir = ROOT / "federated_learning_in_mobile" / "assets"
out_dir.mkdir(parents=True, exist_ok=True)
out_file = out_dir / "movielens_titles.csv"

with source.open("r", encoding="latin-1") as src, out_file.open("w", encoding="utf-8", newline="") as dst:
 writer = csv.writer(dst)
 writer.writerow(["item_id", "title"])
 count = 0
 for line in src:
 parts = line.rstrip("\n").split("|")
```

---

```
 if len(parts) < 2 or not parts[0].isdigit():
 continue
 item_id = int(parts[0]) - 1
 writer.writerow([item_id, parts[1]])
 count += 1

print(f"Wrote {count} movie titles to {out_file}")
```